Chapter 2
Function Spaces and Finite Element Approximations

In this chapter we present function spaces and suitable finite element approximations of them, which we shall use in order to apply the abstract theory of the previous chapters to problems of practical interest.

We do not aim at a general presentation of the subject of a vast literature, but we present the basic properties of the spaces we are going to use in the sequel of this book.

In particular, we consider standard results about the finite element approximation of Sobolev spaces and about approximations of $H(\mathrm{div};\Omega)$ and $H(\underline{\mathrm{curl}};\Omega)$. The results of Sect. 2.1 are technical and may be skipped by a reader interested mostly in numerical results.

Mainly for historical reasons, we present the finite element approximation of the spaces $H^1(\Omega)$, $H(\mathrm{div};\Omega)$, and $H(\underline{\mathrm{curl}};\Omega)$ separately, although they could be seen altogether in the framework of de Rham diagram. We shall briefly comment on it in Sect. 2.1.4.

2.1 Properties of the Spaces $H^m(\Omega)$, $H(\mathrm{div};\Omega)$, and $H(\underline{\mathrm{curl}};\Omega)$

2.1.1 Basic Properties

Some of the results of this section have been already anticipated in Chap. 1. Here, we present them in a unified setting.

- **Sobolev spaces $H^m(\Omega)$.** Given an integer number $m \geq 1$, standard Sobolev spaces read

$$H^m(\Omega) := \{v \mid v \in L^2(\Omega),\ D^\alpha v \in L^2(\Omega),\ |\alpha| \leq m\}, \tag{2.1.1}$$

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where

$$D^{\alpha}v := \frac{\partial^{|\alpha|}v}{\partial x_1^{\alpha_1}\partial x_2^{\alpha_2}\dots\partial x_n^{\alpha_n}}, \quad |\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n. \tag{2.1.2}$$

We shall consider the standard norm

$$\|v\|_{m,\Omega}^2 := \sum_{|\alpha|\le m} \int |D^{\alpha}v|^2 \, dx, \tag{2.1.3}$$

associated with the usual inner product and the semi-norm

$$|v|_{m,\Omega}^2 := \sum_{|\alpha|=m} \int |D^{\alpha}v|^2 \, dx. \tag{2.1.4}$$

The most important of these spaces will be for us $H^1(\Omega)$ (and some of its subspaces) and for fourth-order problems $H^2(\Omega)$.

For the study of Sobolev spaces, we refer the reader to [281, 309], and [3]. It is well-known that if $\Gamma = \partial\Omega$ is smooth enough (for instance Lipschitz continuous), it is possible to define the trace $\gamma u = u|_{\Gamma}$ of $u \in H^1(\Omega)$ on the boundary Γ. The traces of functions in $H^1(\Omega)$ span a Hilbert space, denoted by $H^{\frac{1}{2}}(\Gamma)$, that is a proper dense subspace of $L^2(\Gamma)$. The mapping,

$$\gamma : \ H^1(\Omega) \to H^{\frac{1}{2}}(\Gamma), \tag{2.1.5}$$

is surjective and possesses a continuous lifting. The norm

$$\|\gamma v\|_{\frac{1}{2},\Gamma} := \inf_{\substack{w\in H^1(\Omega)\\ \gamma w = v}} \|w\|_{1,\Omega} \tag{2.1.6}$$

is then equivalent to more standard norms on $H^1(\Omega)$ as defined in [281]. Then we can write

$$\|v\|_{\frac{1}{2},\Gamma} = \|\bar{v}\|_{1,\Omega}, \tag{2.1.7}$$

where $\bar{v}$ is the unique solution in $H^1(\Omega)$ of the Dirichlet problem,

$$\begin{cases} -\triangle\bar{v} + \bar{v} = 0, \\ \qquad \bar{v}|_{\Gamma} = v. \end{cases} \tag{2.1.8}$$

We shall denote by $H^{-\frac{1}{2}}(\Gamma)$ the dual space of $H^{\frac{1}{2}}(\Gamma)$ with the dual norm,

$$\|v^*\|_{-\frac{1}{2},\Gamma} := \sup_{v\in H^{1/2}(\Gamma)} \frac{\langle v, v^*\rangle}{\|v\|_{\frac{1}{2},\Gamma}}, \tag{2.1.9}$$

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where the bracket $\langle\cdot,\cdot\rangle$ denotes duality between $H^{-1/2}(\Gamma)$ and $H^{1/2}(\Gamma)$. It is easily checked that one has

$$\|v^*\|_{-\frac{1}{2},\Gamma} = \|\bar{v}^*\|_{1,\Omega}, \tag{2.1.10}$$

where $\bar{v}^*$ is the solution of the variational Neumann problem,

$$\int_\Omega \underline{\mathrm{grad}}\,\bar{v}^* \cdot \underline{\mathrm{grad}}\, v\, dx + \int_\Omega \bar{v}^* v\, dx = \langle v^*, v\rangle,\ \forall v \in H^1(\Omega). \tag{2.1.11}$$

Remark 2.1.1. We shall sometimes write formally $\int_\Gamma v^* v\, d\sigma$ instead of $\langle v^*, v\rangle$, to denote duality between $H^{\frac{1}{2}}(\Gamma)$ and $H^{-\frac{1}{2}}(\Gamma)$. □

We can define in the same way a trace operator γ on $H^2(\Omega)$. It is now possible to define $v|_\Gamma$ in a space denoted $H^{\frac{3}{2}}(\Gamma)$ but also traces of $\underline{\mathrm{grad}}\, v|_\Gamma \in H^{\frac{1}{2}}(\Gamma)^n$ and thus the trace of the normal derivative $\frac{\partial v}{\partial n}$. We then define

$$H_0^1(\Omega) := \{v \mid v \in H^1(\Omega),\ v|_\Gamma = 0\}, \tag{2.1.12}$$

$$H_0^2(\Omega) := \{v \mid v \in H^2(\Omega),\ v|_\Gamma = 0,\ \frac{\partial v}{\partial n}|_\Gamma = 0\}. \tag{2.1.13}$$

Remark 2.1.2. Spaces $H^{1/2}(\Gamma)$ and $H^{3/2}(\Gamma)$ are particular cases of generalized Sobolev spaces $H^s(\cdot)$ for $s \in \mathbb{R}^+$. The reader should be aware that handling Sobolev spaces $H^s(\cdot)$ where s = integer $+1/2$ requires some caution [281]. In the case of $H^{1/2}(\Gamma)$ it is important to recall some facts. Let Γ_0 be a part of Γ; then $\phi \in H^{1/2}(\Gamma_0)$ cannot be extended by zero outside Γ_0 to a function in $H^{1/2}(\Gamma)$ (even if paradoxically $\mathcal{D}(\Gamma_0)$ is dense in $H^{1/2}(\Gamma_0)$). Dually, if $\Gamma = \Gamma_0 \cup \Gamma_1$, one does not get the whole of $H^{-1/2}(\Gamma)$ by patching functions of $H^{-1/2}(\Gamma_0)$ and $H^{-1/2}(\Gamma_1)$. Unfortunately, spaces $H^{1/2}(\partial K)$ and $H^{-1/2}(\partial K)$ with K an element of a partition of Ω are met very often in the analysis of hybrid and mixed methods and one must be very careful in handling them. □

- **The space $H(\mathbf{div};\Omega)$.** Having considered standard Sobolev spaces, we now present some properties of a space specially adapted to the study of mixed and hybrid methods.

  The mathematical analysis of mixed methods will use constantly

$$H(\mathrm{div};\Omega) := \{\underline{q} \mid \underline{q} \in (L^2(\Omega))^n,\ \mathrm{div}\,\underline{q} \in L^2(\Omega)\} \tag{2.1.14}$$

  with the norm

$$\|\underline{q}\|^2_{\mathrm{div},\Omega} := |\underline{q}|^2_{0,\Omega} + |\,\mathrm{div}\,\underline{q}|^2_{0,\Omega}. \tag{2.1.15}$$

  It is then possible to define $\underline{q}\cdot\underline{n}|_\Gamma$, the normal trace of $\underline{q}$ on Γ.

Lemma 2.1.1. *For $\underline{q} \in H(\mathrm{div},\Omega)$, we can define $\underline{q}\cdot\underline{n}|_\Gamma \in H^{-\frac{1}{2}}(\Gamma)$ and we have Green's formula,*

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$$\int_\Omega \operatorname{div} \underline{q}\ v\ dx + \int_\Omega \underline{q} \cdot \underline{\operatorname{grad}}\, v\, dx = \langle \underline{q} \cdot \underline{n}, v \rangle,\ \forall v \in H^1(\Omega). \tag{2.1.16}$$

Proof. For $\underline{q} \in (\mathcal{D}(\bar{\Omega}))^n$ and $v \in \mathcal{D}(\bar{\Omega})$, we have the standard Green's formula

$$\int_\Gamma \underline{q} \cdot \underline{n}\ v\ d\sigma = \int_\Omega \operatorname{div} \underline{q}\ v\ dx + \int_\Omega \underline{q} \cdot \underline{\operatorname{grad}}\, v\, dx, \tag{2.1.17}$$

and therefore

$$\left| \int_\Gamma \underline{q} \cdot \underline{n}\ v\ d\sigma \right| \le \|\underline{q}\|_{\operatorname{div},\Omega} \|v\|_{1,\Omega}. \tag{2.1.18}$$

Moreover, the expression $\int_\Omega \operatorname{div} \underline{q}\ v\ dx + \int_\Omega \underline{q} \cdot \underline{\operatorname{grad}}\, v\, dx$ depends only on the trace $v|_\Gamma \in H^{\frac{1}{2}}(\Gamma)$. The result follows by density of $\mathcal{D}(\bar{\Omega})$ and $(\mathcal{D}(\bar{\Omega}))^n$ in $H^1(\Omega)$ and $H(\operatorname{div}; \Omega)$, respectively. □

The operator defined above also satisfies a surjectivity property.

Lemma 2.1.2. *The trace operator* $\underline{q} \in H(\operatorname{div}; \Omega) \to \underline{q} \cdot \underline{n}|_\Gamma \in H^{-\frac{1}{2}}(\Gamma)$ *is surjective.*

Proof. Let $g \in H^{-\frac{1}{2}}(\Gamma)$ be given. Then, solving in $H^1(\Omega)$

$$\int_\Omega \underline{\operatorname{grad}}\, \phi \cdot \underline{\operatorname{grad}}\, v\, dx + \int_\Omega \phi v\ dx = \langle g, v \rangle,\ \forall v \in H^1(\Omega), \tag{2.1.19}$$

and making $\underline{q} = \underline{\operatorname{grad}}\, \phi$ implies $\underline{q} \cdot \underline{n}|_\Gamma = g$. □

Let us now suppose a partition $(\Gamma = D \cup N)$ of the boundary Γ. We define

$$H^1_{0,D}(\Omega) := \{ v \mid v \in H^1(\Omega),\ v|_D = 0 \}. \tag{2.1.20}$$

In particular, we have $H^1_{0,D}(\Omega) = H^1_0(\Omega)$ if $D = \Gamma$ and $H^1_{0,D}(\Omega) = H^1(\Omega)$ if $D = \emptyset$. We shall also need the space

$$H_{0,N}(\operatorname{div}; \Omega) := \{ \underline{q} \mid \underline{q} \in H(\operatorname{div} \Omega),\ \langle \underline{q} \cdot \underline{n}, v \rangle = 0,\ \forall v \in H^1_{0,D}(\Omega) \}. \tag{2.1.21}$$

Remark 2.1.3. This space contains functions of $H(\operatorname{div}; \Omega)$ whose normal traces vanish on N. For reasons related to pathological properties of $H^{\frac{1}{2}}(D)$ and $H^{-\frac{1}{2}}(N)$, it is necessary to use definition (2.1.21) and not an expression such as $\underline{q} \cdot \underline{n}|_N = 0$ in $H^{-\frac{1}{2}}(N)$. □

In particular, we denote $H_0(\operatorname{div}; \Omega) = H_{0,N}(\operatorname{div}; \Omega)$ when $N = \Gamma$. Finally, another important subspace of $H(\operatorname{div}; \Omega)$ will be

$$H^0(\operatorname{div}; \Omega) := \{ \underline{q} \mid \underline{q} \in H(\operatorname{div}; \Omega),\ \operatorname{div}\ \underline{q} = 0 \}, \tag{2.1.22}$$

from which we deduce the following result.

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Lemma 2.1.3. *The normal trace operator $\underline{q} \to \underline{q}\cdot\underline{n}|_\Gamma$ is surjective from $H^0(\mathrm{div};\Omega)$ onto $\{\mu^* \mid \mu^* \in H^{-\frac{1}{2}}(\Gamma),\ \langle\mu^*, 1\rangle = 0\}$.*

Proof. By Green's formula (2.1.14), we have $\langle \underline{q}\cdot\underline{n}, 1\rangle = 0$ if $\underline{q} \in N^0(\mathrm{div};\Omega)$. Reciprocally, if $g \in H^{-\frac{1}{2}}(\Gamma)$ is given with $\langle g, 1\rangle = 0$, we can solve in $H^1(\Omega)/\mathbb{R}$ the Neumann problem

$$\int_\Omega \underline{\mathrm{grad}}\,\phi \cdot \underline{\mathrm{grad}}\, v\, dx = \langle g, \phi\rangle, \quad \forall \phi \in H^1(\Omega), \tag{2.1.23}$$

and taking $\underline{q} = \underline{\mathrm{grad}}\,\phi$ yields $\underline{q}\cdot\underline{n} = g$. □

Remark 2.1.4. In applications, D will be the part of Γ where Dirichlet's conditions are given, and N the part with Neumann's conditions. □

- **The space $\boldsymbol{H(\underline{\mathrm{curl}};\Omega)}$.** To conclude this section, we consider the space $H(\underline{\mathrm{curl}};\Omega)$ which will be used, in particular, for the approximation of problems arising from electro-magnetics in Chap. 11.

  For $\Omega \in \mathbb{R}^3$, we define

$$H(\underline{\mathrm{curl}};\Omega) := \{\underline{\chi} \mid \underline{\chi} \in (L^2(\Omega))^3,\ \underline{\mathrm{curl}}\,\underline{\chi} \in (L^2(\Omega))^3\}, \tag{2.1.24}$$

  where the $\underline{\mathrm{curl}}$ operator is as usual defined as

$$\underline{\mathrm{curl}}\,\underline{\chi} = \underline{\nabla} \wedge \underline{\chi} := \det\begin{pmatrix} i & j & k \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ \chi_1 & \chi_2 & \chi_3 \end{pmatrix} \tag{2.1.25}$$

  and where we are using the standard norm

$$\|\underline{\chi}\|^2_{\underline{\mathrm{curl}},\Omega} := |\underline{\chi}|^2_{0,\Omega} + |\,\underline{\mathrm{curl}}\,\underline{\chi}|^2_{0,\Omega}. \tag{2.1.26}$$

Remark 2.1.5. To complete these definitions, we must discuss the definition of $H(\mathrm{curl};\Omega)$ when Ω belongs to $\mathbb{R}^2$. First of all, we recall the possible definitions of the curl operator in this setting. Given a vector $\underline{u}(x_1, x_2) = (u_1, u_2)$, we can evaluate $\mathrm{curl}(u_1, u_2, 0)$, which is a vector oriented in the direction of x_3. This suggests the definition

$$\mathrm{curl}\,\underline{u} := \frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2}. \tag{2.1.27}$$

On the other hand, given a scalar function $\phi(x_1, x_2)$, the vector $\underline{\mathrm{curl}}(0, 0, \phi)$ is perpendicular to the direction of x_3 and drives to the following definition

$$\underline{\mathrm{curl}}\,\phi := \left(\frac{\partial\phi}{\partial x_1}, -\frac{\partial\phi}{\partial x_2}\right). \tag{2.1.28}$$

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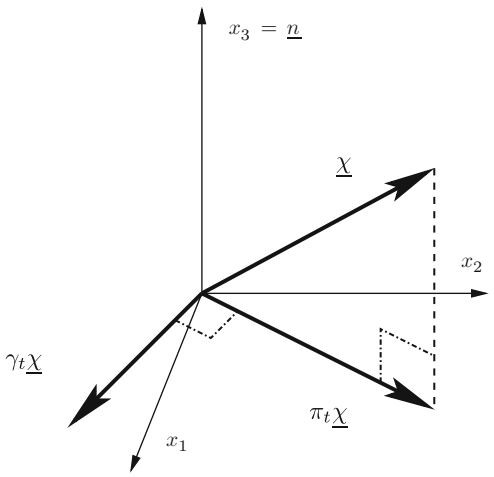


Fig. 2.1 $H(\underline{\mathrm{curl}}; \Omega)$ traces on the flat surface (x_1, x_2)

It is clear that the operators curl and $\underline{\mathrm{curl}}$ introduced above are equivalent to the operators div and $\underline{\mathrm{grad}}$, respectively, after rotation of the vectors by a right angle. More precisely, we have

$$\mathrm{curl}\, \underline{u} = -\,\mathrm{div}(\underline{u}^{\perp}), \tag{2.1.29}$$

$$\underline{\mathrm{curl}}\, \phi = -(\underline{\mathrm{grad}}\, \phi)^{\perp}, \tag{2.1.30}$$

with the standard notation $(v_1, v_2)^{\perp} = (-v_2, v_1)$. It turns out that *in two dimensions* the space $H(\mathrm{curl}; \Omega)$ is isomorphic to $H(\mathrm{div}; \Omega)$. This fact has important consequences for the construction of approximations of $H(\mathrm{curl}; \Omega)$; namely, any finite element space which is a good approximation of $H(\mathrm{div}; \Omega)$ can be turned into a good approximation of $H(\mathrm{curl}; \Omega)$ just by rotating the vectors by a right angle and vice versa. □

We are now going to state results concerning traces of $H(\underline{\mathrm{curl}}; \Omega)$ in three dimensions, in analogy of what has been done in the previous subsection for $H(\mathrm{div}; \Omega)$. Traces of $H(\underline{\mathrm{curl}}; \Omega)$ have been the object of a recent and active research by several authors [6,133–135,143,223,317] and it turns out that the theory is not as straightforward as in the case of $H(\mathrm{div}; \Omega)$; in particular we shall see that different trace definitions can be considered.

In order to get the reader acquainted with the topics which we are going to present, let us start with an easy but significant example.

Example 2.1.1 (Traces of $H(\underline{\mathrm{curl}}; \Omega)$ on a flat surface). Let $\Omega \subset \mathbb{R}^3$ be the half space $x_3 < 0$ and let Γ be the plane $x_3 = 0$. The aim of this example is to consider the basic situation of a flat boundary, i.e., when the tangent plane at any point of Γ coincides with the surface Γ itself. Given the vector $\underline{\chi} = (\chi_1, \chi_2, \chi_3)^T \in H(\underline{\mathrm{curl}}; \Omega)$, we investigate the possible definitions of tangential component of $\underline{\chi}$

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along Γ (see Fig. 2.1). As usual, we denote by $\underline{n} = (0,0,1)^T$ the unit vector pointing in the outward normal direction of Γ with respect to Ω.

We are going to give a meaning to the trace of the tangential component of $\underline{\chi}$ along Γ. Whenever $\underline{\chi} \in C^0(\overline{\Omega})$, it makes sense to consider the vector $\gamma_t \underline{\chi} = (\underline{\chi} \wedge \underline{n})_{|\Gamma}$ which, by definition, is orthogonal to $\underline{n}$ and hence is aligned with the tangent plane $x_3 = 0$. More precisely, it can be easily checked that

$$\gamma_t \underline{\chi} = \begin{pmatrix} \chi_2 \\ -\chi_1 \\ 0 \end{pmatrix}_{|\Gamma} . \tag{2.1.31}$$

On the other hand, we can also consider the projection of $\underline{\chi}$ along the plane Γ, that is the vector $\pi_t \underline{\chi} = \underline{n} \wedge (\underline{\chi} \wedge \underline{n})_{|\Gamma}$, which is again orthogonal to $\underline{n}$ and can be evaluated explicitly as follows:

$$\pi_t \underline{\chi} = \begin{pmatrix} \chi_1 \\ \chi_2 \\ 0 \end{pmatrix}_{|\Gamma} . \tag{2.1.32}$$

It is clear that in this simple example we have that $\gamma_t \underline{\chi}$ is orthogonal to $\pi_t \underline{\chi}$. Moreover, let us denote by div_Γ and $\underline{\mathrm{curl}}_\Gamma$ the divergence and the curl operators in the (x_1, x_2) plane (see Remark 2.1.5 for the definition of $\underline{\mathrm{curl}}$ in two space dimensions and the relation between operators); then, the following relationships are easy to derive

$$\mathrm{div}_\Gamma \, \gamma_t \underline{\chi} = \underline{\mathrm{curl}}_\Gamma \, \pi_t \underline{\chi} = (\underline{\mathrm{curl}} \, \underline{\chi} \cdot \underline{n})_{|\Gamma}. \tag{2.1.33}$$

Although it should be clear from the context, we remark that in (2.1.33) we are formally applying surface operators div_Γ and $\underline{\mathrm{curl}}_\Gamma$ to three dimensional vectors $\gamma_t \underline{\chi}$ and $\pi_t \underline{\chi}$. On the other hand, vectors $\gamma_t \underline{\chi}$ and $\pi_t \underline{\chi}$ are orthogonal to the direction of x_3 so that we can identify them with two dimensional vectors in the tangent space $x_3 = 0$. We shall implicitly make use of this standard abuse of notation in the sequel of this chapter. □

Let us now come back to the general picture and see how the quantities considered in the previous example extend to general domains and boundaries. The following integration by parts, which is valid for smooth functions, constitutes the starting point for the analysis

$$\int_\Omega \underline{\chi} \cdot \underline{\mathrm{curl}} \, \underline{\phi} \, dx - \int_\Omega \underline{\mathrm{curl}} \, \underline{\chi} \cdot \underline{\phi} \, dx = \int_\Gamma (\underline{\chi} \wedge \underline{n}) \cdot \underline{\phi} \, ds. \tag{2.1.34}$$

Green's formula (2.1.34) allows us to define the trace $\gamma_t \underline{\chi}$ of a function $\underline{\chi} \in H(\underline{\mathrm{curl}};\Omega)$ by extending the classical tangential trace $(\underline{\chi} \wedge \underline{n})_{|\Gamma}$.

The following lemma is the analogue of Lemma 2.1.1 and can be proved by a similar technique.

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Lemma 2.1.4. *For $\underline{\chi} \in H(\underline{curl};\Omega)$, we can define the tangential trace $\gamma_t \underline{\chi} = (\underline{\chi} \wedge \underline{n})_{|\Gamma} \in (H^{-1/2}(\Gamma))^3$ and we have Green's formula*

$$\int_\Omega \underline{\chi} \cdot \underline{curl}\, \underline{\phi}\, dx - \int_\Omega \underline{curl}\, \underline{\chi} \cdot \underline{\phi}\, dx = \langle \gamma_t \underline{\chi}, \underline{\phi} \rangle, \quad \forall \underline{\phi} \in (H^1(\Omega))^3. \tag{2.1.35}$$

In particular, it is clear from our construction that, for smooth functions, the tangential trace $\gamma_t \underline{\chi}$ is equal to $(\underline{\chi} \wedge \underline{n})_{|\Gamma}$ in the classical sense.

Let us now consider the range of the trace operator γ_t. The linear and continuous operator γ_t cannot be surjective onto $(H^{-1/2}(\Gamma))^3$ since it has vanishing component in the direction of $\underline{n}$. Moreover, as we already observed in the simplified situation of Example 2.1.1, we remark that for smooth functions it makes sense to evaluate the surface divergence of $\gamma_t \underline{\chi}$ along Γ and that the following identity holds true

$$\mathrm{div}_\Gamma(\gamma_t \underline{\chi}) = (\underline{\mathrm{curl}}\, \underline{\chi} \cdot \underline{n})_{|\Gamma}. \tag{2.1.36}$$

Indeed, the range of γ_t is usually denoted by $H^{-1/2}(\mathrm{div};\Gamma)$ and its dual space by $H^{-1/2}(\underline{\mathrm{curl}};\Gamma)$. In the case of smooth domains, it turns out that we have the following identities

$$\begin{aligned} H^{-1/2}(\mathrm{div};\Gamma) &= \{\underline{q} \in (H^{-1/2}(\Gamma))^3 | \underline{q} \cdot \underline{n} = 0 \text{ a.e. on } \Gamma,\ \mathrm{div}_\Gamma\, \underline{q} \in H^{-1/2}(\Gamma)\} \\ H^{-1/2}(\underline{\mathrm{curl}};\Gamma) &= \{\underline{q} \in (H^{-1/2}(\Gamma))^3 | \underline{q} \cdot \underline{n} = 0 \text{ a.e. on } \Gamma,\ \underline{\mathrm{curl}}_\Gamma\, \underline{q} \in H^{-1/2}(\Gamma)\}, \end{aligned} \tag{2.1.37}$$

where the surface operators div_Γ and $\underline{\mathrm{curl}}_\Gamma$ are applied, by abuse of notation, to the (two-dimensional) tangential component of $\underline{q}$.

In the case of non smooth domains, several insidious aspects have to be taken into account. In general, computational domains are polyhedra and a polyhedron is not a smooth domain. In particular, one major issue is hidden in (2.1.37) and is related to the regularity of the normal vector $\underline{n}$. If Γ is a polyhedral surface, then $\underline{n}$ jumps across the edges of Γ and the product $\underline{q} \cdot \underline{n}$ is not well defined for $\underline{q} \in (H^{-1/2}(\Gamma))^3$ since $\underline{n}$ is not a multiplier for $(H^{1/2}(\Gamma))^3$ where jumps are not allowed. We refer the interested reader to [136] for the general picture and for more details on this issue.

Remark 2.1.6. In the most general situation, the definition of div_Γ and $\underline{\mathrm{curl}}_\Gamma$ is not trivial. Even for smooth domains, we need suitable definitions of differential operators on curved surfaces. This is a typical task of differential geometry and can be performed by means of covariant derivatives. We refer the interested reader, for instance, to [168] for a thorough introduction to this subject. On the other hand, if the considered finite elements have flat faces (as it is the case for usual tetrahedra), surface differential operators reduce to standard two-dimensional ones in a local coordinate system on the face and covariant derivatives along the face are plain directional derivatives. The situation is not trivial (and not completely understood yet for what concerns the construction of good finite element spaces) in the case

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of non-flat faces (as it usually occurs for isoparametric elements or general non-affine hexahedral decompositions). We shall detail some issues for non-affine finite elements in Sect. 2.2.4. □

Looking again at Example 2.1.1, another definition which arises when dealing with the space $H(\underline{\mathrm{curl}};\Omega)$ is the projection of $\underline{\chi}$ along the tangent plane of Γ, also known as the tangential trace $\pi_t\underline{\chi}$, which can be defined as follows for smooth functions:

$$\pi_t\underline{\chi} = \underline{n}\wedge(\underline{\chi}\wedge\underline{n})_{|\Gamma}. \tag{2.1.38}$$

The following Theorem is a consequence of the results of [134–136] and states the well-posedness of the trace operator π_t together with the link between γ_t and π_t.

Theorem 2.1.1. *The trace operator π_t can be extended to an operator from $H(\underline{curl};\Omega)$ to $H^{-1/2}(\underline{curl};\Gamma)$ and the following Green's formula holds*

$$\int_\Omega \underline{curl}\,\underline{\chi}\cdot\underline{\phi}\,dx - \int_\Omega \underline{\chi}\cdot\underline{curl}\,\underline{\phi}\,dx = \langle\pi_t\underline{\phi},\gamma_t\underline{\chi}\rangle, \quad \forall\underline{\chi},\underline{\phi}\in H(\underline{curl};\Omega), \tag{2.1.39}$$

where the brackets denote the duality pairing between $H^{-1/2}(\mathrm{div};\Gamma)$ and $H^{-1/2}(\underline{curl};\Gamma)$.

If the boundary of Ω is split into two parts $\Gamma = D\cup N$, then we can consider the space

$$H_{0,N}(\underline{\mathrm{curl}};\Omega) := \{\underline{\chi}\mid\underline{\chi}\in H(\underline{\mathrm{curl}};\Omega),\ \langle\gamma_t\underline{\chi},\underline{\phi}\rangle = 0,\ \forall\underline{\phi}\in(H^1_{0,D}(\Omega))^3\}, \tag{2.1.40}$$

in analogy to what we have done in the previous subsection. When $D=\emptyset$ (and hence $N=\Gamma$), we shall make use of the notation $H_0(\underline{\mathrm{curl}};\Omega) = H_{0,\Gamma}(\underline{\mathrm{curl}};\Omega)$.

2.1.2 Properties Relative to a Partition of Ω

This section presents a short introduction to properties of some functional spaces. We refer to [331, 366] for more details.

Partitioning Ω into sub-domains is an essential feature of both standard and non-standard methods. Continuity properties at interfaces between sub-domains are an essential part in the definition of a finite element approximation. Moreover, we shall introduce here some notations that will be used throughout the book.

Let $\Omega = \bigcup_{r=1}^m K_r$ be partitioned into a family of sub-domains. In practice, K_r will be a triangle or a quadrilateral (resp., a tetrahedron or a hexahedron in three dimensions) and we shall call it *element*. We shall denote by $\mathcal{T}_h$ a partition of Ω into elements.

The edges of elements will be denoted by e_i ($i=1,2,3$ or $i=1,2,3,4$) in the two-dimensional case. For three-dimensional elements, unless differently stated, we denote again the faces of the elements by e_i ($1\le i\le 4$ or $1\le i\le 6$). We also

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denote by

$$e_{ij} = \partial K_i \cap \partial K_j, \tag{2.1.41}$$

the interface between element K_i and K_j and

$$\mathcal{E}_h = \bigcup_{ij} e_{ij} \bigcup \Gamma_h = \bigcup_K \partial K, \tag{2.1.42}$$

where Γ_h is the set of boundary edges or faces. We only deal with *compatible* meshes in the sense that the intersection between two elements is a common face, side or vertex. The situation when a mesh contains hanging nodes is out of the aims of this work.

Remark 2.1.7. The index h will of course be related to mesh size, that is to the size of elements. With an abuse of notation, we shall also use the symbol h for denoting the *maximum diameter* of the *elements of the decomposition.* □

We introduce the functional spaces

$$X(\Omega) := \{v \mid v \in L^2(\Omega),\ v|_{K_i} \in H^1(K_i),\ \forall i\} = \prod_r H^1(K_r), \tag{2.1.43}$$

with the norm

$$\|v\|^2_{X(\Omega)} := \sum_r \|v\|^2_{1,K_r}, \tag{2.1.44}$$

$$Y(\Omega) := \{\underline{q} \mid \underline{q} \in L^2(\Omega), \underline{q}\,|_{K_i} \in H(\mathrm{div}; K_i),\ \forall i\} = \prod_r H(\mathrm{div}; K_r), \tag{2.1.45}$$

with the norm

$$\|\underline{q}\|^2_{Y(\Omega)} := \sum_r \|\underline{q}\|^2_{\mathrm{div},\Omega} \tag{2.1.46}$$

and

$$W(\Omega) := \{\underline{\chi} \mid \underline{\chi} \in L^2(\Omega),\ \underline{\chi}\,|_{K_i} \in H(\underline{\mathrm{curl}}; K_i),\ \forall i\} = \prod_r H(\underline{\mathrm{curl}}; K_r), \tag{2.1.47}$$

with the norm

$$\|\underline{\chi}\|^2_{W(\Omega)} := \sum_r \|\underline{\chi}\|^2_{\underline{\mathrm{curl}},\Omega}. \tag{2.1.48}$$

We shall now characterize $H^1_{0,D}(\Omega)$, $H_{0,N}(\mathrm{div}; \Omega)$, and $H_{0,N}(\underline{\mathrm{curl}}; \Omega)$ as subspaces of $X(\Omega)$, $Y(\Omega)$, and $W(\Omega)$ respectively. Let us first remark that for $v \in H^1(\Omega)$ and $\underline{q} \in H(\mathrm{div}; \Omega)$, we have, denoting $\underline{n}_r$ the normal to $\Gamma_r = \partial K_r$,

$$\sum_r \langle \underline{q} \cdot \underline{n}_r, v\rangle_{\Gamma_r} = \langle \underline{q} \cdot \underline{n}, v\rangle_{\Gamma}, \tag{2.1.49}$$

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where $\langle\cdot,\cdot\rangle$ denotes duality between $H^{\frac{1}{2}}(\Gamma_r)$ and $H^{-\frac{1}{2}}(\Gamma_r)$. Indeed we can decompose the Green formula as

$$\langle \underline{q}\cdot\underline{n}, v\rangle_{\Gamma} = \sum_r \left\{ \int_{K_r} \mathrm{div}\,\underline{q}\; v\; dx + \int_{K_r} \underline{q}\cdot \underline{\mathrm{grad}}\, v\, dx \right\} \tag{2.1.50}$$

and apply it inside each element. A similar splitting holds for the functions $\underline{\chi} \in H(\underline{\mathrm{curl}};\Omega)$ and $\underline{\phi} \in H^1(\Omega)$ when we consider the tangential trace $\gamma_t\underline{\chi}$, namely

$$\sum_r \langle \gamma_{t_r}\underline{\chi}, \underline{\phi}\rangle_{\Gamma_r} = \langle \gamma_t\underline{\chi}, \underline{\phi}\rangle_{\Gamma}. \tag{2.1.51}$$

We can now state the following proposition.

Proposition 2.1.1. $H^1_{0,D}(\Omega) = \{v | v \in X(\Omega),\ \sum_r \langle \underline{q}\cdot\underline{n}_r, v\rangle = 0, \forall \underline{q} \in H_{0,N}(\mathrm{div};\Omega)\}$.

Proof. It is clear by definition that if $v \in H^1_{0,D}(\Omega)$ we have by (2.1.49) that $\sum_r \langle \underline{q}\cdot\underline{n}_r, v\rangle = 0, \forall \underline{q} \in H_{0,N}(\mathrm{div};\Omega)$. Let us consider the reciprocal. Using Green's formula, we get

$$\int_\Omega v\ \mathrm{div}\,\underline{q}\, dx = -\sum_r \int_{K_r} \underline{\mathrm{grad}}\, v\cdot\underline{q}\, dx,\ \ \forall \underline{q} \in H_{0,N}(\mathrm{div};\Omega). \tag{2.1.52}$$

This implies for all $\underline{q}$, for instance $\underline{q} \in (\mathcal{D}(\Omega))^n$,

$$\left| \int v\ \mathrm{div}\,\underline{q}\, dx \right| \le \Big(\sum_r |v|^2_{1,K_r}\Big)^{\frac{1}{2}}\ \|\underline{q}\|_{0,\Omega}, \tag{2.1.53}$$

and therefore $\underline{\mathrm{grad}}\, v \in (L^2(\Omega))^n$, thus $v \in H^1(\Omega)$. We then have $\langle \underline{q}\cdot\underline{n}, v\rangle = 0$, $\forall \underline{q} \in H_{0,N}(\mathrm{div};\Omega)$, so that $v \in H^1_{0,D}(\Omega)$. □

The same kind of proof would yield the following analogous results for $H(\mathrm{div};\Omega)$ and $H(\underline{\mathrm{curl}};\Omega)$.

Proposition 2.1.2. $H_{0,N}(\mathrm{div};\Omega) = \{\underline{q} \mid \underline{q} \in Y(\Omega),\ \sum_r \langle \underline{q}\cdot\underline{n}_r, v\rangle = 0,\ \forall v \in H^1_{0,D}(\Omega)\}$. □

Proposition 2.1.3. $H_{0,N}(\underline{curl};\Omega) = \{\underline{\chi} \mid \underline{\chi} \in W(\Omega),\ \sum_r \langle \gamma_t\underline{\chi}, \underline{\phi}\rangle = 0,\ \forall \underline{\phi} \in H^1_{0,D}(\Omega)\}$. □

The last results state that functions of $Y(\Omega)$ (resp. $W(\Omega)$) belong to $H(\mathrm{div};\Omega)$ (resp. $H(\underline{\mathrm{curl}};\Omega)$) if and only if their normal (resp. tangential) traces are "continuous" at the interfaces. This will be an essential point for finite element approximations.

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2.1.3 *Properties Relative to a Change of Variables*

The use of a *reference element*, and therefore of coordinate changes, is an essential ingredient of finite element methods, whether for convergence studies or for practical implementation. We must therefore study the effect of a change of variables on our function spaces. We refer to [147, 148] for a more complete presentation.

Let $\hat{K} \subset \mathbb{R}^n$. We denote by $\partial\hat{K}$ its boundary, by $\underline{\hat{n}}$ the outward oriented normal, by $d\hat{x}$ the Lebesgue measure on $\hat{K}$ and by $d\hat{\sigma}$ the superficial measure induced by it on $\partial\hat{K}$.

Let F be a smooth (at least C^1) mapping from $\mathbb{R}^n$ into $\mathbb{R}^n$. We define $K = F(\hat{K})$. We suppose that the Jacobian matrix $DF(\hat{x})$ is invertible for any $\hat{x}$ and that F is globally invertible on K. We then have

$$DF^{-1}(x) = (DF(\hat{x}))^{-1}. \tag{2.1.54}$$

An important case is $F(\hat{x}) = x_0 + B\hat{x}$, that is if F is an affine mapping. Then $DF(\hat{x}) = B$ is a constant matrix. We denote

$$\|DF\|_\infty := \sup_{\hat{x}\in K}\left(\sup_{\xi\in\mathbb{R}^n}\frac{|DF(\hat{x})\xi|}{|\xi|_{\mathbb{R}^n}}\right), \tag{2.1.55}$$

the norm in $L^\infty(\hat{K})$ of function $\hat{x} \to \|DF(\hat{x})\|$, that is, matrix norm of $DF(\hat{x})$. In the same way, we have

$$\|DF^{-1}\|_\infty := \sup_{x\in K}\left(\sup_{\xi\in\mathbb{R}^n}\frac{|(DF^{-1}(x))\xi|}{|\xi|_{\mathbb{R}^n}}\right). \tag{2.1.56}$$

We write

$$J(\hat{x}) := |\det DF(\hat{x})| \tag{2.1.57}$$

and, for $\hat{x} \in \partial\hat{K}$,

$$J_{\underline{\hat{n}}}(\hat{x}) := J(\hat{x})\ \|(DF^{-1})^t\ \underline{\hat{n}}\|_{\mathbb{R}^n}. \tag{2.1.58}$$

- **Sobolev spaces $H^s(\Omega)$.** If $\hat{v}(\hat{x})$ is a function on $\hat{K}$, we define $v(x)$ on K by

$$v := \hat{v} \circ F^{-1}, \tag{2.1.59}$$

and we denote this by $v = \mathcal{F}(\hat{v})$. We then have the classical formulas,

$$\underline{\text{grad}}\, v = (DF^{-1})^t\ \underline{\text{grad}}\, \hat{v} \circ F^{-1} = \mathcal{F}((DF^{-1})^t\ \underline{\text{grad}}\, \hat{v}) \tag{2.1.60}$$

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and

$$\int_K \mathcal{F}(\hat{v})\, dx = \int_{\hat{K}} \hat{v}\ J\ d\hat{x}, \tag{2.1.61}$$

$$\int_{\partial K} \mathcal{F}(\hat{v})\, d\sigma = \int_{\partial \hat{K}} \hat{v}\ J_{\underline{\hat{n}}}\, d\hat{\sigma}. \tag{2.1.62}$$

From this, it is immediate to deduce the following lemma.

Lemma 2.1.5. *The mapping $\mathcal{F}$ is an isomorphism from $L^2(\hat{K})$ onto $L^2(K)$ and from $H^1(\hat{K})$ onto $H^1(K)$, satisfying,*

$$|v|_{0,K} \leq \left(\sup_{\hat{x}} J(\hat{x})\right)^{1/2} |\hat{v}|_{0,\hat{K}}, \tag{2.1.63}$$

$$|\hat{v}|_{0,\hat{K}} \leq \left(\inf_{\hat{x}} J(\hat{x})\right)^{-1/2} |v|_{0,K}, \tag{2.1.64}$$

$$|v|_{1,K} \leq \left(\sup_{\hat{x}} J(\hat{x})\right)^{1/2} \|DF^{-1}\|_\infty\, |\hat{v}|_{0,\hat{K}}, \tag{2.1.65}$$

$$|\hat{v}|_{1,\hat{K}} \leq \left(\inf_{\hat{x}} J(\hat{x})\right)^{-1/2} \|DF\|_\infty\, |v|_{1,K}. \tag{2.1.66}$$

Remark 2.1.8. If F is an affine mapping, we also have [146]

$$|v|_{m,K} \leq c\ (\det B)^{\frac{1}{2}}\ \|B^{-1}\|^m\ |\hat{v}|_{m,\hat{K}} \tag{2.1.67}$$

and similarly,

$$|\hat{v}|_{m,\hat{K}} \leq c\ (\det B)^{-\frac{1}{2}}\ \|B\|^m\ |v|_{m,K}, \tag{2.1.68}$$

where the constant c depends only on m and on the space dimension n. □

In the general case, one must use Leibnitz's formula and the final result is much more complex. For some comments in this directions, see Sect. 2.2.4.

- **The space $\boldsymbol{H}(\mathbf{div};\boldsymbol{\Omega})$.** When building approximations of $H(\mathrm{div};\Omega)$ in Sect. 2.3, we shall be led to use the normal component of vectors as degrees of freedom. This is pretty natural according to Proposition 2.1.2. The above transformation obviously does not preserve normal components. It does neither map $H(\mathrm{div};\hat{K})$ into $H(\mathrm{div};K)$. To overcome this problem, we have to introduce a special (contravariant) transformation known as *Piola's transformation*.

Let, as before, $DF(\hat{x})$ be the Jacobian matrix of the transformation $F(\hat{x})$. We consider, for $\underline{\hat{q}} \in (L^2(\hat{K}))^n$, the mapping,

$$\mathcal{G}(\underline{\hat{q}})(x) := \frac{1}{J(\hat{x})} DF(\hat{x})\ \underline{\hat{q}}(\hat{x}),\ x = F(\hat{x}). \tag{2.1.69}$$

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It is then elementary to check that one has (in $\mathbb{R}^2$, but the result holds for $\mathbb{R}^n$)

$$\begin{pmatrix} \frac{\partial q_1}{\partial x} & \frac{\partial q_1}{\partial y} \\ \frac{\partial q_2}{\partial x} & \frac{\partial q_2}{\partial y} \end{pmatrix} = \frac{1}{J}(DF) \begin{pmatrix} \frac{\partial \hat{q}_1}{\partial \hat{x}} & \frac{\partial \hat{q}_1}{\partial \hat{y}} \\ \frac{\partial \hat{q}_2}{\partial \hat{x}} & \frac{\partial \hat{q}_2}{\partial \hat{y}} \end{pmatrix} (DF^{-1}). \tag{2.1.70}$$

As the trace of a matrix is invariant by a change of variables, we have

$$\operatorname{div} \underline{q} = \frac{1}{J} \operatorname{div} \hat{\underline{q}}. \tag{2.1.71}$$

More generally, we have [366]

Lemma 2.1.6. *Let $v = \mathcal{F}(\hat{v})$ and $\underline{q} = \mathcal{G}(\hat{\underline{q}})$, then*

$$\int_K \underline{q} \cdot \underline{\operatorname{grad}}\, v \, dx = \int_{\hat{K}} \hat{\underline{q}} \cdot \underline{\operatorname{grad}}\, \hat{v} \, d\hat{x}, \tag{2.1.72}$$

$$\int_K v \cdot \operatorname{div} \underline{q} \, dx = \int_{\hat{K}} \hat{v} \, \operatorname{div} \hat{q} \, d\hat{x}, \tag{2.1.73}$$

$$\int_{\partial K} \underline{q} \cdot \underline{n} v \, d\sigma = \int_{\partial \hat{K}} \hat{\underline{q}} \cdot \hat{\underline{n}} \, \hat{v} \, d\hat{\sigma}. \tag{2.1.74}$$

We refer to [366] and [331] for the proof of this result and most of the following ones.

From (2.1.74), we see that $\mathcal{G}$ preserves the normal trace in $H^{-\frac{1}{2}}$ and enables us to define subspaces of $H(\operatorname{div}; K)$ through the reference element $\hat{K}$. More precisely we have,

Lemma 2.1.7. *The mapping $\mathcal{G}$ is an isomorphism of $H(\operatorname{div}; \hat{K})$ onto $H(\operatorname{div}; K)$ and of $H^0(\operatorname{div}; \hat{K})$ onto $H^0(\operatorname{div}; K)$. Moreover we have:*

$$|\underline{q}|_{0,K} \le \left(\inf_{\hat{x}} \, J(\hat{x}) \right)^{-\frac{1}{2}} \|DF\|_\infty \, |\underline{q}|_{0,\hat{K}}, \tag{2.1.75}$$

$$|\hat{\underline{q}}|_{0,\hat{K}} \le \left(\sup_{\hat{x}} \, J(\hat{x}) \right)^{\frac{1}{2}} \|DF^{-1}\|_\infty \, |\underline{q}|_{0,K}, \tag{2.1.76}$$

$$|\operatorname{div} \underline{q}|_{0,K} \le \left(\inf_{\hat{x}} \, J(\hat{x}) \right)^{-\frac{1}{2}} |\operatorname{div} \hat{\underline{q}}|_{0,\hat{K}}, \tag{2.1.77}$$

$$|\operatorname{div} \hat{\underline{q}}|_{0,\hat{K}} \le \left(\sup_{\hat{x}} \, J(\hat{x}) \right)^{\frac{1}{2}} |\operatorname{div} \underline{q}|_{0,K}. \tag{2.1.78}$$

It is also possible to obtain relations between $|\underline{q}|_{m,K}$ and $|\hat{\underline{q}}|_{m,\hat{K}}$ or between $|\operatorname{div} \underline{q}|_{m,K}$ and $|\operatorname{div} \hat{\underline{q}}|_{m,\hat{K}}$. We refer to [366] for details. The next lemma deals with the case where F is an affine transformation and $\underline{q} \in H^m(\operatorname{div}; \Omega)$, where

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$$H^m(\mathrm{div};\Omega) := \{\underline{q} \mid \underline{q} \in (H^m(\Omega))^n,\ \mathrm{div}\,\underline{q} \in H^m(\Omega)\}. \tag{2.1.79}$$

Lemma 2.1.8. *If the mapping F is affine and if $\underline{q} \in H^m(\mathrm{div};\Omega)$, the following estimates hold, with $B = DF$,*

$$|\underline{q}|_{m,K} \le (det\ B)^{-\frac{1}{2}} \|B^{-1}\|^m\ \|B\|\ |\underline{\hat{q}}|_{m,\hat{K}}, \tag{2.1.80}$$

$$|\,\mathrm{div}\,\underline{q}|_{m,K} \le (det\ B)^{-\frac{1}{2}} \|B^{-1}\|^m\ |\,\mathrm{div}\,\underline{\hat{q}}|_{m,\hat{K}}. \tag{2.1.81}$$

The reverse inequalities also hold by a simple exchange of roles between K and $\hat{K}$. Such results are of course essential in the proofs of error estimates. The Piola transformation can be extended to tensor-valued functions with similar properties (cf. for instance [127, 295] or [146]).

- **The space $H(\underline{\mathrm{curl}};\Omega)$.** According to Remark 2.1.5, in this subsection we restrict the discussion to the three-dimensional situation where $\hat{K} \subset \mathbb{R}^3$ since the two-dimensional case can be easily deduced from the result of the previous subsection about approximations of $H(\mathrm{div};\Omega)$. In general, when dealing with reference elements, $\hat{K}$ will be a simple geometric entity like a cube or a tetrahedron. In particular, it will contain vertices, edges and faces. We shall denote by $\hat{e}$ and $\hat{f}$ generic edges and faces, respectively, and by e and f their corresponding images under the action of F. We shall then refer to an edge e or a face f of K. It is clear that, for general isomorphisms F, an edge of K might be curved and a face of K might not be contained in a plane. On the other hand, if F is a trilinear map, then edges of K will be straight and if, moreover, F is affine, then faces of K will be flat.

  In order to deal with the approximation of $H(\underline{\mathrm{curl}};\Omega)$, we shall make use of the following (covariant) transformation:

$$\mathcal{H}(\underline{\hat{\chi}})(x) := [DF(\hat{x})]^{-T}\,\underline{\hat{\chi}}(\hat{x}), \quad x = F(\hat{x}). \tag{2.1.82}$$

  We notice that formula (2.1.82) is formally identical to (2.1.60); i.e., we have chosen to transform vectors of $H(\underline{\mathrm{curl}};\Omega)$ like gradients. In particular, one of the main features of transformation (2.1.82) is that it preserves the tangential components in a sense that will soon be made clear.

  Before stating the general results, let us go back to the setting of Example 2.1.1.

Example 2.1.2 (Case when Γ is flat.). We recall that, in our example, $\Omega \subset \mathbb{R}^3$ is the half space $x_3 < 0$ and that Γ is the plane $x_3 = 0$. Let us consider a linear mapping $F : \mathbb{R}^3 \to \mathbb{R}^3$ defined as follows

$$F(\hat{x}_1, \hat{x}_2, \hat{x}_3) = (\alpha\hat{x}_1 + \beta\hat{x}_2, \gamma\hat{x}_1 + \delta\hat{x}_2, \hat{x}_3)^T, \tag{2.1.83}$$

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so that

$$DF = \begin{pmatrix} \alpha & \beta & 0 \\ \gamma & \delta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{2.1.84}$$

Basically, we are considering a mapping that corresponds to a motion of Ω only in the direction of x_1 and x_2. In particular, the restriction of F to Γ maps linearly the plane Γ into itself. According to the notation introduced before, we shall denote by $\hat{\Omega}$ the reference domain and by Ω the transformed domain; analogously, Γ is the image of the reference boundary $\hat{\Gamma}$. Let us consider a vector field $\hat{\underline{\chi}} \in H(\underline{\mathrm{curl}}; \hat{\Omega})$; we are interested in investigating how the traces of $\hat{\underline{\chi}}$ transform under the effects of the covariant map (2.1.82).

It is immediate to evaluate the determinant of DF, given by the constant $J = \alpha\delta - \beta\gamma$, and the matrix $[DF]^{-T}$ given by

$$[DF]^{-T} = \frac{1}{J} \begin{pmatrix} \delta & -\gamma & 0 \\ -\beta & \alpha & 0 \\ 0 & 0 & J \end{pmatrix}. \tag{2.1.85}$$

Hence, transformation (2.1.82) reads

$$\underline{\chi} = \begin{pmatrix} \chi_1 \\ \chi_2 \\ \chi_3 \end{pmatrix} = \mathcal{H}(\hat{\underline{\chi}}) = \frac{1}{J} \begin{pmatrix} \delta\hat{\chi}_1 - \gamma\hat{\chi}_2 \\ \alpha\hat{\chi}_2 - \beta\hat{\chi}_1 \\ J\hat{\chi}_3 \end{pmatrix}. \tag{2.1.86}$$

The first important result which we are going to check is that the trace $\hat{\gamma}_t\hat{\underline{\chi}}$ transforms like vectors in $H(\mathrm{div}; \hat{\Gamma})$. More precisely, let $\mathcal{G}_\Gamma$ denote the Piola transformation defined in (2.1.69) related to the plane Γ, i.e.,

$$\mathcal{G}_\Gamma(\hat{\underline{q}}) := \frac{1}{J} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \hat{\underline{q}}. \tag{2.1.87}$$

From (2.1.31) we obtain

$$\mathcal{G}_\Gamma(\hat{\gamma}_t\hat{\underline{\chi}}) = \frac{1}{J} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \hat{\chi}_2 \\ -\hat{\chi}_1 \end{pmatrix} = \frac{1}{J} \begin{pmatrix} \alpha\hat{\chi}_2 - \beta\hat{\chi}_1 \\ \gamma\hat{\chi}_2 - \delta\hat{\chi}_1 \end{pmatrix}. \tag{2.1.88}$$

On the other hand, comparing (2.1.31), (2.1.86), and (2.1.88), we easily get

$$\mathcal{G}_\Gamma(\hat{\gamma}_t\hat{\underline{\chi}}) = \gamma_t(\mathcal{H}(\hat{\underline{\chi}})) = \gamma_t\underline{\chi}. \tag{2.1.89}$$

In a similar way, we can find how the trace $\hat{\pi}_t\hat{\underline{\chi}}$ transforms when $\hat{\underline{\chi}}$ is mapped like in (2.1.82). It turns out that $\hat{\pi}_t\hat{\underline{\chi}}$ transforms like vectors in $H(\underline{\mathrm{curl}}; \hat{\Gamma})$. In this case, we can make use of the mapping $\mathcal{H}_\Gamma$, defined as

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$$\mathcal{H}_\Gamma\left(\begin{pmatrix}\chi_1\\ \chi_2\end{pmatrix}\right) := \frac{1}{J}\begin{pmatrix}\delta & -\gamma\\ -\beta & \alpha\end{pmatrix}\begin{pmatrix}\chi_1\\ \chi_2\end{pmatrix}, \tag{2.1.90}$$

in order to obtain the result

$$\mathcal{H}_\Gamma(\hat{\pi}_t\,\underline{\hat{\chi}}) = \pi_t(\mathcal{H}(\underline{\hat{\chi}})) = \pi_t\,\underline{\chi}. \tag{2.1.91}$$

□

Let us now come back to the general situation. For properties of the covariant transformation (2.1.82) and for the proofs of most of the results presented in the sequel of this subsection, we refer the interested reader to [222], to the abstract theory of differential forms presented in [248], to the book [302], and to the comprehensive review [33].

We shall make use, in particular, of the following relationship between the curl of $\underline{\chi}$ and the curl of $\underline{\hat{\chi}}$

$$\underline{\mathrm{curl}}\,\underline{\chi}(x) = \frac{1}{J(\hat{x})}DF(\hat{x})\,\underline{\mathrm{curl}}\,\underline{\hat{\chi}}(\hat{x}), \quad x = F(\hat{x}), \tag{2.1.92}$$

so that we have $\underline{\mathrm{curl}}\,\underline{\chi} = \mathcal{G}(\underline{\mathrm{curl}}\,\underline{\hat{\chi}})$. Formula (2.1.92) is a consequence of the more general transformation rule (see [310, Theorem 2] and [223])

$$\underline{\mathrm{curl}}\,\underline{\chi}(x) = [DF]^{-T}\,\underline{\underline{\mathrm{curl}}}\,\underline{\hat{\chi}}(\hat{x})[DF]^{-1}, \quad x = F(\hat{x}), \tag{2.1.93}$$

where the tensor is defined by $\underline{\underline{\mathrm{curl}}}\,\underline{\chi} = \left(\frac{\partial\chi_j}{\partial x_i} - \frac{\partial\chi_i}{\partial x_j}\right)_{i,j}$.

Before stating results which relate quantities evaluated on the reference element to corresponding quantities evaluated on the actual element (in the spirit of Lemma 2.1.6), we need to recall how tangent and normal vectors behave under the action of the mapping F. Namely, if $\underline{\hat{n}}$ is a normal vector at a given point of $\partial\hat{K}$, then

$$\underline{n}(x) = \frac{[DF]^{-T}\cdot\underline{\hat{n}}(\hat{x})}{\|[DF]^{-T}\cdot\underline{\hat{n}}(\hat{x})\|} \tag{2.1.94}$$

is the corresponding normal unit vector on ∂K. If $\underline{\hat{t}}$ is a tangent unit vector, then

$$\underline{t}(x) = \frac{DF\cdot\underline{\hat{t}}(\hat{x})}{\|DF\cdot\underline{\hat{t}}(\hat{x})\|} \tag{2.1.95}$$

is the corresponding tangent unit vector on ∂K. Moreover, if $\underline{\hat{t}}$ is tangent to an edge of $\hat{K}$, then $\underline{t}$ defined in (2.1.95) is tangent to the corresponding edge of K.

Lemma 2.1.9. *Let $\underline{\chi} := \mathcal{H}(\underline{\hat{\chi}})$. Let additional functions on K be constructed from reference functions as follows: $v := \mathcal{F}(\hat{v})$, $\underline{q} := \mathcal{G}(\underline{\hat{q}})$, and $\underline{\tau} := \mathcal{H}(\underline{\hat{\tau}})$. Then*

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$$\int_e \underline{\chi} \cdot \underline{t}\, v \, ds = \int_{\hat{e}} \hat{\underline{\chi}} \cdot \hat{\underline{t}} \hat{v} \, d\hat{s}, \tag{2.1.96}$$

$$\int_f \pi_t \underline{\chi} \cdot \gamma_t \underline{\tau} \, d\sigma = \int_{\hat{f}} \hat{\pi}_t \hat{\underline{\chi}} \cdot \hat{\gamma}_t \hat{\underline{\tau}} \, d\hat{\sigma}, \tag{2.1.97}$$

$$\int_K \underline{\chi} \cdot \underline{q} \, dx = \int_{\hat{K}} \hat{\underline{\chi}} \cdot \hat{\underline{q}} \, d\hat{x}. \tag{2.1.98}$$

Remark 2.1.9. Formula (2.1.97) is worth some comments. For simplicity, let us consider the case where f is flat. From Example 2.1.2, it follows that, when $\underline{\tau} := \mathcal{H}(\hat{\underline{\tau}})$, then $(\gamma_t \underline{\tau})|_f$ is related to $(\gamma_t \hat{\underline{\tau}})|_{\hat{f}}$ by means of the two-dimensional Piola transform (2.1.69) on the face f. If we call $\mathcal{G}_f$ such a transform, then we have $\gamma_t \underline{\tau} = \mathcal{G}_f(\hat{\gamma}_t \hat{\underline{\tau}})$. On the other hand, in an analogous way we have that $\pi_t \underline{\chi} = H_f(\hat{\pi}_t \hat{\underline{\chi}})$ on the face f. Then, Eq. (2.1.97) might be rephrased in terms of functions defined on the face f as follows: let $\hat{\underline{\phi}}$ and $\hat{\underline{\psi}}$ be vector fields on $\hat{f}$, then

$$\int_{\hat{f}} \hat{\underline{\psi}} \cdot \hat{\underline{\phi}} \, d\hat{\sigma} = \int_f H_f(\hat{\underline{\psi}}) \cdot \mathcal{G}_f(\hat{\underline{\phi}}) \, d\sigma. \tag{2.1.99}$$

□

Finally, the following analogue of Lemma 2.1.7 holds true.

Lemma 2.1.10. *The mapping $\mathcal{H}$ is an isomorphism of $H(\underline{curl}; \hat{K})$ onto $H(\underline{curl}; K)$. Moreover, we have:*

$$\|\underline{\chi}\|_{0,K} \leq \left(\sup_{\hat{x}} J(\hat{x}) \right)^{\frac{1}{2}} \|DF^{-1}\|_\infty \|\hat{\underline{\chi}}\|_{0,\hat{K}}, \tag{2.1.100}$$

$$\|\hat{\underline{\chi}}\|_{0,\hat{K}} \leq \left(\inf_{x} J(x) \right)^{-\frac{1}{2}} \|DF\|_\infty \|\underline{\chi}\|_{0,K}, \tag{2.1.101}$$

$$\|\underline{\mathrm{curl}}\, \underline{\chi}\|_{0,K} \leq \left(\inf_{\hat{x}} J(\hat{x}) \right)^{-\frac{1}{2}} \|DF\|_\infty \|\underline{\mathrm{curl}}\, \hat{\underline{\chi}}\|_{0,\hat{K}}, \tag{2.1.102}$$

$$\|\underline{\mathrm{curl}}\, \hat{\underline{\chi}}\|_{0,\hat{K}} \leq \left(\sup_{x} J(x) \right)^{\frac{1}{2}} \|DF^{-1}\|_\infty \|\underline{\mathrm{curl}}\, \underline{\chi}\|_{0,K}. \tag{2.1.103}$$

2.1.4 De Rham Diagram

The topics introduced in this chapter can be presented in a general unified approach by means of tools of exterior algebra. The De Rham complex, in particular, has been rediscovered recently as a very convenient tool in order to provide a general setting for handling function spaces and their finite element approximations. In this

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framework, the commuting diagram property which will be presented throughout this chapter can be seen as particular cases of a more general picture. Even though it is out of the aims of this book to give a detailed presentation of this topic, we would like to provide the reader with a short introduction to this subject, which proves to be a very useful technique not only for the analysis of known finite elements, but also for designing new ones.

The importance of the de Rham complex in the analysis of finite element schemes has been detected independently by several authors in different fields. The commuting diagram properties for finite elements in $H(\mathrm{div};\Omega)$ (see [177, 178]) has been the driving force for an active research in the approximation of second order elliptic problems in mixed form. Bossavit [102] was the first one who used the full form of the de Rham complex for the approximation of problems arising from electromagnetism. His pioneer idea has been exploited by several authors (see, for instance, [75, 170, 247, 248]). The general framework has been designed in [31] and successfully applied to the construction of new finite element spaces for the elasticity equations (see [32]). References [33, 34] present an excellent review on the state of the art of this active research field.

Here we recall the de Rham complex related with the spaces considered in this chapter. Let us suppose that $\Omega \subset \mathbb{R}^3$ is a simply connected domain, then the following sequence is exact

$$\mathbb{R} \hookrightarrow H^1(\Omega) \xrightarrow{\underline{\mathrm{grad}}} H(\underline{\mathrm{curl}};\Omega) \xrightarrow{\underline{\mathrm{curl}}} H(\mathrm{div};\Omega) \xrightarrow{\mathrm{div}} L^2(\Omega) \to 0\cdot \tag{2.1.104}$$

We shall present in the sequel some discrete variants of (2.1.104), which will be useful to understand interconnections between different approximations.

2.2 Finite Element Approximations of $H^1(\Omega)$ and $H^2(\Omega)$

This section will be mainly devoted to the approximation of $H^1(\Omega)$ and its subspace of the form $H^1_{0,D}(\Omega)$. We shall moreover sketch some results concerning the approximation of $H^2(\Omega)$. Standard approximations of Sobolev spaces can be subdivided into two classes: conforming and nonconforming methods. Even though nonconforming methods will be studied in the context of hybrid finite element methods, their importance makes it useful to introduce them here. We refer to [41, 146] or [334] for a detailed presentation of the following results.

2.2.1 Conforming Methods

Conforming methods are the most natural finite element methods. They yield *internal approximations* in the sense that they enable us to build finite dimensional subspaces of the function space that we want to approximate.

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Given a partition of the domain Ω into polygonal or polyhedral elements, a conforming approximation of $H^1(\Omega)$ is a space of *continuous functions* defined by a finite number of parameters (or degrees of freedom).

The last condition is *usually* met by using a space of piecewise polynomial functions or functions obtained from polynomials by a change of variables like (using the notation of Sect. 2.1.3)

$$v_h|_K = \hat{v} \circ F^{-1}, \tag{2.2.1}$$

where $K = F(\hat{K})$ and $\hat{v}$ is a polynomial function on $\hat{K}$. Continuity is obtained by a clever choice of degrees of freedom.

Remark 2.2.1. For triangular elements, it is usual and convenient to use piecewise polynomial functions on K. For quadrilaterals, it is essential to use (2.2.1). It must then be noted that $v_h|_K$ is not in general a polynomial on K. This will be the case only for affine transformations. More comments on this issue will be discussed in Sect. 2.2.4. □

To give a more precise definition of our finite element approximations, we shall need a few definitions. Let us define on an element K

$$P_k(K) := \text{ the space of polynomials of degree } \leq \text{k}. \tag{2.2.2}$$

The dimension of $P_k(K)$ is $\frac{1}{2}(k+1)(k+2)$ for $n = 2$ and, for $n = 3$, it is $\frac{1}{6}(k+1)(k+2)(k+3)$. For a rectangular element, it will be convenient to define (for $n = 2$)

$$P_{k_1,k_2}(K) := \left\{ p(x_1, x_2) \mid p(x_1, x_2) = \sum_{\substack{i \leq k_1 \\ j \leq k_2}} a_{ij}\, x_1^i\, x_2^j \right\} \tag{2.2.3}$$

the space of polynomials of degree $\leq k_1$ in x_1 and $\leq k_2$ in x_2. In the same way, we can define on a rectangular hexahedron $P_{k_1,k_2,k_3}(K)$ for $n = 3$. The dimension of these spaces is $(k_1+1)(k_2+1)$ and $(k_1+1)(k_2+1)(k_3+1)$, respectively. We then define

$$Q_k(k) := \begin{cases} P_{k,k}(K), & \text{for } n = 2, \\ P_{k,k,k}(K), & \text{for } n = 3. \end{cases} \tag{2.2.4}$$

We shall also need polynomial spaces on the edges (or faces) of the elements. Using the notations of Sect. 2.1.2, we define

$$R_k(\partial K) := \{\phi \mid \phi \in L^2(\partial K),\ \phi|_{e_i} \in P_k(e_i),\ \forall e_i\}, \tag{2.2.5}$$

$$T_k(\partial K) := \{\phi \mid \phi \in R_k(\partial K) \cap C^0(\partial K)\}. \tag{2.2.6}$$

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We define a subspace of P_k, which could have traces of a lower degree on the boundary of K

$$P_k^r := \{p \mid p \in P_k|_{\partial K},\ p \in T_r(\partial K),\ r \leq k\}. \tag{2.2.7}$$

Functions of $R_k(\partial K)$ are polynomials of degree $\leq k$ on each side (or face) of K. They do not have to be continuous at vertices. The dimensions of $R_k(\partial K)$ and $T_k(\partial K)$ are respectively for $k \geq 1$:

— $3(k+1)$ and $3k$ for triangles,
— $4(k+1)$ and $4k$ for quadrilaterals,
— $2(k+1)(k+2)$ and $2(k^2+1)$ for tetrahedra.

For hexahedra, it will *usually* be more convenient to consider functions in $Q_k(e_i)$ in the definition of $R_k(\partial K)$ and $T_k(\partial K)$.

In order to define a finite element, following [146], we need to specify three things.

— The geometry: we choose a reference element $\hat{K}$, a change of variables $F(\hat{x})$ and we set $K = F(\hat{K})$.
— A set $\hat{P}$ of polynomials on $\hat{K}$. For $\hat{p} \in \hat{P}$ we define on K, $p = \hat{p} \circ F^{-1}$.
— A set of degrees of freedom $\hat{\Sigma}$, that is, a set of linear forms $\{\hat{\ell}_i\}_{1 \leq i \leq \dim \hat{P}}$ on $\hat{P}$. We say that this set is unisolvent when these linear forms are linearly independent, i.e. the knowledge of $\hat{\ell}_i(\hat{p})$ for all i completely defines $\hat{p}$.

A finite element is of *Lagrange type* if its degrees of freedom are *point values*, that is, if one is given a set $\{\hat{a}_i\}_{1 \leq i \leq \dim \hat{P}}$ of points in $\hat{K}$ and one defines

$$\hat{\ell}_i(\hat{p}) = \hat{p}(\hat{a}_i),\ 1 \leq i \leq \dim \hat{P}. \tag{2.2.8}$$

For the approximation of $H^1(\Omega)$, Lagrange type elements will be sufficient but approximating $H^2(\Omega)$ requires Hermite type elements, that is degrees of freedom involving derivatives.

Remark 2.2.2. The reader should be aware that not any choice of points will yield a unisolvent set of degrees of freedom. Moreover, the points have to be chosen in order to ensure inter-element continuity. □

Example 2.2.1 (Affine Finite Elements). This is the most classical family of finite elements. The reference element is the triangle $\hat{K}$ of Fig. 2.2 and we use the affine transformation

$$F(\hat{x}) = x_0 + B\hat{x}. \tag{2.2.9}$$

The element K is still a triangle and it is not degenerate provided $\det(B) \neq 0$. We now take $\hat{P} = P_k(\hat{K})$ and choose an appropriate set of degrees of freedom. The standard choices for $k \leq 3$ are presented in Fig. 2.3.

It can be easily observed that this choice of points ensures continuity at interfaces. □

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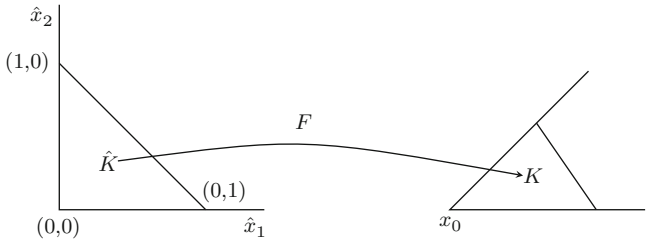


Fig. 2.2 An affine tranformation

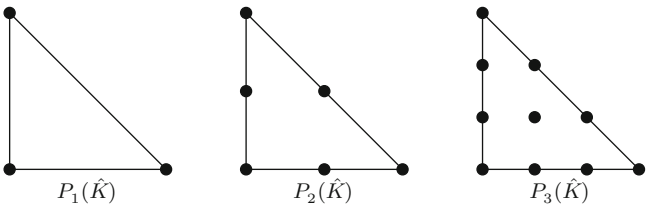


Fig. 2.3 Standard conforming elements

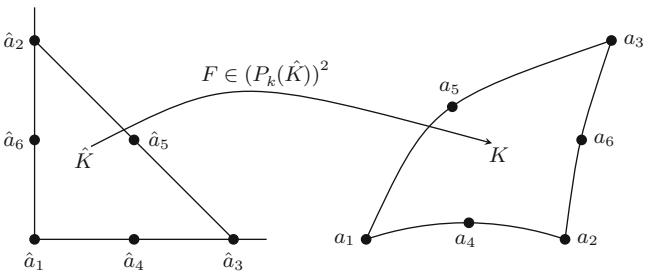


Fig. 2.4 Isoparametric triangle of degree 2

Example 2.2.2 (Isoparametric triangular elements). We use the same reference element and the same set $\hat{P}$ as in the previous example. We now take the transformation $F(\hat{x})$ such that each of its components F_i belongs to $P_k(\hat{K})$. For $k = 1$, nothing is changed but for $k \geq 2$, the element K now has curved boundaries. We present the case $k = 2$ in Fig. 2.4.

Using such curved triangles enables us to obtain a better approximation of curved boundaries. It must be noted that the curvature of boundaries introduces additional terms in the approximation error and the curved elements should be used only when they are really necessary [152] or [146]. □

Example 2.2.3 (Isoparametric quadrilateral elements). This is also a very classical family of finite elements. The reference element is the square $\hat{K} =]0,1[\times]0,1[$. We take $\hat{P} = Q_k(\hat{K})$ and a transformation F with each component in $Q_k(\hat{K})$.

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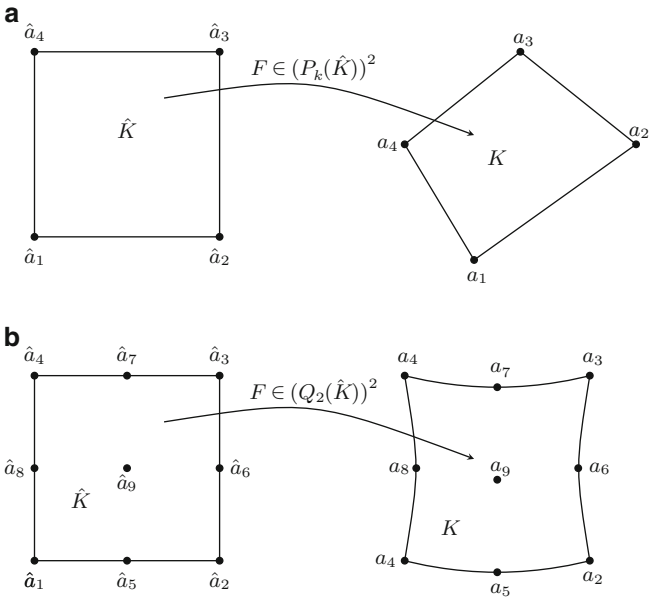


Fig. 2.5 (**a**) The Q1 isoparametric element. (**b**) The Q2 isoparametric element

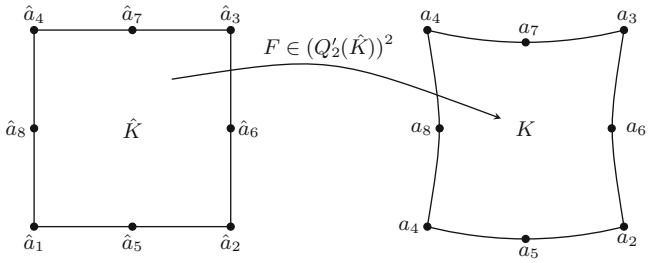


Fig. 2.6 Serendipity element

We present the standard choice of degrees of freedom for $k \leq 2$ in Fig. 2.5. It must be noted that we need $F \in (Q_1(\hat{K}))^2$ to define a general straight-sided quadrilateral.

Finally we recall that it is possible to eliminate internal nodes to get the so-called serendipity finite elements. For instance, if we take

$$\hat{P} = Q_2'(\hat{K}) := \{\hat{p} \mid \hat{p} \in Q_2(\hat{K}),\ 4\hat{p}(\hat{a}_9) + \sum_{i=1}^{4} \hat{p}(\hat{a}_i) - 2\sum_{i=5}^{8} \hat{p}(\hat{a}_i) = 0\}$$

$$= P_3(\hat{K}) \cap Q_2(\hat{K}) \tag{2.2.10}$$

we obtain the element of Fig. 2.6.

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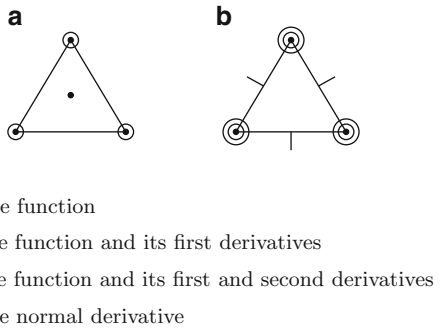


Fig. 2.7 (**a**) P_3 triangle. (**b**) Argyris' triangle

As before, the degrees of freedom have been chosen in order to ensure continuity between elements. The use of serendipity elements should be avoided on distorted (non-affine) meshes. More details on quadrilateral elements will be given in Sect. 2.2.4. □

Example 2.2.4 (Hermite type elements). Approximating $H^2(\Omega)$ will require continuity of derivatives at inter-element boundaries and leads to the introduction of elements in which values of the derivatives are used as degrees of freedom. The simplest Hermite type element is the P_3 triangle of Fig. 2.7a.

Here, the degrees of freedom are values of the function and its derivatives at vertices plus the function value at the barycentre. This element *does not* enable us to build an approximation of $H^2(\Omega)$. To do so, one must use Argyris' triangle (see Fig. 2.7.b) where polynomials of degree 5 are used. (Composite elements can also been used.) For quadrilaterals, the analogues are easily built. The difficulty of building approximations of $H^2(\Omega)$ by standard methods was one of the major reasons for the introduction of various kinds of mixed or hybrid methods for plate problems (cf. Sects. 10.2 or 10.3). □

The remaining part of this section is devoted to the analysis of the approximation of a given function v by the finite element spaces just described or similar ones. We shall not give proofs for which we refer to [146, 154, 358].

For a general set of degrees of freedom $\{\hat{\ell}_i\}$ on $\hat{K}$, we define the *interpolate* $\hat{r}_h\hat{v}$ of v by

$$\hat{\ell}_i(\hat{r}_h\hat{v}) := M(\hat{v}),\ 1 \le i \le \dim\ \hat{P}. \tag{2.2.11}$$

The operator M must be a well-defined continuous form. When the linear forms $\hat{\ell}_i$ are defined by (2.2.5), it is natural to set

$$(\hat{r}_h\hat{v})(\hat{a}_i) = \hat{v}(\hat{a}_i). \tag{2.2.12}$$

This definition makes sense only when $\hat{v}$ is a *continuous function* which is not the case when $v \in H^1(\Omega)$. For Lagrange type elements in $\mathbb{R}^2$ or $\mathbb{R}^3$, $\hat{v} \in H^2(\hat{K})$ is a sufficient condition for (2.2.12) to be justified and $\hat{r}_h\hat{v}$ is just the

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Lagrange interpolate, in the classical sense, of $\hat{v}$. For $v \in H^1(\Omega)$, [154] *has defined a continuous interpolate* $\hat{r}_h$ *using averages of* u *instead of point values.* This also implies a more elaborate use of reference elements. In particular, the operator $\hat{r}_h\hat{v}$ is no longer defined on one single element. In fact the nodal values of $\hat{r}_h\hat{v}$ depend on the value of $\hat{v}$ on the adjacent elements through an averaging process.

Once $\hat{r}_h\hat{v}$ is defined, we can define on K

$$r_h v := (\hat{r}_h(v \circ F)) \circ F^{-1} = (\hat{r}_h\hat{v}) \circ F^{-1}. \tag{2.2.13}$$

We rapidly recall a few classical results. We refer the reader to [146] for a detailed presentation. We first consider the case of *affine elements*, assuming first r_h to be defined by a usual interpolate (2.2.12).

Proposition 2.2.1. *If the mapping* F *is affine, that is* $F(\hat{x}) = x_0 + B\hat{x}$*, and if* $r_h p_k = p_k$ *for any* $p_k \in P_k(K)$*, we have for* $v \in H^s(\Omega)$*,* $m \le s$*,* $1 < s \le k+1$*,*

$$|v - r_h v|_{m,K} \le c\ \|B^{-1}\|^m\ \|B\|^s\ |v|_{s,K}. \tag{2.2.14}$$

The proof uses (2.1.67), its reciprocal, and the classical results stated below.

Lemma 2.2.1. $|\cdot|_{k+1,\Omega}$ *is a norm on* $H^{k+1}(\Omega)/P_k(\Omega)$*, equivalent to the standard quotient norm.*

From this lemma, the following classical result can be deduced.

Lemma 2.2.2 (Bramble-Hilbert's lemma). *Let* L *be a continuous linear form on* $H^{k+1}(\Omega)$ *such that* $L(p_k) = 0$ *for any* $p_k \in P_k(\Omega)$*, then there exists a constant* c *(depending on* L *and* Ω*) such that one has*

$$|L(v)| \le c\ |v|_{k+1,\Omega}. \tag{2.2.15}$$

Results similar to (2.2.14), although more complex, can be obtained for general isoparametric elements [146, 151, 152]. Let then h_K be the diameter of K. Provided some classical conditions on the *shape of elements* forbidding degeneracy are fulfilled [146], relation (2.2.14) can be converted into a relation involving a power of h_K. For affine elements, one defines for instance

$$\sigma_K := \frac{h_K}{\rho_K}, \tag{2.2.16}$$

where ρ_K is the diameter of the largest inscribed disk (or sphere) in K.

We shall, in the following, always assume that the interpolation operator r_h is defined by the method of [154], that is, by a local projection instead of a point-wise interpolate. This allows us to get rid of the condition $s > 1$ of Proposition 2.2.1. To state this result, we define

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$$\triangle K := \{K' |\ \bar{K}' \cap \bar{K} \neq \emptyset\}, \tag{2.2.17}$$

$$h_{\triangle K} := \sup_{K' \in \triangle K} h_{K'}, \tag{2.2.18}$$

$$\sigma_{\triangle K} := \sup_{K' \in \triangle K} \sigma_{K'}. \tag{2.2.19}$$

We then have the following proposition [154].

Proposition 2.2.2. *If the mapping F is affine and if $r_h p_k = p_k$ for any $p_k \in P_k(K)$, then there is a constant, depending on k and σ, such that for $m \leq s$, $1 \leq s \leq k+1$,*

$$|v - r_h v|_{m,K} \leq c \sigma_{\triangle K} h_{\triangle K}^{s-m} |v|_{s,\triangle K}. \tag{2.2.20}$$

□

We then say that *a family of triangulations $(\mathcal{T}_h)_{h \geq 0}$ is regular* if

$$\sigma_K < \sigma,\ \forall K \in \mathcal{T}_h,\ \forall h. \tag{2.2.21}$$

For the geometrical meaning of this condition, we refer to [146]. We may recall however that (2.2.21) can be written as a condition on angles excluding degenerate elements. For general curved elements, there is also a condition on the curvature of the sides.

We then have the following approximation result.

Corollary 2.2.1. *If $(\mathcal{T}_h)_{h \geq 0}$ is regular family of affine partitions, there exists a constant c depending on k and σ such that*

$$|v - r_h v|_{1,K} \leq c h^k |v|_{k+1,\triangle K}. \tag{2.2.22}$$

For more general partitions including general isoparametric elements, the result is qualitatively the same: we have an $O(h^k)$ approximation provided the family of partitions is regular in a sense to be specified (see Sect. 2.2.4 for more details).

We also refer the reader to [260] where some degenerate cases are analysed.

From the elements described above, we can build approximations of $H^1(\Omega)$ and $H^2(\Omega)$. The idea is of course to use functions whose restriction to an element belongs to a set of polynomial (or image S of polynomial) functions. Let $S_k(K)$ be a subspace of $P_k(K)$. We define, for a partition $\mathcal{T}_h$ of Ω,

$$\mathcal{L}^s(S_k, \mathcal{T}_h) := \{v \mid v \in H^s(\Omega),\ v|_K \in S_k(K)\}. \tag{2.2.23}$$

Remark 2.2.3. Since the elements of $\mathcal{L}^s(S_k, \mathcal{T}_h)$ are piecewise polynomials, we have $\mathcal{L}^s(S_k, \mathcal{T}_h) \subset C^{s-1}(\bar{\Omega})$ although $H^s(\Omega) \not\subset C^{s-1}(\bar{\Omega})$. □

We shall reduce this notation when no confusion is to be feared and write

$$\mathcal{L}^s(S_k) \tag{2.2.24}$$

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when no ambiguity is possible as to $\mathcal{T}_h$ and still more compactly

$$\mathcal{L}_k^s = \mathcal{L}^s(P_k, \mathcal{T}_h), \tag{2.2.25}$$

when $\mathcal{T}_h$ is built from triangles and $S_k = P_k$, the space of polynomials of degree $\leq$k. In the same way, we shall write

$$\mathcal{L}_{[k]}^s = \mathcal{L}^s(Q_k, \mathcal{T}_h), \tag{2.2.26}$$

when $\mathcal{T}_h$ is built from quadrilaterals.

Remark 2.2.4 (Bubble functions). We shall often use in our constructions *bubble functions*. We consider them here in the case of H^1. For an element K, a bubble function is a function vanishing on ∂K. Thus, we say that S_k is a set of bubble functions if $S_k \subset H_0^1(K)$. We then denote

$$B(S_k) = \mathcal{L}^1(S_k, \mathcal{T}_h) = \mathcal{L}^0(S_k, \mathcal{T}_h) \tag{2.2.27}$$

and we shall use the compact notation,

$$\begin{cases} B_k = B(P_k \cap H_0^1(K)), \\ B_{[k]} = B(Q_k \cap H_0^1(K)), \end{cases} \tag{2.2.28}$$

when no ambiguity will be possible. Spaces of bubble functions will be used to build enriched spaces. For instance, the space $\mathcal{L}_2^1 \oplus B_3$ will be useful in Chap. 8 for the approximation of Stokes' problem. This space could also be written as $\mathcal{L}^1(P_3^2)$. □

When approximating a standard elliptic problem, the finite element spaces introduced up to now can be used directly in the variational formulation of the problem and error estimates follow from interpolation error estimates [146]. In many cases, however, nonconforming methods have proved to yield accurate (and sometimes easier to handle) approximations.

2.2.2 Explicit Basis Functions on Triangles and Tetrahedra

In the case of affine elements, it is often possible to define explicitly the basis functions associated to a choice of degrees of freedom. This is done using the following classical result.

Lemma 2.2.3. *Let K be an affine element of dimension k with vertices $\underline{x}_i$, $(1 \leq i \leq k)$. There exists a set $\lambda_i(\underline{x})$, $(1 \leq i \leq k)$ of linear functions on K, called barycentric coordinates, satisfying*

$$\lambda_i(\underline{x}_j) = \delta_{ij}$$

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and

$$\sum_i \lambda_i(\underline{x}) = 1.$$

- It is then immediate that the barycentric coordinates are the basis functions of the affine $P_1(K)$ element.
- For the $P_2(K)$ element of Fig. 2.3 the basis function associated to the vertex $\underline{x}_i$ is $\lambda_i(2\lambda_i - 1)/2$ while for the node at midpoint of the edge $\underline{t}_{ij}$ between $\underline{x}_i$ and $\underline{x}_j$, the basis function is $4\lambda_i\lambda_j$.

The bubbles of lower degree are on affine elements

$$\begin{cases} b_{3,K} = \lambda_1\lambda_2\lambda_3 & \text{in the two dimensional case ,} \\ b_{4,K} = \lambda_1\lambda_2\lambda_3\lambda_4 & \text{in the three-dimensional case.} \end{cases} \tag{2.2.29}$$

We shall also define nonconforming bubbles in (2.2.39).
Barycentric coordinates will also be employed in Sect. 2.6.

2.2.3 *Nonconforming Methods*

We shall meet later nonconforming methods when studying hybrid finite element methods. In many cases, it will however be more convenient to see them in the framework of *external approximations*.

Let us consider a variational problem (with $f \in V'$),

$$a(u, v) = \langle f, v\rangle_{V'\times V}, \ \forall v \in V, \ u \in V, \tag{2.2.30}$$

where V is some Hilbert space and $a(u, v)$ is a bilinear (coercive) form on $V \times V$.

Suppose that we can find a larger space $S \supset V$ such that there exists a canonical extension $\tilde{a}(\cdot, \cdot)$ of a to $S \times S$, satisfying

$$\tilde{a}(u, v) = a(u, v), \ \forall u, \ v \in V. \tag{2.2.31}$$

Moreover, let $V_h \subset S$ be a family of finite-dimensional subspaces of S such that, given $v_h \in V_h$,

$$v = \lim_{h\to 0} v_h \Rightarrow v \in V. \tag{2.2.32}$$

When (2.2.32) is satisfied, V_h is said to be an external approximation to V. Assuming that f can be extended to an element $\tilde{f}$ in S', we can now approximate problem (2.2.30) by: find $u_h \in V_h$ such that

$$\tilde{a}(u_h, v_h) = \langle \tilde{f}, v_h\rangle_{S'\times S}, \ \forall v_h \in V_h. \tag{2.2.33}$$

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Using standard coerciveness and continuity assumptions, one gets from (2.2.30) and (2.2.33) a result known as Strang's lemma [146, 358].

$$\|u-u_h\|_S \le c \inf_{v_h\in V_h} \|u-v_h\|_S + \sup_{v_h\in V_h} \frac{|\tilde{a}(u,v_h)-\langle \tilde{f}, v_h\rangle|}{\|v_h\|_S}$$
$$= c \inf_{v_h\in V_h} \|v-v_h\| + E_h(u,v_h). \tag{2.2.34}$$

The last term can be seen as a *consistency* term: it measures how well the exact solution satisfies the discrete equation. This term vanishes when $V_h \subset V$ and we get the standard result for the conforming case.

In classical situations we have $V = H^1(\Omega)$ or $V = H^2(\Omega)$ (or one of their subspaces). Introducing a partition of the domain into m sub-domains K_r, and assuming $V = H^1(\Omega)$, we take $S = X(\Omega)$ as defined in Sect. 2.1.3. Any bilinear form of the type

$$\int_\Omega a(x)\, \underline{\mathrm{grad}}\, u \cdot \underline{\mathrm{grad}}\, v\, dx \tag{2.2.35}$$

can immediately be extended to $X(\Omega)$ by writing

$$\tilde{a}(u,v) = \sum_{r=1}^{m} \int_{K_r} a(x)\, \underline{\mathrm{grad}}\, u \cdot \underline{\mathrm{grad}}\, v\, dx. \tag{2.2.36}$$

We now want to find a subspace of $X(\Omega)$ approximating $H^1(\Omega)$ such that error estimates obtained from (2.2.27) are "optimal". Optimality is here relative to the degree of the polynomials from which the approximation is built: we would like to get $O(h^k)$ estimates when using polynomials of degree k. We are thus led to study the second term in the right-hand side of (2.2.34). We shall make this analysis later in the context of hybrid finite element methods; we shall therefore merely state the result which is quite classical [142, 165, 211, 259], which was discovered on empirical grounds, known as the *Céa patch test*: *the moments up to degree* $k-1$ *of* u_h *on any interface of the partition must be continuous, that is,*

$$\int_e u_h\, p_{k-1}\, dx, \ \forall\, p_{k-1} \in P_{k-1}(e) \tag{2.2.37}$$

is continuous across any interface e between two adjacent elements.

A more general form was given by Lascaux and Lesaint [275]. It states that the consistency term $E_h(u,v_h)$ must vanish whenever $u \in P_r(\Omega)$. For plate problems, this implies a condition similar to (2.2.37) for u_h and its derivatives. To fix ideas we recall a few classical examples.

In conformity to notation (2.2.23), we denote by $\mathcal{L}^{1,\,NC}(S_k, T_h)$ a nonconforming approximation of $H^1(\Omega)$ built from functions of $S_k(K)$. We shall simplify this notation whenever possible as in the following example.

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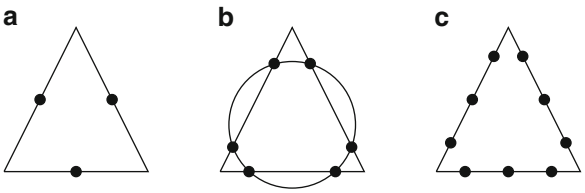


Fig. 2.8 Continuity points for nonconforming elements. **(a)** $k = 1$. **(b)** $k = 2$. **(c)** $k = 3$

Example 2.2.5 (Nonconforming elements on the triangle). Let us consider a partition of Ω into straight-sided triangles and an approximation

$$\mathcal{L}_k^{1,NC} := \{v_h \mid v_h \in L^2(\Omega),\ v_h|_K \in P_k(K),\ \forall K \in \mathcal{T}_h,$$
$$\sum_K \int_{\partial K} v_h \phi \, ds = 0,\ \forall \phi \in R_k(\partial K)\}. \quad (2.2.38)$$

It is then easy to see that the patch test implies that the functions of $\mathcal{L}_k^{1,NC}$ should be continuous at the k Gauss-Legendre points on every side of the triangles (see Fig. 2.8).

For k odd, those points, with internal points for $k \geq 3$, can be used as degrees of freedom, but for k even it is not so and the values at these points are not independent. We shall come back to this point in Sect. 7.1.4, in particular in Example 7.1.4. The trouble is that the six Gaussian points of the $k = 2$ case lie on an ellipse centered at the barycentre. This ellipse is easily expressed, in terms of barycentric coordinates by defining the *nonconforming bubble*, an ellipsoid taking value one at the barycentre and vanishing at the Gaussian points of the edges

$$b_{NC}(K) = 1 - \frac{4}{3}(\lambda_1^2 + \lambda_2^2 + \lambda_3^2). \quad (2.2.39)$$

It was however shown in [209] that this element can nevertheless be used in a very simple way. This was extended to the three-dimensional case in [200]. In this case the nonconforming bubble becomes

$$b_{NC}(K) = 1 - 2(\lambda_1^2 + \lambda_2^2 + \lambda_3^2 + \lambda_4^2). \quad (2.2.40)$$

It must be noted that, in three-dimensional nonconforming elements, the patch test implies, in general, no point continuity. However, there exists a six-point quadrature formula on the triangle, exact for polynomials of degree 3, of which all points lie on the ellipse defined by (2.2.40) and this ensures the patch-test. □

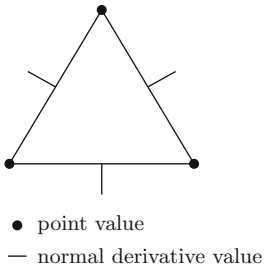
Nonconforming approximations of $H^2(\Omega)$, [147, 148], have been widely used because of the difficulty to obtain conforming elements. We refer the reader to [275] where several examples are given. We shall however use in Chaps. 8 and 10 the following nonconforming approximation of $H^2(\Omega)$.

Example 2.2.6 (Morley's triangle). In plate problems, where an approximation of $H^2(\Omega)$ is needed, an important nonconforming element is Morley's triangle (Fig. 2.9).

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Fig. 2.9 Morley's triangle



The functions v_h are supposed to be in $P_2(K)$ for every K. The degrees of freedom are point values at the vertices of the triangle and normal derivatives at mid-side points. It can be shown by the method of [275] that this provides a consistent approximation that will converge as $O(h)$ in a *discrete* $H^2(\Omega)$-norm. We shall denote by $\mathcal{L}_2^{2,NC}$ the approximation of $H^2(\Omega)$ built from such elements. □

Example 2.2.7 (Nonconforming elements on the rectangle). We consider a partition of Ω into straight-sided quadrilaterals, and an approximation of $H^1(\Omega)$ defined by

$$\mathcal{L}_{[k]}^{1,NC} := \{u_h \mid u_h \in L^2(\Omega),\ u_h|_{K_r} = \hat{u}_h \circ F_r^{-1},\ \hat{u}_h \in Q_k(\hat{K}) \text{ and } (2.2.37) \text{ holds}\}. \quad (2.2.41)$$

Here again the patch test implies continuity at the Gauss-Legendre points of the interfaces. It is never possible to use these points as degrees of freedom. For $k = 1$, the function $(\hat{x} - \frac{1}{2})(\hat{y} - \frac{1}{2}) \in Q(\hat{K})$ vanishes at the four Gauss-Legendre points of the sides that are indeed midpoints in this case. For $k = 2$, the points lie on an ellipse, and so on. It is however possible to extend the method of [209] to these cases. Another example of a nonconforming quadrilateral element is presented in [330].

□

The above examples are in no way exhaustive: many other nonconforming approximations can be built and some are indeed effectively used [244, 279]. As we shall see in the sequel, nonconforming methods are strongly related, and often equivalent, to hybrid methods or mixed methods. We think it is preferable to delay further examples until they are met in a proper context.

2.2.4 *Quadrilateral Finite Elements on Non Affine Meshes*

We present here some general results about finite elements on non affine meshes. A mesh $\mathcal{T}_h$ is called *affine* when for all elements $K \in \mathcal{T}_h$ the mapping F_K from the reference element $\hat{K}$ to K is an affine function. If this is the case, the Jacobian matrix DF_K is constant and this property has important consequences for the theoretical analysis.

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On the other hand, non affine meshes are met very frequently in real world applications. They may occur in the approximation of general domains, when isoparametric elements are used (see, for instance, Example 2.2.2). Moreover, non affine meshes are generated when quadrilateral or hexahedral elements are used. The fact that the Jacobian matrix DF_K may not be constant not only makes the analysis of non affine elements more difficult, but may also lead to substantial degeneracy of their approximation properties.

The aim of this section is to recall recent results which hold for the quadrilateral finite element approximation of scalar functions (see [20]) and of vector-valued functions in $H(\mathrm{div};\Omega)$ (see [21]).

The possible sub-optimality of some quadrilateral finite elements has been observed by several authors, often as a result of numerical experiments; a non exhaustive list of relevant references is [270, 297, 330, 384, 386].

In order to understand where the trouble come from, let us make some comments on Corollary 2.2.1 in the case of non affine partitions. We consider, for instance, the reference triangle $\hat{K}$ and its image $K = F_K(\hat{K})$. Let us take a smooth function $\hat{v} : \hat{K} \to \mathbb{R}$, the corresponding mapped function $v = \hat{v} \circ F_K^{-1} : K \to \mathbb{R}$, and its linear interpolant $r_h v : K \to \mathbb{R}$. When F_K is affine, then K is a triangle with straight sides, the Jacobian matrix $DF(\hat{x}) = B$ is constant and Corollary 2.2.1 reads

$$|v - r_h v|_{1,K} \le c h_K |v|_{2,K}. \tag{2.2.42}$$

A fundamental ingredient for such an estimate is the scaling (2.1.68), which in this particular situation reads

$$|\hat{v}|_{2,\hat{K}} \le c|\det(B)|^{-1/2}\|B\|^2 |v|_{2,K} \le c|\det(B)|^{-1/2} h_K^2 |v|_{2,K}. \tag{2.2.43}$$

When the Jacobian matrix is not constant, we cannot use (2.1.68) and the chain rule gives

$$|\hat{v}|_{2,\hat{K}} \le c \left|\inf_{\hat{x}}(J(\hat{x}))\right|^{-1/2} \left(\|DF\|^2_{L^\infty(\hat{K})} |v|_{2,K} + \|D^2 F\|_{L^\infty(\hat{K})} |v|_{1,K}\right). \tag{2.2.44}$$

The term $\|DF\|^2_{L^\infty(\hat{K})}$ in the right hand side is typically $O(h_K^2)$, while the norm of the Hessian matrix D^2F might be a lower order term. This fact is clearly a potential source of trouble for optimal order approximation.

Remark 2.2.5. Besides the situations presented in Sects. 2.2.5 and 2.5.5, there are other cases which could be studied but for which the analysis is not yet completed. Three dimensional $H(\mathrm{div},\Omega)$ and $H(\underline{\mathrm{curl}};\Omega)$ approximations on general hexahedral meshes, for instance, do not have a complete analysis yet (see, for instance, [19, 191]), while it is known that standard finite elements are suboptimal in several situations [64, 307]. □

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2.2.5 *Quadrilateral Approximation of Scalar Functions*

Necessary and sufficient conditions for optimal order approximation by quadrilateral finite elements have been investigated in [20]. The theory applies to finite elements defined on a reference square element $\hat{K}$ and mapped to the actual quadrilateral element K by the standard transformation (2.1.59). The generic mapping F_K is bilinear in each component, so that K is a quadrilateral with straight sides.

Before stating the main results, let us recall a measure of shape regularity for quadrilateral meshes.

Definition 2.2.1. Let $\{\mathcal{T}_h\}$ be a family of partitions of *convex* quadrilaterals. For each K, consider the four triangles obtained by the possible choices of three vertices from the vertices of K and denote by ρ_K the smallest diameter of the circles inscribed in the four triangles. Define $\sigma_K = h_K/\rho_K$, where h_K is as usual the diameter of K. The **family of partitions** $\{\mathcal{T}_h\}$ is said **shape-regular** if

$$\sigma_K \leq C \quad \forall K,$$

uniformly in h.

The shape regularity is equivalent to a uniform bound on the ratio of any two sides of the elements and also to a bound away from 0 and π for the element angles.

Given a smooth function $u : \Omega \to \mathbb{R}$ and a finite element space family V_h, we are interested in the following optimal approximation properties (k is the polynomial degree of the reference finite element space and $\underline{\mathrm{grad}}_h$ is the element-by-element gradient):

$$\inf_{v_h \in V_h} \|u - v_h\|_{0,\Omega} = O(h^{k+1}), \tag{2.2.45}$$

$$\inf_{v_h \in V_h} \|\underline{\mathrm{grad}}_h(u - v_h)\|_{0,\Omega} = O(h^k). \tag{2.2.46}$$

It is well known (see, for instance, [146, 223]) that a sufficient condition for (2.2.46) to hold is that *the reference finite element space contains* $Q_k(\hat{K})$, the space of polynomials of degree less than or equal to k in each variable, separately. In this case, the following estimates are known to hold

$$\inf_{v_h \in V_h} \|u - v_h\|_{0,\Omega} \leq c h^{k+1} |u|_{k+1,\Omega}, \tag{2.2.47}$$

$$\inf_{v_h \in V_h} \|\underline{\mathrm{grad}}_h(u - v_h)\|_{0,\Omega} \leq c h^{k+1} |\underline{\mathrm{grad}}\, u|_{k+1,\Omega}. \tag{2.2.48}$$

Indeed, *this is also a necessary condition*. This result has been proved in [20] by exhibiting a very simple (and far from pathological) counterexample: the domain is a square, the mesh sequence, sketched in Fig. 2.10, is made of self similar trapezoids and the function to be approximated is as smooth as possible (a polynomial of degree k).

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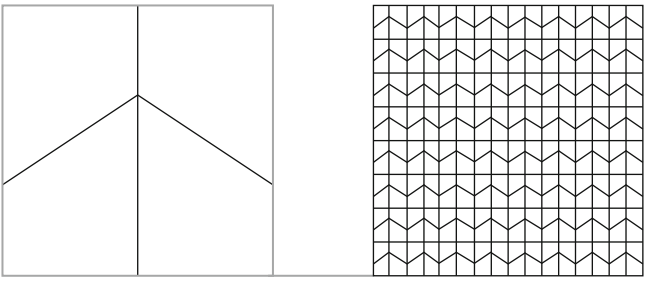


Fig. 2.10 Distorted quadrilateral mesh based on a trapezoid macro-element

Remark 2.2.6. The presented result has important consequences for some commonly used finite element space families. In particular, serendipity elements, obtained from standard Q_k elements by eliminating some internal degrees of freedom, cannot achieve optimal approximation order (2.2.46) on general quadrilateral meshes. □

Remark 2.2.7. The presented results extend to three space dimensions in a straightforward way [296]. □

Remark 2.2.8. The degeneracy of the approximation properties is related to the mesh distortion. In particular, on asymptotically affine partitions, the optimal approximation order is achieved. We refer the interested reader to [20] for the definition of an asymptotically affine mesh and for the proof of this result. □

2.2.6 Non Polynomial Approximations

2.2.6.1 Spaces $\mathcal{L}^s_k(\mathcal{E}_h)$

In the applications involving hybrid methods, it will be useful to consider approximation spaces built from functions that have a polynomial trace on ∂K but which are not necessarily polynomials inside K. These spaces will be useful whenever only the trace is computationally important: they can be thought of as defined only on $\mathcal{E}_h = \bigcup_K \partial K$ (cf. (2.1.42)). We thus define for $s \geq 1$

$$\mathcal{L}^s_h(\mathcal{E}_h) := \{v \mid v \in H^s(\Omega)\ v|_{\partial K} \in T_k(\partial K),\ \forall K\} \tag{2.2.49}$$

and for $s = 0$

$$\mathcal{L}^0_k(\mathcal{E}_h) := \{v \mid v \in L^2(\mathcal{E}_h),\ v|_{\partial K} \in R_k(\partial K),\ \forall K\}. \tag{2.2.50}$$

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For $s \geq 1$, functions of $\mathcal{L}_k^s(\mathcal{E}_h)$ are evidently approximations of $H^1(\Omega)$ of optimal order with respect to k. It is also possible to get error estimates on the traces.

Virtual element methods. The spaces $\mathcal{L}_k^s(\mathcal{E}_h)$ can easily be defined on polygons of arbitrary shape, but (as we already pointed out) they can be used only on the boundary of elements, and, even restricted to a single element, are *infinite dimensional*. To reach a finite dimensional version, one has to *prolongate* them inside the elements. This can be done in several ways (see e.g. [367, 377] or [306] and the reference therein).

Recently a variant of these methods has been introduced, called *Virtual Elements*, in which the extension is made as the solution of a partial differential equation, but the construction is such that one does not need to know the solution of this PDE in order to compute the stiffness matrix.

Let us see a simple example, just to give the flavour of the idea. On an element K (a polygon of a practically arbitrary shape), we define $VM_1(K)$ as the space of functions that are linear on each edge of K and harmonic inside. The dimension of such a space is clearly equal to the number of vertices of K. If we want to compute

$$a^K(v, p) := \int_K \underline{\mathrm{grad}}\, v \cdot \underline{\mathrm{grad}}\, p \, dx, \tag{2.2.51}$$

where v is generic in $VM_1(K)$ and p is a polynomial of degree ≤ 1, we have

$$a^K(u, p) \equiv \int_K \underline{\mathrm{grad}}\, v \cdot \underline{\mathrm{grad}}\, p \, dx = \int_{\partial K} v \, \frac{\partial p}{\partial n} \, ds, \tag{2.2.52}$$

which is easily computable from the knowledge of v on ∂K. This allows to compute a projection operator (that we denote by Π_1^∇) from $VM_1(K)$ to $P_1(K)$ by

$$\int_{\partial K} (v - \Pi_1^\nabla v)\, ds = 0 \ \text{ and } \ a^K(v - \Pi_1^\nabla v, q) = 0 \ \forall q \in P_1(K). \tag{2.2.53}$$

Then we can take as *approximate local stiffness matrix* a_h^K the following expression:

$$a_h^K(u, v) := a^K(\Pi_1^\nabla u, \Pi_1^\nabla v) + S^K(u - \Pi_1^\nabla u, v - \Pi_1^\nabla v), \tag{2.2.54}$$

where S^K is any bilinear form acting on the vertex values and scaling like 1 (for instance, for a polygon with, say, five vertices, the usual scalar product in $\mathbb{R}^5$ will do). This will provide an optimal error bound (see [57]).

This can be extended to higher orders to improve the accuracy. Taking $k = 2$, for simplicity, we now define $VM_2(K)$ as the space of functions that are quadratic on each edge of K and whose Laplacian, inside K, is a constant (in general: a polynomial in $P_{k-2}(K)$). The dimension of such a space is clearly equal to twice the number of vertices of K plus one. As degrees of freedom we take the values

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at the vertices, the values at the midpoints of the edges, and the mean value on K. Now, if we want to compute

$$a^K(v, p) := \int_K \underline{\mathrm{grad}}\, v \cdot \underline{\mathrm{grad}}\, p \, dx, \tag{2.2.55}$$

where v is generic in $VM_2(K)$ and p is a polynomial of degree ≤ 2, we have

$$\int_K \underline{\mathrm{grad}}\, v \cdot \underline{\mathrm{grad}}\, p \, dx = -\int_K v \, \Delta p \, dx + \int_{\partial K} v \, \frac{\partial p}{\partial n} \, ds, \tag{2.2.56}$$

which is easily computable, as we know v on ∂K and its average on K. This allows to compute a projection operator (that we now denote by Π_2^∇) from $VM_2(K)$ to $P_2(K)$ by

$$\int_K (v - \Pi_2^\nabla v)\, ds = 0 \ \text{ and } \ a^K(v - \Pi_2^\nabla v, q) = 0 \ \forall q \in P_2(K). \tag{2.2.57}$$

Following (2.2.54) we can then take as *approximate local stiffness matrix* a_h^K the following one:

$$a_h^K(u, v) := a^K(\Pi_2^\nabla u, \Pi_2^\nabla v) + S^K(u - \Pi_2^\nabla u, v - \Pi_2^\nabla v), \tag{2.2.58}$$

where again S^K is any bilinear form acting on the vertex and inside values and scaling like 1 (for instance, for a polygon with, say, five vertices, the usual scalar product in $\mathbb{R}^{11}$ will do). We still refer to [57] for more details.

We observe that another possibility would be to consider the space $\widetilde{VM}_1$ of functions that are linear on each edge, whose Laplacian is constant, and such that

$$\int_K (v - \Pi_1^\nabla v)\, dx = 0, \tag{2.2.59}$$

where Π_1^∇ is still defined by (2.2.53) using only the boundary values of v. The advantage of $\widetilde{VM}_1$ over VM_1 is that in $\widetilde{VM}_1$ we can compute exactly the mean value of a function using only its boundary values; this can be useful, for instance, to define the Virtual Element spaces on a polygonal *face* of a polyhedron, in particular, if thereafter we want to use a formula like (2.2.56). For more information in this direction see [4].

2.2.7 Scaling Arguments

We shall briefly recall here the basic idea of the scaling arguments of [180]. We shall do it on a very simple example, but it will be clear how the idea applies to more

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general cases. Assume that we want to prove the following *inverse inequality* for elements $v_h \in \mathcal{L}_k^s$: there exists a constant c depending only on k and on the minimum angle θ_0 in $\mathcal{T}_h$ such that, on every element K, we have

$$|v_h|_{1,K} \leq c h_K^{-1} \, |v_h|_{0,K}. \tag{2.2.60}$$

We construct first a new element $\hat{K}$ such that the mapping $F : \hat{K} \to K$ is simply given by

$$\underline{x} = h_K \underline{\hat{x}} + \underline{b} \tag{2.2.61}$$

and K has a vertex at the origin. Formulas (2.1.67) and (2.1.68) then simply become (in two dimensions)

$$|\hat{v}|_{m,\hat{K}} = h_K^{m-1} |v|_{m,K} \tag{2.2.62}$$

and we easily get

$$|v_h|_{1,K} = |\hat{v}|_{1,\hat{K}} \leq c(k, \hat{K}) \, |\hat{v}|_{0,\hat{K}} \leq c(k, \hat{K}) \, h_K^{-1} \, |v|_{0,K}. \tag{2.2.63}$$

Now we remark that $c(k, \hat{K})$ actually depends *continuously* on the shape of $\hat{K}$ (a similar argument was already used in [127]). In particular, if one considers the family K_{θ_0} of all the triangles having diameter $= 1$, one vertex at the origin and a minimum angle $\geq \theta_0$, one easily gets

$$\sup_{\hat{K} \in K_{\theta_0}} c(k, \hat{K}) \leq c(k, \theta_0) \tag{2.2.64}$$

by compactness [180]. Hence from (2.2.63) and (2.2.64) we get

$$|v_h|_{1,K} \leq c(k, \theta_0) \, h_K^{-1} \, |v|_{0,K}, \tag{2.2.65}$$

that is (2.2.60).

Note that, in this particular case, it would have been equally easy (or even easier) to derive directly (2.2.60) by using (2.1.67) and (2.1.68) and a fixed $\hat{K}$ = unit triangle. However, (2.2.62) is easier to use and the continuity argument (2.2.64) is always essentially the same in many different applications, so that using the scaling (2.2.61) actually results in a simplification. For instance, one can get by this method the inequality

$$\int_{\partial K} |v_h| \, d\sigma = h_K \int_{\partial \hat{K}} |\hat{v}_h| \, d\hat{\sigma} \leq c(k, \theta_0) \, h_K \, |\hat{v}_h|_{o,\hat{K}} = c(k, \theta_o) \, |v_h|_{0,K}. \tag{2.2.66}$$

In the same way, one can guess, for instance, that one has

$$\|\partial v_h / \partial n\|_{L^\infty(\partial K)} \leq c(k, \theta_0) \, h_k^{-2} \, |v_h|_{0,K}, \tag{2.2.67}$$

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because both sides scale like h_K^{-1} in the transformation (2.2.61) and the inequality holds on a fixed element of size $= 1$. However, note that an inequality of the type

$$\|v_h\|_{L^\infty(\partial K)} \leq c(k, \theta_0)\ |v_h|_{1,K} \tag{2.2.68}$$

is still hopeless (take $v_h = 1$!) unless we specify, for instance, that v_h has zero mean value in K.

2.3 Simplicial Approximations of $H(\mathrm{div}; \Omega)$ and $H(\underline{\mathrm{curl}}; \Omega)$

Although this section and the following one are important by themselves (we shall use $H(\mathrm{div}; \Omega)$ or $H(\underline{\mathrm{curl}}; \Omega)$ in many applications throughout this book), its importance also lies in its value as a model. The techniques introduced for the approximation of $H(\mathrm{div}; \Omega)$ can indeed be applied to other situations and similar constructions have been employed in the discretisation of the Hellan-Hermann-Johnson mixed formulation for which we refer to [318] and [319]. The approximations that we shall present derive from the original work of [366], and [331] later generalized and extended to the three-dimensional case by Nédélec [310]. We shall also use the results of [118, 120], and [311] for the definition of elements that contain (for simplicial elements) the elements of [310] and [331]. In the case of rectangles, we introduce a general element containing the elements of [331], the elements of [120] and the ones of [119], thus clarifying the relation between those two. As the simplicial case is simple and more intuitive, we shall first consider it in detail. Quadrilateral and hexahedral elements will be treated afterwards.

2.3.1 Simplicial Approximations of $H(\mathrm{div}; \Omega)$

In this section, the element K will be either a triangle ($n = 2$) or a tetrahedron ($n = 3$) and we will suppose that we have a mesh $\mathcal{T}_h$ built from such elements. We denote by e_i ($i = 1, 2, 3$ or $i = 1, 2, 3, 4$) the sides (or the faces) of K. We then start from the general space of piecewise polynomial vectors. It will be convenient to denote

$$\underline{P}_k(K) := (P_k(K))^n. \tag{2.3.1}$$

In analogy with (2.2.23), we want to build approximations of $H(\mathrm{div}; \Omega)$ of the form

$$\mathcal{L}^{\mathrm{div}}(\underline{S}_k, \mathcal{T}_h) := \{\underline{p} \in H(\mathrm{div}; \Omega) : \underline{p}|_K \in \underline{S}_k(K)\}, \tag{2.3.2}$$

which will evidently imply continuity of the normal traces. We shall thus proceed to build suitable subspaces of to ensure this continuity. We first define, using (2.2.5)

$$\underline{P}_k^{n,s}(K) := \{\underline{p} \in \underline{P}_k(K) : \underline{p} \cdot \underline{n} \in R_s(\partial K)\} \tag{2.3.3}$$

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and

$$\underline{P}^{n,s}_{\underline{k}+\underline{x}k}(K) := \{\underline{p} \in (\underline{P}_k(K) + \underline{x}P_k(K)) : \underline{p}\cdot\underline{n} \in R_s(\partial K)\}, \tag{2.3.4}$$

where $\underline{x} = (x_1, x_2, \ldots, x_n)$. For these spaces, we shall write (2.3.2) in a compact way:

$$\mathcal{L}^{\mathrm{div}}_{k,s}(\mathcal{T}_h) = \mathcal{L}^{\mathrm{div}}(\underline{P}^{n,s}_k, \mathcal{T}_h) \tag{2.3.5}$$

$$\mathcal{L}^{\mathrm{div}}_{\underline{k}+\underline{x}k,s}(\mathcal{T}_h) = \mathcal{L}^{\mathrm{div}}(\underline{P}^{n,s}_{\underline{k}+\underline{x}k}, \mathcal{T}_h) \tag{2.3.6}$$

or even $\mathcal{L}^{\mathrm{div}}_{k,s}$ and $\mathcal{L}^{\mathrm{div}}_{\underline{k}+\underline{x}k,s}$ when there will be no ambiguity as to the choice of $\mathcal{T}_h$. The case $s = k$ is the most natural and widely used. Taking $s \leq k-1$ defines what we shall call *reduced* spaces. We introduce special names for a few classical cases. For $s = k$, the space $\underline{P}^{n,k}_k(K)$ was introduced in [120] (for $n = 2$) and [118] (for $n = 3$). We shall thus call it the Brezzi-Douglas-Marini space and write,

$$\mathcal{BDM}_k(K) := \underline{P}^{n,k}_k(K). \tag{2.3.7}$$

The dimension of $\mathcal{BDM}_k(K)$ is

$$\dim \mathcal{BDM}_k(K) = \begin{cases} (k+1)+(k+2), & \text{for } n = 2, \\ \frac{1}{2}(k+1)(k+2)(k+3), & \text{for } n = 3. \end{cases} \tag{2.3.8}$$

The space $\underline{P}^{n,k}_{\underline{k}+\underline{x}k}$ was defined in [310] following [331]. We shall call it the Raviart-Thomas space and write

$$\mathcal{RT}_k(K) := \underline{P}^{n,k}_{\underline{k}+\underline{x}k}. \tag{2.3.9}$$

Finally the reduced case $\underline{P}^{n,k-1}_k$ was considered in [119] and we shall write

$$\mathcal{BDFM}_k(K) := \underline{P}^{n,k-1}_k(K). \tag{2.3.10}$$

Remark 2.3.1. The original work of [331] used an expression equivalent to (2.3.4) with $s = k$ on the *reference element* $\hat{K}$ and defined $\mathcal{RT}_k(K)$ by the change of variable $\mathcal{G}$ of (2.1.69). It must be noted that this definition is not equivalent to the definition of $\mathcal{RT}_k(K)$ given above: it depends on the orientation of space. For simplicial elements, definition (2.3.4) is more natural and easier to handle. □

For the simplicial case, we thus have the following inclusions between the spaces just defined:

$$\mathcal{RT}_{k-1} \subset \mathcal{BDFM}_k \subset \mathcal{BDM}_k \subset \mathcal{RT}_k \subset \mathcal{BDFM}_{k+1} \subset \mathcal{BDM}_{k+1} \subset \mathcal{RT}_{k+1}. \tag{2.3.11}$$

We now have to define suitable degrees of freedom. For $\underline{q} \in \underline{P}^{n,s}_k(K)$, we evidently have $\mathrm{div}\,\underline{q} \in P_{k-1}(K)$. Moreover, the normal trace $\underline{q}\cdot\underline{n}$ on ∂K belongs

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to $R_s(\partial K)$. In order to build from $\underline{P}_k^{n,s}(K)$ an approximation of $H(\mathrm{div};\Omega)$, it will be necessary to ensure continuity of $\underline{q}\cdot\underline{n}$ at the interfaces. This will be made possible by the choice of appropriate degrees of freedom.

Proposition 2.3.1. *For $k \geq 1$ and for any $\underline{q} \in \underline{P}_k^{n,s}(K)$, the following relations imply $\underline{q} = 0$*

$$\int_{\partial K} \underline{q}\cdot\underline{n}\; p_s\, ds = 0, \qquad \forall\, p_s \in R_s(\partial K), \tag{2.3.12}$$

$$\int_K \underline{q}\cdot \underline{\mathrm{grad}}\; p_{k-1}\, dx = 0, \qquad p_{k-1} \in P_{k-1}(K), \tag{2.3.13}$$

$$\int_K \underline{q}\cdot\underline{p}_k\, dx = 0, \qquad \forall \underline{p}_k \in \underline{\mathbb{H}}_k(K), \tag{2.3.14}$$

where

$$\underline{\mathbb{H}}_k(K) := \{\underline{q}_k \mid \underline{q}_k \in \underline{P}_k(K),\ \mathrm{div}\,\underline{q}_k = 0,\ \underline{q}_k\cdot\underline{n}|_{\partial K} = 0\}. \tag{2.3.15}$$

Indeed, it is easy to check that (2.3.12) and (2.3.13) are equivalent to $\underline{q} \in \underline{\mathbb{H}}_k(K)$ as (2.3.12) implies $\underline{q}_k\cdot\underline{n}|_{\partial K} = 0$. Moreover,

$$\int_K \mathrm{div}\,\underline{q}\; p_{k-1}\; dx = -\int_K \underline{q}\cdot \underline{\mathrm{grad}}\; p_{k-1}\, dx + \int_{\partial K} \underline{q}\cdot\underline{n}\; p_{k-1}\; ds. \tag{2.3.16}$$

Thus (2.3.12) and (2.3.13) imply $\mathrm{div}\,\underline{q} = 0$. Reciprocally, it is trivial that (2.3.12) and (2.3.13) hold for $\underline{q}_k \in \underline{\mathbb{H}}_k(K)$. □

To prove that (2.3.12)–(2.3.14) can be used to define degrees of freedom for $\underline{P}_k^{n,s}(K)$ by choosing bases for $R_s(\partial K)$, $P_{k-1}(K)$, and $\underline{\mathbb{H}}_k(K)$, there remains to check that the set obtained from (2.3.12) and (2.3.13) is linearly independent. This is the object of the next lemma.

Lemma 2.3.1. *Let $g \in R_s(\partial K)$ and $f \in P_{k-1}(K)$ be such that*

$$\int_{\partial K} g\underline{q}\cdot\underline{n}\; d\sigma + \int_K \underline{q}\cdot \underline{\mathrm{grad}}\; f\; dx = 0,\ \forall \underline{q} \in \underline{P}_k^{n,s}(K). \tag{2.3.17}$$

Then, $g = 0$ and $f =$ constant.

Proof. Using the change of variables (2.1.69) and Lemma 2.1.6, it is sufficient to prove the result on the reference element (see Fig. 2.11). We give the construction for $n = 3$ as the case $n = 2$ is a simple restriction of it. One first uses in (2.3.17)

$$q_1 = x\frac{\partial f}{\partial x}\lambda_4,\ q_2 = y\frac{\partial f}{\partial y}\lambda_4,\ q_3 = z\frac{\partial f}{\partial z}\lambda_4, \tag{2.3.18}$$

where λ_4 is the fourth barycentric coordinate, that is $\lambda_4 = 1 - x - y - z$.

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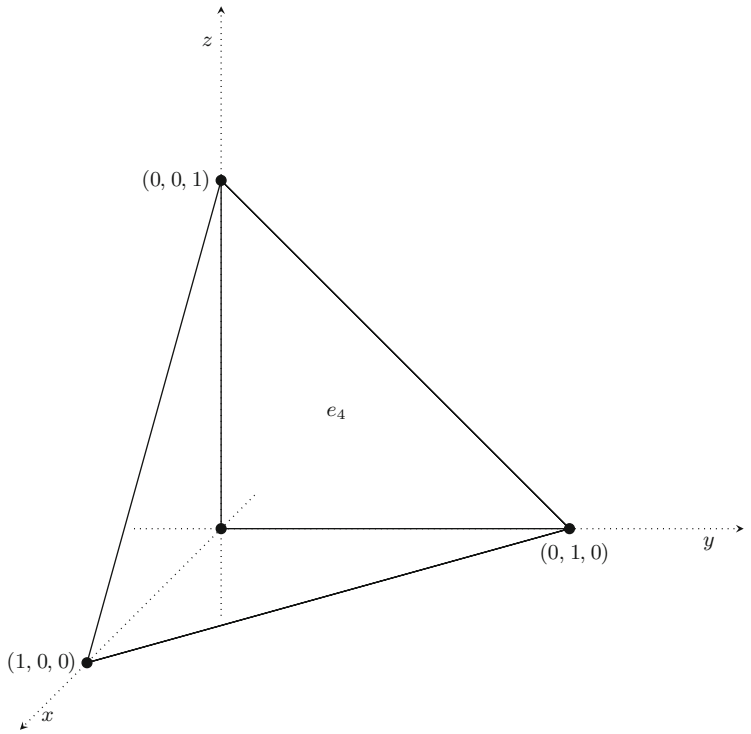


Fig. 2.11 The reference element

Then $\underline{q} \in \underline{P}_k(K)$ and $\underline{q} \cdot \underline{n}|_{\partial K} = 0$ and we get from (2.3.17)

$$\int_K \Big[x\Big(\frac{\partial f}{\partial x}\Big)^2 + y\Big(\frac{\partial f}{\partial y}\Big)^2 + z\Big(\frac{\partial f}{\partial z}\Big)^2 \Big] \lambda_4 = 0, \tag{2.3.19}$$

which implies $\underline{\mathrm{grad}}\, f = 0$ since all terms in the integral are positive. We now take $q_1 = x p_{s-1}$, $q_2 = q_3 = 0$. From this we obtain $\int_{e_4} xg \;\; p_{s-1} \; ds = 0$. In the same way we get $\int_{e_4} yg \;\; p_{s-1} \; ds = \int_{e_4} zg \;\; p_{s-1} \; ds = 0$ and, as $x + y + z = 1$ on e_4, $\int_{e_4} g \;\; p_{s-1} \; ds = 0$. All these conditions imply $g|_{e_4} = 0$. Finally, we take $q_i = g|_{e_i} = g_i$ and (2.3.17) implies $\sum_{i=1}^{3} \int_{e_i} (g_i)^2 \, ds = 0$, hence $g = 0$. □

Let us now count the number of conditions thus induced for $\mathcal{BDM}_k(K)$:

$$\dim R_k(\partial K) + \dim P_{k-1}(K) - 1 = \begin{cases} \frac{1}{2}k^2 + 7k + 4 & \text{for } n = 2, \\ \frac{1}{6}k^3 + 15k^2 + 38k + 18 & \text{for } n = 3. \end{cases} \tag{2.3.20}$$

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From this we can deduce, by standard arguments of linear algebra,

$$\dim \underline{\mathbb{H}}_k = \begin{cases} \dfrac{1}{2}k(k-1) = \dim\, P_{k-2}(K) \text{ for } n = 2,\ k \geq 2, \\ \dfrac{1}{6}3k^3 - 3k - \dfrac{1}{6}(k-2)(k-1)k = \begin{cases} \dim\,[(P_{k-2})^3] - \dim\,(P_{k-3}), \\ \qquad\qquad n = 3,\ k \geq 3, \\ \dim\,[(P_{k-2})^3],\ n=3,\ k=2. \end{cases} \end{cases} \tag{2.3.21}$$

In the two-dimensional case, the space $\underline{\mathbb{H}}_k(K)$ can easily be characterized.

Lemma 2.3.2. *For $n = 2$, we have,*

$$\underline{\mathbb{H}}_k(K) = \{\underline{p}_k \mid \underline{p}_k = \underline{\mathrm{curl}}(b_K\ p_{k-2}),\ p_{k-2} \in P_{k-2}(K)\}, \tag{2.3.22}$$

where $b_K = \lambda_1\lambda_2\lambda_3 \in B_3(K)$ is the bubble function on K.

Proof. Any $\underline{p}_k \in \underline{\mathbb{H}}_k(K)$ is the curl of a polynomial of degree $k+1$. A simple count of degrees of freedom concludes the proof. □

In the three-dimensional case, the construction of $\underline{\mathbb{H}}_k(K)$ is less direct. It is still true that $\underline{p}_k \in \underline{\mathbb{H}}_k(K)$ implies that $\underline{p}_k$ is the $\underline{\mathrm{curl}}$ of a vector function polynomial of degree $k+1$. To characterise $\underline{\mathbb{H}}_k(K)$, we need the polynomial spaces that will be introduced in the next section for the approximation of $H(\underline{\mathrm{curl}};\Omega)$.

The next result shows, in particular, that the internal degrees of freedom coming from (2.3.13) and (2.3.14) can be replaced by a term involving the space $\mathcal{N}_{k-2}(K)$ which will be introduced in (2.3.37) for the approximation of $H(\underline{\mathrm{curl}};\Omega)$. The reader is referred to Sect. 2.3.2 for more details.

Proposition 2.3.2. *For $k \geq 1$ and for any $\underline{q} \in \underline{P}_k^{n,s}(K)$, the following relations imply $\underline{q} = 0$,*

$$\int_{\partial K} \underline{q}\cdot\underline{n}\ p_s\, ds = 0, \qquad \forall\ p_s \in R_s(\partial K), \tag{2.3.23}$$

$$\int_K \underline{q}\cdot\underline{w}_{k-2}\, dx = 0, \qquad \forall \underline{w}_{k-2} \in \mathcal{N}_{k-2}(K), \tag{2.3.24}$$

where $\mathcal{N}_{k-2}(K)$ is defined in (2.3.37).

Proof. Space $\mathcal{N}_{k-2}(K)$ contains the gradients of all polynomials of degree $k-1$ so that condition (2.3.24) is stronger than (2.3.13). We thus get $\underline{q} \in \underline{\mathbb{H}}_k(K)$. By Proposition 2.3.4, a polynomial of degree k which is divergence-free is the curl of $\underline{\phi} \in \mathcal{N}_k^+(K)$ and $\underline{q}\cdot\underline{n} = 0$ implies that we must take the tangential trace $\pi_t\underline{\phi}$ to vanish on ∂K. From Proposition 2.3.5 it follows that

$$\int_K \underline{\phi}\cdot\underline{p}_{k-2} dx = 0, \qquad \forall \underline{p}_{k-2} \in \underline{P}_{k-2} \tag{2.3.25}$$

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which implies $\underline{\phi} = 0$. On the other hand, (2.3.24) implies

$$\int_K \underline{\mathrm{curl}}\,\underline{\phi}\cdot\underline{w}_{k-2}\,dx = \int_K \underline{\phi}\cdot\underline{\mathrm{curl}}\,\underline{w}_{k-2}\,dx = 0, \qquad \forall \underline{w}_{k-2}\in\mathcal{N}_{k-2}(K) \tag{2.3.26}$$

and $\underline{\mathrm{curl}}\,\mathcal{N}_{k-2}(K)$ spans $\mathcal{BDM}^0_{k-2}(K)$ (see Proposition 2.3.4). From Lemma 2.3.3, the complement of $\mathcal{BDM}^0_{k-2}(K)$ is made of gradients and the result follows. □

One must also say that the degrees of freedom described above have mainly *a theoretical importance*, for instance in building a B-compatible interpolation operator for proving the *inf-sup* condition. In practice, as we shall see in the applications of Chap. 7, any basis of $\underline{P}_k$ will be convenient and standard degrees of freedom can be used.

We can now consider the case of $\mathcal{RT}_k(K)$ as defined in (2.3.9). It can easily be checked that the dimension of $\mathcal{RT}_k(K)$ is given by

$$\dim \mathcal{RT}_k(K) = \begin{cases} (k+1)(k+3) & \text{for } n=2,\\ \frac{1}{2}(k+1)(k+2)(k+4) & \text{for } n=3, \end{cases} \tag{2.3.27}$$

and that only the part of $\underline{x}P_k(K)$ involving homogeneous polynomials of degree k is important. We now prove some basic results about $\mathcal{RT}_k(K)$ spaces. These spaces have indeed been tailor designed in order to satisfy the properties which we now state in the following proposition.

Proposition 2.3.3. *For any n-simplicial element K we have for $\underline{q}\in\mathcal{RT}_k(K)$,*

$$\begin{cases} \mathrm{div}\,\underline{q}\in P_k(K),\\ \underline{q}\cdot\underline{n}|_{\partial K}\in R_k(\partial K). \end{cases} \tag{2.3.28}$$

Moreover, the divergence operator is surjective from $\mathcal{RT}_k(K)$ onto $P_k(K)$.

Proof. $\underline{q}\in\mathcal{RT}_k(K)$ can be written as $\underline{q} = \underline{q}_0 + \underline{x}p_k$ with $\underline{q}_0\in\underline{P}_k(K)$. It is then clear that $\mathrm{div}\,\underline{q}$ is a polynomial of degree k. This proves the results about $\mathrm{div}\,\underline{q}$. On the other hand, let $\underline{n} = \{n_1, n_2\}$ be the normal to a side (we consider the two-dimensional case for simplicity)

$$\underline{q}\cdot\underline{n} = \underline{q_0}\cdot\underline{n} + p_k(x_1n_1 + x_2n_2). \tag{2.3.29}$$

Along a side, $x_1n_1 + x_2n_2$ is constant, so that $\underline{q}\cdot\underline{n}$ is a polynomial of degree k. The same argument works in $\mathbb{R}^n$. To end the proof, we observe that in $\mathbb{R}^n$

$$\int_K \mathrm{div}(\underline{x}\,p)\,p\,dx = \frac{n}{2}\int_K |p|^2\,dx + \frac{1}{2}\int_{\partial K}(\underline{x}\cdot\underline{n})|p|^2\,ds, \tag{2.3.30}$$

so that $\mathrm{div}(\underline{x}\,p_k) = 0$ implies $p_k = 0$. Hence, $\mathrm{div}(\underline{x}P_k)$ has the same dimension as P_k. □

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Proposition 2.3.4. *For $k \geq 0$ and for any $\underline{q} \in \mathcal{RT}_k(K)$, the following relations imply $\underline{q} = 0$*

$$\int_{\partial K} \underline{q} \cdot \underline{n}\, p_k \, ds = 0, \ \forall p_k \in R_k(\partial K), \tag{2.3.31}$$

$$\int_k \underline{q} \cdot \underline{p}_{k-1} \, dx = 0, \ \forall \underline{p}_{k-1} \in (P_{k-1}(K))^n. \tag{2.3.32}$$

This is a variant of Proposition 2.3.1 and the proof is left as an exercise.

Let us now define

$$\mathcal{RT}_k^0(K) := \{\underline{q} \mid \underline{q} \in \mathcal{RT}_k(K), \operatorname{div} \underline{q} = 0\}. \tag{2.3.33}$$

We can define in the same way $\mathcal{BDM}_k^0(K)$ and $\mathcal{BDFM}_k^0(K)$. From (2.3.4), we can easily deduce the following result.

Corollary 2.3.1. $\mathcal{RT}_k^0(K) \subset (P_k(K))^n$.

Therefore $\mathcal{RT}_k^0(K) = \mathcal{BDM}_k^0(K)$ while $\mathcal{BDFM}_k^0(K) = \mathcal{RT}_{k+1}^0$ contains the same divergence-free vectors.

Corollary 2.3.2.

- *For $n = 2$, any $\underline{q}_0 \in \mathcal{RT}_k^0(K) = \mathcal{BDM}^0(K)$ is the curl of a stream-function $\psi_{k+1} \in P_{k+1}(K)/\mathbb{R}$.*
  The dimension of $\mathcal{RT}_k^0(K)$ is equal to dim $(P_{k+1}(K) - 1) = \frac{1}{2}(k+1)(k+4)$.
- *For $n = 3$, from Lemma 2.3.4, any $\underline{q}_0 \in \mathcal{RT}_k^0(K)$ is the curl of a vector $\underline{\psi} \in \mathcal{N}_k^+(K) = \mathcal{N}_k(K)/\operatorname{\underline{grad}} P_{k+1} = \underline{P}_{k+1}/\operatorname{\underline{grad}} P_{k+2}$.*

Finally, one can obtain a complement of $\mathcal{RT}_k^0(K)$ by the following construction.

Lemma 2.3.3. *Any $\underline{p}_k \in \underline{P}_k$ can be written as the sum*

$$\underline{p}_k = \underline{p}^0 + \operatorname{\underline{grad}}(b_{nc}(K) p_{k-1}) \tag{2.3.34}$$

with $\underline{p}^0 \in \mathcal{BDM}^0(K)$ where the nonconforming bubble is defined by (2.2.39).

Proof. It is clear that $b_{nc}(K) P_{k-1}$ vanishes on the ellipse $b_{nc} = 0$ and hence contains no harmonic functions. □

Before considering the case of $H(\operatorname{\underline{curl}}; \Omega)$ we present a few examples.

Example 2.3.1 (The spaces $\mathcal{RT}_0$, $\mathcal{RT}_1$, $\mathcal{BDM}_1$). From the results above, we know that $\mathcal{RT}_0$ is a space of dimension 3 containing polynomials of the form

$$\begin{cases} q_1(x, y) = a + cx, \\ q_2(x, y) = b + cy. \end{cases} \tag{2.3.35}$$

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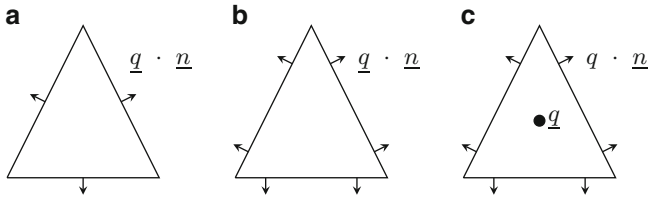


Fig. 2.12 (**a**) RT_0. (**b**) BDM_1. (**c**) RT_1

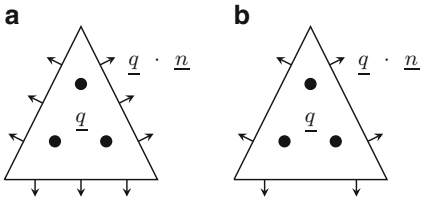


Fig. 2.13 (**a**) BDM_2. (**b**) $BDFM_2$

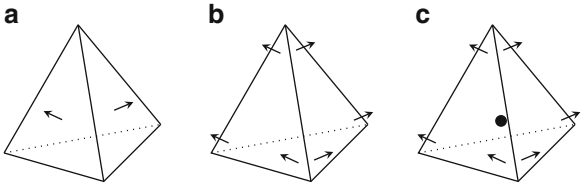


Fig. 2.14 (**a**) RT_0. (**b**) BDM_1. (**c**) RT_1

We can specify it by the three normal components of $\underline{q}$ on ∂K as sketched in Fig. 2.12. Space $\mathcal{BDM}_1$ is of dimension 6 and $\mathcal{RT}_1$ is of dimension 8. It must be noted that $\mathrm{div}\ \mathcal{BDM}_1 = \mathrm{div}\,\mathcal{RT}_0 = P_0$. In the same way, $\mathcal{BDM}_1$ is the subset of $\mathcal{RT}_1$ such that $\mathrm{div}\,\underline{q} \in P_0$ instead of P_1. The same concepts can be extended to $\mathcal{BDM}_2$ and $\mathcal{BDFM}_2$ as shown in Fig. 2.13. □

Example 2.3.2 (Three-dimensional elements: $\mathcal{RT}_0$, $\mathcal{BDM}_1$, $\mathcal{RT}_1$). The simplest cases of three-dimensional elements are depicted in Fig. 2.14. □

Remark 2.3.2 (Two-dimensional approximations of $H(\mathrm{curl};\Omega)$). Due to Remark 2.1.5, in the two dimensional case, approximations of $H(\mathrm{curl};\Omega)$ can be derived directly from the approximations of $H(\mathrm{div};\Omega)$ which we just presented since in two dimensions $H(\mathrm{curl};\Omega)$ is isomorphic to $H(\mathrm{div};\Omega)$ through a rotation by the angle $\pi/2$. Vector fields approximating $H(\mathrm{div};\Omega)$ will provide an approximation of $H(\mathrm{curl};\Omega)$ after a suitable rotation. □

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2.3.2 Simplicial Approximation of $H(\underline{\mathrm{curl}};\Omega)$

As stated in Remark 2.3.2, two dimensional approximations of $H(\underline{\mathrm{curl}};\Omega)$ can be derived directly from those of $H(\mathrm{div};\Omega)$. For this reason, we restrict our presentation to the three-dimensional case. In the following, K will therefore be a tetrahedron. The problem of defining a subspace of $\underline{P}_k(K)$ which would lead naturally to the continuity of the tangential trace is more complex than for $H(\mathrm{div};\Omega)$. To understand this, we may consider the trace π_t defined in (2.1.32). From (2.1.37), this trace is in $H^{-\frac{1}{2}}(\underline{\mathrm{curl}};\partial K)$ which will imply continuity along the edges of K. This will also imply in the definition of degrees of freedom the appearance of the duality pairing with $H^{-\frac{1}{2}}(\mathrm{div};\partial K)$. Finite element spaces approximating $H(\underline{\mathrm{curl}};\Omega)$ have therefore often been referred to as *edge* finite elements, in analogy to *face* finite elements approximating $H(\mathrm{div};\Omega)$ and to *nodal* finite element approximations of $H^1(\Omega)$ as degrees of freedom will be associated to edges (see, in particular, Example 2.3.3), faces, and vertices, respectively.

Edge elements are also known as Nédélec elements since they have been originally presented in [310, 311]. Important contributions to the analysis can be found in [8, 102, 170, 223], in [248, 302] and in the references therein. In the simplicial case, a comprehensive list can be deduced from the presentation of [33] (see, in particular, Table 5.2) where they have been discussed in a more general setting.

Remark 2.3.3. As we did for approximations of $H(\mathrm{div},\Omega)$, we shall define, for $\underline{S}_k \subset \underline{P}_k$,

$$\mathcal{L}^{\mathrm{curl}}(\underline{S}_k,\mathcal{T}_h) := \{\underline{p} \mid \underline{p} \in H(\underline{\mathrm{curl}};\Omega), \quad \underline{p}|_K \in \underline{S}_k(K)\}, \tag{2.3.36}$$

which will imply continuity of the tangential traces as defined in the previous sections. We now proceed to present classical ways of building some suitable $\underline{S}_k$.

□

The most popular choice, which has been introduced by Nédélec [310] and further analysed in [222] is often referred to as first kind Nédélec family.

Let $K \subset \mathbb{R}^3$ be a tetrahedron and let e_i, $i = 1,\dots,6$, denote its edges and f_i, $i = 1,\dots,4$, its faces. Given an integer $k \geq 0$, we define

$$\mathcal{N}_k(K) := (P_k(K))^3 \oplus [\underline{x} \wedge (P_k^h(K))^3], \tag{2.3.37}$$

where $\underline{x} = (x,y,z)$ and P_k^h denotes the space of homogeneous polynomials of degree k.

It is worth observing that definition (2.3.37) might also be given in the following equivalent way. Let us define first

$$\mathcal{S}^k := \{\underline{\chi} \in (P_k(K))^3 | \underline{\chi} \cdot \underline{x} = 0\}; \tag{2.3.38}$$

then we have $\mathcal{N}_k(K) := (P_k(K))^3 \oplus \mathcal{S}^{k+1}$.

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Proposition 2.3.5. *For $k \geq 0$ and for any $\underline{\chi} \in \mathcal{N}_k(K)$, the degrees of freedom of $\mathcal{N}_k(K)$ can be defined by*

$$\int_{e_i} \underline{\chi} \cdot \underline{t}\ p_k\, ds \qquad \forall p_k \in P_k(e_i),\ \forall e_i \tag{2.3.39}$$

$$\int_{f_j} \pi_t \underline{\chi} \cdot \underline{\phi}_{k-1}\, d\sigma \qquad \forall \underline{\phi}_{k-1} \in (P_{k-1}(f_j))^2,\ \forall f_j \tag{2.3.40}$$

$$\int_K \underline{\chi} \cdot \underline{p}_{k-2}\, dx \qquad \forall \underline{p}_{k-2} \in (P_{k-2}(K))^3. \tag{2.3.41}$$

For the proof of the previous proposition, we refer the interested reader to [222]. In particular, let us check that the number of conditions involved with (2.3.39)–(2.3.41) is the same as the dimension of the space $\mathcal{N}_k(K)$. Indeed, (2.3.39) imposes $k+1$ conditions on each edge, (2.3.40) gives $k(k+1)/2$ conditions in each of the two components of the tangential component on each face, and (2.3.41) adds $(k-1)k(k+1)/6$ more conditions in each of the three components of the vector field. As it can be easily seen, the sum of the three contributions is equal to

$$\dim(\mathcal{N}_k(K)) = (k+1)(k+3)(k+4)/2. \tag{2.3.42}$$

Following (2.3.36), in analogy with what has been done for approximations of $H(\mathrm{div};\Omega)$, we define

$$\mathcal{L}^{\mathrm{curl}}_{\underline{k}+\underline{x}\wedge\underline{k}}(\mathcal{T}_h) := \mathcal{L}^{\mathrm{curl}}(\mathcal{N}_k, \mathcal{T}_h). \tag{2.3.43}$$

Remark 2.3.4. It would be possible to introduce, if needed, reduced spaces where the tangential trace is of a lower degree. We can, for example, introduce a reduced space, $\mathcal{N}^r_k(K)$, which is the subspace of $\mathcal{N}_k(K)$ where the degree of the trace on the faces of K is lowered by one. We thus have for $k \geq 1$ the following degrees of freedom

$$\int_{e_i} \underline{\chi} \cdot \underline{t}\ p_{k-1}\, ds \qquad \forall p_{k-1} \in P_{k-1}(e_i),\ \forall e_i \tag{2.3.44}$$

$$\int_{f_j} \pi_t \underline{\chi} \cdot \underline{\phi}_{k-2}\, d\sigma \qquad \forall \underline{\phi}_{k-2} \in (P_{k-2}(f_j))^2,\ \forall f_j \tag{2.3.45}$$

$$\int_K \underline{\chi} \cdot \underline{p}_{k-2}\, dx \qquad \forall \underline{p}_{k-2} \in (P_{k-2}(K))^3. \tag{2.3.46}$$

□

Proposition 2.1.3 states that in order to construct an approximation of $H(\underline{\mathrm{curl}};\Omega)$, given $\underline{\chi} \in \mathcal{N}_k(k)$, we need to ensure continuity of the tangential

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component $\gamma_t \underline{\chi}$ across elements. The following proposition implies that the degrees of freedom (2.3.39) and (2.3.40) guarantee such continuity.

Proposition 2.3.6. *Assume that* $\underline{\chi} \in \mathcal{N}_k(K)$ *is such that* (2.3.39) *and* (2.3.40) *vanish on a given face* $f \subset K$ *and on all edges* e *of* f. *Then* $\underline{\chi} \wedge \underline{n} = 0$ *on* f. □

Another possible choice for constructing approximations of $H(\underline{\mathrm{curl}}; \Omega)$ has been introduced in [311] and is often referred to as the second kind Nédélec family. On a tetrahedron K, for $k \geq 1$, we consider the full polynomial space

$$\mathcal{NC}_k(K) := \underline{P}_k. \tag{2.3.47}$$

This is the same set of polynomials which we used to define $\mathcal{BDM}_k$. However, we must now build a set of degrees of freedom which would ensure the continuity of the tangential trace. Given $\underline{\chi} \in \mathcal{NC}_k(K)$, we introduce the following moments:

$$\int_{e_i} \underline{\chi} \cdot \underline{t}\, p_k \, ds, \qquad \forall\, p_k \in P_k(e_i),\ \forall e_i \tag{2.3.48}$$

$$\int_{f_j} \pi_t \underline{\chi} \cdot \underline{\phi}_{k-2} \, d\sigma, \qquad \forall \underline{\phi}_{k-2} \in \mathcal{RT}_{k-2}(f_j),\ \forall f_j \tag{2.3.49}$$

$$\int_K \underline{\chi} \cdot \underline{q}_{k-3} \, dx, \qquad \forall \underline{q}_{k-3} \in \mathcal{RT}_{k-3}(K). \tag{2.3.50}$$

It is not difficult to check that the total number of degrees of freedom introduced in (2.3.48)–(2.3.50) is equal to

$$\dim(\mathcal{NC}_k(K)) = \frac{(k+1)(k+2)(k+3)}{2}. \tag{2.3.51}$$

Indeed, moments in (2.3.48)–(2.3.50) correspond respectively to $6(k+1)$, $4(k+1)(k-1)$, and $(k+1)(k-1)(k-2)/2$ conditions. Lemma 2.1.9 guarantees that the moments are compatible with the mapping $\mathcal{H}$ and, finally, we refer the interested reader to [311] for the proof of the unisolvence.

We shall denote as in (2.3.43)

$$\mathcal{L}^{\mathrm{curl}}_{\underline{k}}(\mathcal{T}_h) := \mathcal{L}^{\mathrm{curl}}(\mathcal{NC}_k, \mathcal{T}_h). \tag{2.3.52}$$

Remark 2.3.5. We shall say that the vectors in $\mathcal{N}_k(K), \mathcal{N}^r_k(K)$ or $\mathcal{NC}_k(K)$ for which $\underline{\chi} \wedge \underline{n} = 0$ on all faces of K and which are thus defined by the degrees of freedom (2.3.41), (2.3.46), and (2.3.50) are $H(\mathrm{curl})$-bubbles. It is thus clear that $\mathcal{N}_k(K)$ and $\mathcal{N}^r_k(K)$ contain exactly the same bubbles while $\mathcal{NC}_k(K)$ has more. □

Remark 2.3.6. Another important property is that the spaces $\mathcal{N}_k(K)$, $\mathcal{N}^r_k(K)$ and $\mathcal{NC}_k(K)$ are invariant under the action of the covariant transform $\mathcal{H}$ in case of

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affine mappings (as it can be checked directly). Moreover, from Lemma 2.1.9 if follows that the degrees of freedom introduced, for example, in (2.3.39)–(2.3.41) are compatible with respect to $\mathcal{H}$. □

Our last comment is about the characterization of the kernel of the curl operator in $\mathcal{N}_k(K), \mathcal{N}_k^r(K)$ and $\mathcal{NC}_k(K)$. Using a similar notation to what we had introduced in the previous section, we define

$$\mathcal{N}_k^0(K) := \{\underline{\chi} \in \mathcal{N}_k(K) |\, \underline{\mathrm{curl}}\, \underline{\chi} = 0\}, \tag{2.3.53}$$

$$\mathcal{N}_k^{r0}(K) := \{\underline{\chi} \in \mathcal{N}_k^r(K) |\, \underline{\mathrm{curl}}\, \underline{\chi} = 0\}, \tag{2.3.54}$$

$$\mathcal{NC}_k^0(K) := \{\underline{\chi} \in \mathcal{NC}_k(K) |\, \underline{\mathrm{curl}}\, \underline{\chi} = 0\}. \tag{2.3.55}$$

One easily sees that

$$\mathcal{N}_k^0(K) := \underline{\mathrm{grad}}\, P_{k+1}(K), \tag{2.3.56}$$

$$\mathcal{N}_k^{r0}(K) := \underline{\mathrm{grad}}\, P_k(K) + \underline{\mathrm{grad}}\, B_4(K) P_{k-3}^h(K), \tag{2.3.57}$$

$$\mathcal{NC}_k^0(K) := \underline{\mathrm{grad}}\, P_{k+1}(K), \tag{2.3.58}$$

where $B_4(K)$ is the bubble defined in (2.2.28) and $P_k^h(K)$ denotes, as above, the space of homogeneous polynomials of degree k. Let us define

$$\mathcal{N}_k^+(K) := \mathcal{N}_k(K)/\, \underline{\mathrm{grad}}\, P_{k+1}(K), \tag{2.3.59}$$

$$\mathcal{N}_k^{r+}(K) := \mathcal{N}_k^r/\mathcal{N}_k^{r0}(K), \tag{2.3.60}$$

$$\mathcal{NC}_k^+(K) := \mathcal{NC}_k/\, \underline{\mathrm{grad}}\, P_{k+1}(K). \tag{2.3.61}$$

We then have $\mathcal{N}_k^+(K) = \mathcal{NC}_{k+1}^+(K)$ while $\mathcal{N}_k^{r+}(K)$ is smaller. The important point is that we have

$$\underline{\mathrm{curl}}\, \mathcal{N}_k(K) = \underline{\mathrm{curl}}\, \mathcal{NC}_{k+1}(K) = \underline{\mathrm{curl}}\, \mathcal{N}_k^+(K) = \mathcal{BDM}_k^0(K) = \mathcal{RT}_K^0(K) \tag{2.3.62}$$

$$\underline{\mathrm{curl}}\, \mathcal{N}_k^r(K) = \underline{\mathrm{curl}}\, \mathcal{N}_k^{r+}(K) = \mathcal{BDFM}_k^0(K). \tag{2.3.63}$$

We can, for example, check the following result.

Lemma 2.3.4. *The* $\underline{\mathrm{curl}}$ *operator is surjective from* $\mathcal{N}_k^+(K)$ *onto* $\mathcal{BDM}_k^0(K) = \mathcal{RT}_k^0(K)$.

Proof. A simple count shows that the dimensions of $\mathcal{N}_k^+(K)$ and the dimension of $\mathcal{BDM}_k^0(K)$ are equal. □

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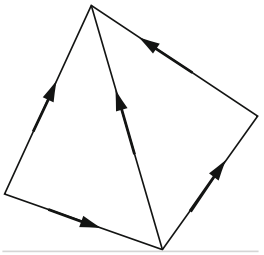


Fig. 2.15 Lowest order edge elements on a tetrahedron

This can be summarised in the exact sequences,

$$\mathcal{L}^1_{k+1} \xrightarrow{\underline{\text{grad}}} \mathcal{N}_k \xrightarrow{\underline{\text{curl}}} \mathcal{BDM}_k \xrightarrow{\text{div}} \mathcal{L}^0_k \tag{2.3.64}$$

$$\mathcal{L}^1_{k+1} \xrightarrow{\underline{\text{grad}}} \mathcal{N}^r_k \xrightarrow{\underline{\text{curl}}} \mathcal{BDFM}_k \xrightarrow{\text{div}} \mathcal{L}^0_k \tag{2.3.65}$$

This relation is part of a more general *commuting diagram property* which will be discussed in Sect. 2.1.4.

Example 2.3.3 (Lowest order edge elements). We conclude this section by explaining in more detail the case $k = 0$. This is probably the most used edge finite element and it is also known as the Whitney element, since it has been used by Whitney in a different context [379]. The case $k = 0$ is very particular, since the only meaningful degrees of freedom are those presented in (2.3.39) (and this is also a good explanation for the name *edge* elements). The space $\mathcal{N}_0$ is simply given by

$$\begin{aligned}\mathcal{N}_0 :&= (P_0(K))^3 \oplus [(\underline{x}) \wedge (P_0(K))^3] \\ &= \text{span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}, \begin{pmatrix}0\\z\\-y\end{pmatrix}, \begin{pmatrix}-z\\0\\x\end{pmatrix}, \begin{pmatrix}y\\-x\\0\end{pmatrix}\right\}.\end{aligned} \tag{2.3.66}$$

The moments (shown in Fig. 2.15), are given by the six degrees of freedom (2.3.39), that is

$$\int_e \underline{\chi} \cdot \underline{t}\, ds \tag{2.3.67}$$

and it can be checked that the quantity $\underline{\chi} \cdot \underline{t}$ is constant along the edges. □

2.4 Approximations of $H(\text{div}; K)$ on Rectangles and Cubes

We now consider the extension of the previous construction to rectangular elements. The general quadrilateral case is not straightforward; some considerations about it are presented in Sect. 2.2.4. In the present case, the use of a reference element

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is essential and *we shall build our spaces on* $\hat{K} =]-1, +1[^n$. Contrarily to the simplicial case, it will be simpler here to first introduce the approximations of Raviart and Thomas. The extension to the three-dimensional case can be found in [310].

2.4.1 Raviart-Thomas Elements on Rectangles and Cubes

We first consider a simple extension of the $\mathcal{RT}_k$ approximation introduced above for the simplicial case. Let us define

$$\mathcal{RT}_{[k]} := \begin{cases} P_{k+1,k} \times P_{k,k+1} & \text{for } n = 2, \\ P_{k+1,k,k} \times P_{k,k+1,k} \times P_{k,k,k+1} & \text{for } n = 3. \end{cases} \tag{2.4.1}$$

It is then easy to check that

$$\dim \mathcal{RT}_{[k]} = \begin{cases} 2(k+1)(k+2) & \text{for } n = 2, \\ 3(k+1)^2(k+2) & \text{for } n = 3. \end{cases} \tag{2.4.2}$$

These spaces have been defined in order to have

$$\mathrm{div}\, \underline{q}_k \in Q_k. \tag{2.4.3}$$

Moreover, we have

$$\begin{cases} \underline{q} \cdot \underline{n}_k|_{e_i} \in P_k(e_i) \text{ on the edges for } n = 2, \\ \underline{q} \cdot \underline{n}_k|_{f_i} \in Q_k(e_i) \text{ on the faces for } n = 3. \end{cases} \tag{2.4.4}$$

Defining as in the simplicial case

$$RT^0_{[k]} := \{\underline{q} \mid \underline{q} \in \mathcal{RT}_{[k]},\ \mathrm{div}\, \underline{q} = 0\}, \tag{2.4.5}$$

we have the following result.

Lemma 2.4.1. *For $n = 2$, if $\underline{q} \in RT^0_{[k]}(\hat{K})$, there exists $\psi \in Q_{k+1}(\hat{K})$ such that $\underline{q} = \underline{curl}\, \psi$. The dimension of $RT^0_{[k]}(\hat{K})$ is $(k+1)(k+3)$.*

In order to choose an approximate set of degrees of freedom, we define

$$\Psi_k(K) := \begin{cases} P_{k-1,k}(K) \times P_{k,k-1}(K) & \text{for } n = 2, \\ P_{k-1,k,k}(K) \times P_{k,k-1,k}(K) \times P_{k,k,k-1}(K) & \text{for } n = 3. \end{cases} \tag{2.4.6}$$

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Proposition 2.4.1. *For any* $\underline{q} \in \mathcal{RT}_{[k]}(\hat{K})$*, the relations*

$$\begin{aligned} &\int_{e_i} \phi_i \underline{q} \cdot \underline{n}\, ds = 0 \quad && \forall \phi_i \in Q_k(e_i) \text{ for } n = 3 \\ & && \forall \phi_i \in P_k(e_i) \text{ for } n = 2, \\ &\int_{\hat{K}} \underline{\phi} \cdot \underline{q}\, dx = 0 && \forall \underline{\phi} \in \Psi_k(\hat{K}) \end{aligned} \tag{2.4.7}$$

imply $\underline{q} = 0$.

For $n = 2$ the proof is analogous to the proof of Proposition 2.3.4. For $n = 3$ see [310]. Note that, for $n = 2$, the sides e_i are one-dimensional, so that actually $Q_k(e_i) = P_k(e_i)$ in (2.4.7).

The $\mathcal{RT}_{[k]}$ spaces just described are based on the idea that a finite element approximation on the rectangle should use a space of type Q_k. This is however by no means necessary in the present case.

2.4.2 *Other Approximations of* $H(\mathrm{div}; K)$ *on Rectangles*

In the following, we discuss rectangular finite element approximations of $H(\mathrm{div}; K)$, which are based on P_k polynomial spaces instead of Q_k. The original idea of the construction was introduced in [120] and a suitable modification was presented in [118]. Here we follow such approaches for $n = 2$. For the case when $n = 3$, we use the definitions given in [18], which are more natural and provide spaces which are independent of interchange of coordinate directions.

Let us define following [120] and [118], for $n = 2,\ k \geq 1$,

$$\begin{aligned} \mathcal{BDM}_{[k]} := \{\underline{q} \mid \underline{q} = \underline{p}_k(x, y) + r\, \underline{\mathrm{curl}}(x^{k+1} y) \\ + s\, \underline{\mathrm{curl}}(x y^{k+1}),\ \underline{p}_k \in (P_k)^2\}. \end{aligned} \tag{2.4.8}$$

These spaces have been carefully defined in order to have

$$\begin{cases} \mathrm{div}\, \underline{q} \in P_{k-1}(K), \\ \underline{q} \cdot \underline{n}|_{e_i} \in P_k(e_i). \end{cases} \tag{2.4.9}$$

It must be remarked that these last conditions are rather unusual for a rectangular approximation. We have by a simple count,

$$\dim \mathcal{BDM}_{[k]} = (k+1)(k+2) + 2 = k^2 + 3k + 4 \qquad (n = 2). \tag{2.4.10}$$

For the choice of degrees of freedom, we have the following proposition.

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Proposition 2.4.2. *For $k \geq 1$, the following conditions imply $\underline{q} = 0$,*

$$\int_{e_i} \underline{q} \cdot \underline{n}\ p_k\ ds,\ \forall p_k \in P_k(e_i), \tag{2.4.11}$$

$$\int_{\hat{K}} \underline{q} \cdot \underline{p}_{k-2}\ dx = 0,\ \forall \underline{p}_{k-2} \in (P_{k-2})^n. \tag{2.4.12}$$

Proof. It is sufficient to prove that (2.4.11) implies $\underline{q} \in (P_k)^2$, that is, all terms introduced through $\underline{\mathrm{curl}}$ vanish. Then, we have that if $\underline{q} \in (P_k)^2$, then $\underline{q}_0 \cdot \underline{n}|_{e_i} = 0$ implies $q_1 = (1 - x^2)p_{1,k-2}$ and $q_2 = (1 - y^2)p_{2,k-2}$, so that (2.4.12) implies $\underline{q} = 0$. Indeed, from (2.4.8) we have

$$\begin{aligned} q_1 &= p_{1,k}(x, y) - r x^{k+1} + s(k + 1)x y^k \\ q_2 &= p_{2,k}(x, y) + r(k + 1)x^k y - s y^{k+1}. \end{aligned} \tag{2.4.13}$$

In order to have $q_1 = 0$ for $x = \pm 1$ and $q_2 = 0$ for $y = \pm 1$, we have that r and s must vanish, so that $\underline{q} \in (P_k)^2$. □

Remark 2.4.1. Definition (2.4.8) has been designed in order to keep $\mathrm{div}\ \underline{q}$ in P_{k-1} by adding divergence-free functions to $(P_k)^n$ while providing terms with a normal component in $P_k(e_i)$ on each side or face e_i. □

We would now like to see what are the relations between $\mathcal{BDM}_{[k]}(K)$ and $\mathcal{RT}_{[k]}(K)$. First, one obviously has $\mathcal{BDM}_{[k]} \subset \mathcal{RT}_{[k]}$. However, the space obtained by restricting the normal component of $\mathcal{BDM}_{[k]}$ to belong to $P_{k-1}(e_i)$ on each side has no direct relation to $\mathcal{RT}_{[k-1]}$ and is a much smaller space (providing an approximation of the same accuracy). In order to get a pattern of inclusions, we define the space

$$S_{[k+1]} := \mathcal{RT}_{[k]} + \{\underline{\mathrm{curl}}\, x^{k+2} y,\ \underline{\mathrm{curl}}\, y x^{k+2},\ \underline{\mathrm{curl}}\, x^{k+2},\ \underline{\mathrm{curl}}\, y^{k+2}\}. \tag{2.4.14}$$

This space obviously contains $\mathcal{RT}_{[k]}$ but also contains $\mathcal{BDM}_{[k+1]}$.

We can also define the space $\mathcal{BDFM}_{[k+1]}$ by restricting the normal component of $\underline{q} \in \mathcal{BDM}_{[k+1]}$ to belong to $P_k(e_i)$ instead of $P_{k+1}(e_i)$ on each side [119].

It can easily be checked that $\mathcal{BDFM}_{[1]} = \mathcal{RT}_{[0]}$. To make things clear, let us consider a diagram in Fig. 2.16.

We can then summarize the previous facts in Fig. 2.17 in which arrows indexed by b represent a reduction in boundary degrees of freedom and arrows indexed by i represent a reduction in internal degrees of freedom. Space $\mathcal{RT}_{[0]}$ plays a special role in this set of spaces. It is the simplest possible space and it is related to the MAC space [240] that has been extensively used in fluid mechanical computations. It is clear from Fig. 2.17 that both $\mathcal{RT}_{[k]}$ and $\mathcal{BDFM}_{[k+1]}$ are a generalization of

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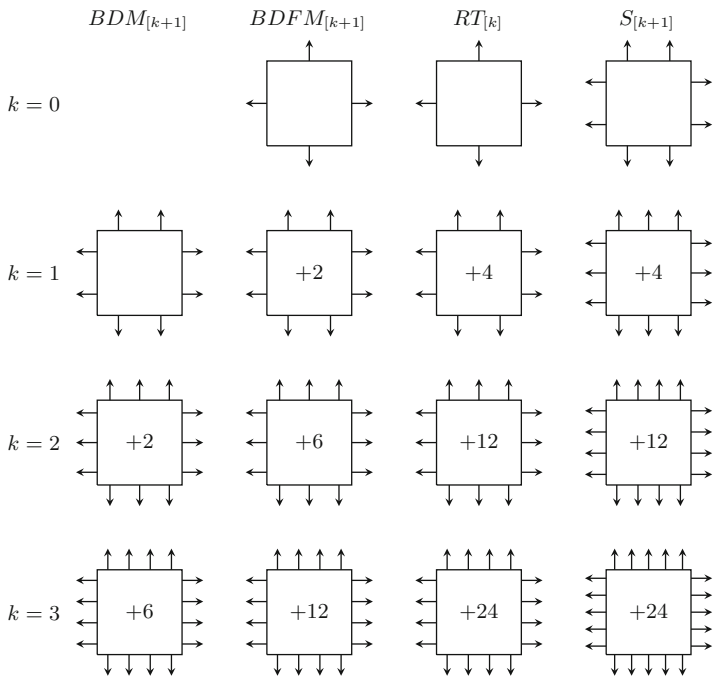


Fig. 2.16 Two dimensional approximations of $H(\mathrm{div}; K)$

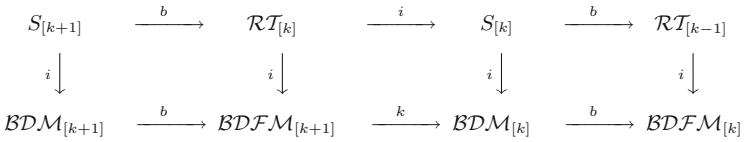


Fig. 2.17 Relations between elements approximating $H(\mathrm{div}; K)$: operators b represent reduction in boundary degrees of freedom and i reduction in internal degrees of freedom

this space with the same order of accuracy. One uses $\mathcal{RT}_{[k]}$ whenever one wants $\mathrm{div}\, \underline{q} \in Q_k$ and $\mathcal{BDFM}_{[k+1]}$ if $\mathrm{div}\, \underline{q} \in P_k$ is sufficient. It is thus worth considering $\mathcal{BDFM}_{[k+1]}$ in more details. It is easy to check that

$$\mathcal{BDFM}_{[k+1]} = (P_{k+1})^2 \backslash \{0, x^{k+1}\} \backslash \{y^{k+1}, 0\}. \tag{2.4.15}$$

This shows that it is natural to move to $\mathcal{BDM}_{[k+1]}$ and get an extra order of accuracy whenever one is ready to pay for extra boundary nodes.

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2.4.3 *Other Approximations of* $H(\mathbf{div}; K)$ *on cubes*

To make our presentation complete, we now consider the extension of the elements of the previous section to the three-dimensional case. In [18] a general framework for designing finite element spaces on tensor product elements has been introduced. It turns out that this new construction is more natural than the original one. More precisely, we are considering the space denoted $\mathcal{S}_k \Lambda^2$ in this article: it turns out that, while in two-dimensions this space coincides with $\mathcal{BDM}_{[k]}$, in three-dimensions it provides a finite element with the same degrees of freedom as the original $\mathcal{BDM}_{[k]}$ space, but with no arbitrariness in the choice of the shape functions with respect to the order of the variables.

Following [18], we thus define, for $n = 3$, $k \geq 1$,

$$\begin{aligned}\mathcal{BDM}_{[k]} := (P_k(\hat{K}))^3 + \mathrm{span}(\mathrm{curl}\{&yz(w_2(x,z) - w_3(x,y)),\\ &zx(w_3(x,y) - w_1(y,z)),\\ &xy(w_1(y,z) - w_2(x,z))\}),\end{aligned} \tag{2.4.16}$$

where each w_i belongs to P_k. We have

$$\dim \mathcal{BDM}_{[k]}(K) = (k+1)(k^2 + 5k + 12)/2. \tag{2.4.17}$$

The following proposition is the natural extension of Proposition 2.4.2 and has been proved in a more general setting in [18, Theorem 3.6].

Proposition 2.4.3. *For $k \geq 1$, the following conditions imply $\underline{q} = 0$,*

$$\begin{aligned}&\int_{e_i} \underline{q} \cdot \underline{n}\, p_k \, ds, && \forall p_k \in P_k(e_i),\\ &\int_{\hat{K}} \underline{q} \cdot \underline{p}_{k-2} \, dx = 0, && \forall \underline{p}_{k-2} \in (P_{k-2})^n.\end{aligned} \tag{2.4.18}$$

We could also consider the three-dimensional $\mathcal{BDFM}_{[k]}$ case by restricting the degrees of the traces on the boundary. We leave this as an exercise for the reader.

2.4.4 *Approximations of* $H(\underline{\mathrm{curl}}; K)$ *on Cubes*

Surprisingly enough, the construction of edge finite elements on cubes is less studied than the corresponding spaces on tetrahedrons. Namely, only one finite element family was basically known to provide a good approximation of problems involving the space $H(\underline{\mathrm{curl}}; \Omega)$ before the recent paper [18]. We shall describe the associated space, also known as the Nédélec first kind space (see [310]). Given a cube K and an integer $k \geq 0$, we introduce the polynomial space

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$$\mathcal{N}_{[k]} := P_{k,k+1,k+1}(K) \times P_{k+1,k,k+1}(K) \times P_{k+1,k+1,k}(K) \tag{2.4.19}$$

and the following degrees of freedom

$$\int_e \underline{\chi} \cdot \underline{t}\, p_k \, ds, \qquad \forall\, p_k \in P_k(e),\ \forall e \tag{2.4.20}$$

$$\int_f \pi_t \underline{\chi} \cdot \underline{\phi}_{k-1} \, d\sigma, \qquad \forall \underline{\phi}_{k-1} \in \mathcal{RT}_{[k-1]}(f),\ \forall f \tag{2.4.21}$$

$$\int_K \underline{\chi} \cdot \underline{q}_{k-1} \, dx, \qquad \forall \underline{q}_{k-1} \in \mathcal{RT}_{[k-1]}(K). \tag{2.4.22}$$

The number of degrees of freedom introduced in (2.4.20)–(2.4.22) is $12(k+1)$, $12k(k+1)$, and $3k^2(k+1)$, respectively, which sums up to

$$\dim \mathcal{N}_{[k]} = 3(k+1)(k+2)^2. \tag{2.4.23}$$

We refer the interested reader to [310] for the proof of unisolvence.

Remark 2.4.2. Sometimes people refer to second kind Nédélec finite elements on cubes as well. Although such an element has been introduced in [311], it should be noted that it does not seem to be a good choice for the approximation of problems arising from electromagnetism. See, for instance, [163] and [86]. □

A second discretisation of $H(\underline{\mathrm{curl}}; \Omega)$ on cubes comes from the general framework of [18]. More precisely, the space $\mathcal{S}_k(\Lambda^1)$ can be defined as follows:

$$\begin{aligned} \mathcal{S}_k(\Lambda^1)(K) := (P_k(K))^3 + \operatorname{span}(\{&yz(w_2(x,z) - w_3(x,y)),\\ &zx(w_3(x,y) - w_1(y,z)),\\ &xy(w_1(y,z) - w_2(x,z))\}\\ &+ \underline{\mathrm{grad}}\, s(x,y,z)), \end{aligned} \tag{2.4.24}$$

where each w_i belongs to P_k and s is a polynomial on K with *superlinear degree* at most $k+1$. The superlinear degree of a polynomial was defined in [18] as the ordinary degree ignoring variables which appear linearly. We have

$$\dim \mathcal{S}_k(\Lambda^1)(K) = (k+1)(k^2 + 5k + 18)/2 \tag{2.4.25}$$

and a set of degrees of freedom is given by

$$\int_e \underline{\chi} \cdot \underline{t}\, p_k \, ds, \qquad \forall\, p_k \in P_k(e),\ \forall e \tag{2.4.26}$$

$$\int_f \pi_t \underline{\chi} \cdot \underline{\phi}_{k-2} \, d\sigma, \qquad \forall \underline{\phi}_{k-2} \in P_{k-2}(f),\ \forall f \tag{2.4.27}$$

$$\int_K \underline{\chi} \cdot \underline{q}_{k-4} \, dx, \qquad \forall \underline{q}_{k-4} \in P_{[k-4]}(K). \tag{2.4.28}$$

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2.5 Interpolation Operator and Error Estimates

2.5.1 *Approximations of* $H(\mathrm{div}; K)$

Let now $\underline{q}$ be some function of $H(\mathrm{div}; K)$. Using for each of the spaces the degrees of freedom previously described, it is possible to define an interpolation operator $\rho_K \underline{q}$, *provided* $\underline{q}$ *is slightly smoother than merely belonging* to $H(\mathrm{div}; K)$. Indeed the degrees of freedom used always involve the moments of $\underline{q}$ on the faces (or sides) of an element. However, functions $p_k \in R_k(\partial K)$ do not belong to $H^{\frac{1}{2}}(\partial K)$ and it is not possible in general to compute expressions like $\int_{\partial K} \underline{q} \cdot \underline{n}\ p_k\ ds$ since $\underline{q} \cdot \underline{n}$ is only defined in $H^{-\frac{1}{2}}(\partial K)$.

However, it is easy to check that if $\underline{q}$ belongs to the space

$$W(K) := \{\underline{q} \in (L^s(K))^n \mid \mathrm{div}\ \underline{q} \in L^2 \in (\Omega)\} \tag{2.5.1}$$

(for a fixed $s > 2$), then such a construction is possible.

Remark 2.5.1. Readers less familiar with functional analysis will normally wonder why, given a triangle T and a function χ belonging to $H^{-1/2}(\partial T)$, even if we are allowed to take

$$\int_{\partial T} \chi$$

by interpreting it as a duality pairing

$$\int_{\partial T} \chi := \langle \chi, \varphi \rangle \text{ with } \varphi \equiv 1,$$

we cannot take the integral over an edge ℓ of ∂T. The typical answer, at this point, is: "Because the function identically equal to 1 on the whole boundary ∂T belongs to $H^{1/2}(\partial T)$, while the function that is equal to 1 on the edge ℓ and 0 on the rest of ∂T *does not* belong to $H^{1/2}(\partial T)$". The above answer is perfectly correct but, in general, leaves the person who asked the question *totally unhappy*. Let us see, therefore, some example which might help in shedding some more light.

Consider, to start with, the open circle

$$\Omega := \{(x, y) |\ x^2 + y^2 < (1/e)^2\},$$

(where $e = 2,718\ldots$ as usual), and the function

$$u(x, y) := \ln(|\ln(\sqrt{x^2 + y^2})|). \tag{2.5.2}$$

An easy computation (taking the derivatives of u and integrating their square) would show to everybody that $u \in H_0^1(\Omega)$. As a consequence, its restriction to the upper quarter

$$Q := \{(x, y) |\ x > 0,\ y > 0,\ x^2 + y^2 < (1/e)^2\},$$

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will belong to $H^1(Q)$ and its trace on ∂Q will belong to $H^{1/2}(\partial Q)$. So far so good. Now we define χ as *the (anticlockwise) tangential derivative* on ∂Q of the trace of u. This will be a distribution over ∂Q, and it will belong to our puzzling space $H^{-1/2}(\partial Q)$. Now we have (finally!) in our hands an element of $H^{-1/2}(\partial Q)$ that is *irregular enough* to show some of the pathologies of the space. Let us see it.

To start with, the action of the distribution χ on a smooth function φ is easily described by

$$\langle \chi, \varphi \rangle := -\int_{\partial Q} u \frac{\partial \varphi}{\partial t} \tag{2.5.3}$$

where t is the anticlockwise tangent direction on ∂Q. Using the expression of u in (2.5.2) and taking the derivative, we easily see that in every open interval $]a, b[\subset]0, 1/e[$ of the x axis we have

$$\chi(x) = -\frac{1}{x \ln x},$$

while in every open interval $]a, b[\subset]0, 1/e[$ of the y axis we have

$$\chi(y) = \frac{1}{y \ln y}$$

and in every open subset of the curved part of ∂Q we have $\chi = 0$ (being the tangential derivative of the zero function).

The first (and now rather easy) fact that we can observe is that both

$$\int_0^{1/e} \chi(x) \mathrm{d}x$$

and

$$\int_0^{1/e} \chi(y) \mathrm{d}y$$

are diverging, so that neither of them can be properly defined. Hence, so to speak, *forget about taking the integral of an element of* $H^{-1/2}(\partial Q)$ *over* **a piece** *of* ∂Q.

One might still wonder, however, why we can take, instead, the integral *on the whole boundary* of the product of χ times a smooth enough function φ (including the function identically equal to 1). To do so, we parametrise the part of the boundary of Q where u does not vanish by a parameter $\sigma \in]-1/e, 1/e[$ such that

$$x(\sigma) = 0, y(\sigma) = |\sigma| \quad \text{if } \sigma \le 0,$$
$$x(\sigma) = \sigma, y(\sigma) = 0 \quad \text{if } \sigma \ge 0.$$

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Then, taking into account that $u \equiv 0$ on the remaining part of ∂S, (2.5.3) can be written as

$$\langle \psi, \varphi \rangle := \int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi}{\partial \sigma} \mathrm{d}\sigma. \tag{2.5.4}$$

Moreover, integrating by parts, we easily see that in every open interval $]a, b[$ of the x axis included in $]-1/e, 1/e[$ and *not containing* 0 we have

$$\chi(x) = -\frac{1}{\sigma \ln|\sigma|}.$$

Hence (as we can see), it is still forbidden to compute, for instance, the integral

$$\int_0^{1/e} \chi(\sigma)\, \mathrm{d}\sigma = \int_0^{1/e} \frac{1}{\sigma \ln|\sigma|}\, \mathrm{d}\sigma = +\infty.$$

Let us see, however, what happens if we consider the integral

$$\int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi}{\partial \sigma}\, \mathrm{d}\sigma. \tag{2.5.5}$$

As a first step, we introduce the *even* and *odd* parts of φ

$$\varphi_{even} := \frac{\varphi(\sigma) + \varphi(-\sigma)}{2} \qquad \varphi_{odd} := \frac{\varphi(\sigma) - \varphi(-\sigma)}{2}.$$

It is obvious that $\varphi = \varphi_{even} + \varphi_{odd}$, so that now (2.5.5) becomes

$$\int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial(\varphi_{even} + \varphi_{odd})}{\partial \sigma}\, \mathrm{d}\sigma = \int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi_{even}}{\partial \sigma}\, \mathrm{d}\sigma + \int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi_{odd}}{\partial \sigma}\, \mathrm{d}\sigma.$$

Clearly the derivative of φ_{even} is an odd function, and $\ln(|\ln|\sigma||)$ is even. Hence

$$\int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi_{even}}{\partial \sigma}\, \mathrm{d}\sigma = 0,$$

and we only have to deal with

$$\int_{-1/e}^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi_{odd}}{\partial \sigma}\, \mathrm{d}\sigma = 2 \int_0^{1/e} \ln(|\ln|\sigma||) \frac{\partial \varphi_{odd}}{\partial \sigma}\, \mathrm{d}\sigma. \tag{2.5.6}$$

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Now the discussion becomes delicate, as the regularity of φ plays a crucial role. Roughly speaking, if φ is, say, in $H^1(\ ]-1/e, 1/e[\ )$, then $\varphi_{odd}(0) = 0$ and moreover:

$$|\varphi_{odd}(\sigma)| \le C\,|\sigma|^{1/2} \qquad \text{for } \sigma \to 0, \qquad \text{with } C = \|\varphi_{odd}\|_{H^1(\ ]0,1/e[\ )}. \tag{2.5.7}$$

This allows the integration by parts in (2.5.6), yielding

$$-\int_0^{1/e} \frac{1}{\sigma \ln|\sigma|}\,\varphi_{odd}(\sigma)\,\mathrm{d}\sigma,$$

that can be easily seen to be convergent due to (2.5.7). On the other hand, if φ is simply in $H^{1/2}(\ ]-1/e, 1/e[\ )$, then φ_{odd} will "vanish at 0" only in a very weak sense, namely

$$\int_0^{1/e} \sigma^{-1}(\varphi_{odd}(\sigma))^2\,\mathrm{d}\sigma < +\infty.$$

That however will be enough to make the integral in (2.5.6) convergent. □

For the convenience of the reader we consider a few spaces introduced in this section and, for each of them, we shall define the corresponding operator ρ_K that we will always assume to be defined in $W(K)$ (see (2.5.1)).

Example 2.5.1 (Interpolation operator for $\underline{P}_k^{n,s}$). ρ_K: $W(K) \to \underline{P}_k^{n,s}$ is defined by

$$\begin{cases} \displaystyle\int_{\partial K} (\underline{q} - \rho_K \underline{q}) \cdot \underline{n}\ p_k\ ds = 0,\ \forall p_k \in R_k(\partial K), \\ \displaystyle\int_K (\underline{q} - \rho_K \underline{q}) \cdot \underline{w}_{k-2}\ dx = 0,\ \forall w_{k-2} \in \mathcal{N}_{k-2}(K), (k \ge 2). \end{cases} \tag{2.5.8}$$

□

Example 2.5.2 (Interpolation operator for $\mathcal{BDM}_{[k]}(K)$, case $n = 2$, $K =$ unit square). We recall that $\mathcal{BDM}_{[K]}(K) = (P_k(k))^2 \oplus \underline{\mathrm{curl}}(x^{k+1}y) \oplus \underline{\mathrm{curl}}(xy^{k+1})$, $(k \ge 1)$. ρ_K : $W(K) \to \mathcal{BDM}_{[k]}(K)$ is defined by

$$\begin{cases} \displaystyle\int_{\partial K} (\underline{q} - \rho_K \underline{q}) \cdot \underline{n}\ p_k\,d\sigma = 0, \quad \forall p_k \in R_k(\partial K), \\ \displaystyle\int_K (\underline{q} - \rho_K \underline{q}) \cdot \underline{p}_{k-2}\,dx = 0, \quad \forall \underline{p}_{k-2} \in (P_{k-2}(K))^2,\ (k \ge 2). \end{cases} \tag{2.5.9}$$

□

Example 2.5.3 (Interpolation operator for $\mathcal{RT}_k(K) = \underline{P}_{\underline{k}+\underline{x}k}^{n,k}(K)$). $\rho_K : W(K) \to \mathcal{RT}_k(K)$ is defined by

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$$\begin{cases} \displaystyle\int_{\partial K} (\underline{q} - \rho_K \underline{q}) \cdot \underline{n}\ p_k\, d\sigma = 0, \quad \forall p_k \in R_k(\partial K), \\ \displaystyle\int_K (\underline{q} - \rho_K \underline{q}) \cdot \underline{p}_{k-1}\, dx = 0, \quad \forall \underline{p}_{k-1} \in \underline{P}_{k-1}(K). \end{cases} \tag{2.5.10}$$

□

Note that for rectangular elements we used the unit square for K (or the unit cube for $n = 3$). For a general K, the spaces and the interpolation operators ρ_K have to be modified by means of the contravariant mapping $\mathcal{G}$ of (2.1.69). In particular, $\rho_K \underline{q} = \mathcal{G}(\rho_{\hat{K}} \hat{\underline{q}})$ where $\hat{\underline{q}} = \mathcal{G}^{-1}(\underline{q})$ and $\hat{K}$ is the unit square or the unit cube. As we already noted in Sect. 2.2.4, everything works in the case of affine elements while some complications may arise for general elements.

In the following, whenever it may be convenient, we will denote by the symbol $\underline{M}(K)$ anyone of the above approximations of $H(\mathrm{div}; K)$. Since, as we shall see, the accuracy of these approximations in the L^2-norm is particularly relevant, we shall denote by $\underline{M}_k(K)$ anyone of the above spaces such that $\underline{P}_k(K) \subseteq \underline{M}_k(K)$ but $\underline{P}_{k+1}(K) \not\subset \underline{M}_k(K)$. Hence, in the following, $\underline{M}_k(K)$ might be, for example, one the following spaces: $\mathcal{BDM}_k(K)$, $\mathcal{BDM}_{[k]}(K)$, $\mathcal{RT}_k(K)$, $\mathcal{RT}_{[k]}(K)$, $\mathcal{BDFM}_{k+1}(K)$, $\mathcal{BDFM}_{[k+1]}(K)$.

Using Lemmas 2.1.7 and 2.1.8 and usual techniques [146] we have immediately the following result.

Proposition 2.5.1. *Let K be an affine element and ρ_K be the interpolation operator $W(K) \to \underline{M}_k(K)$. There exists a constant c depending only on k and on the shape of K, such that, for $1 \le m \le k+1$, for $s = 0$ or 1 and for any $\underline{q}$ in $(H^m(K))^n$, we have*

$$\|\underline{q} - \rho_K \underline{q}\|_{s,K} \le c h_K^{m-s}\ |\underline{q}|_{m,K}. \tag{2.5.11}$$

□

We now want to analyse the behaviour of the error in $H(\mathrm{div}; K)$. We need to characterize the space of the divergences of the vectors in $\underline{M}_k(K)$. Let

$$D_k(K) := \mathrm{div}(\underline{M}_k(K)). \tag{2.5.12}$$

For affine elements, we have, for example,

$$\mathrm{div}(\mathcal{BDM}_k(K)) = \mathrm{div}(\mathcal{BDM}_{[k]}(K)) \qquad = P_{k-1}(K), \tag{2.5.13}$$

$$\mathrm{div}(\mathcal{BDFM}_{k+1}(K)) = \mathrm{div}(\mathcal{BDFM}_{[k+1]}(K)) = P_k(K), \tag{2.5.14}$$

$$\mathrm{div}(\mathcal{RT}_k(K)) = P_k, \tag{2.5.15}$$

$$\mathrm{div}(\mathcal{RT}_{[k]}(K)) = \mathcal{F}(Q_k(K)), \tag{2.5.16}$$

where the definition of $\mathcal{F}$ is given immediately after (2.1.59) (note that Q_k is not invariant under affine transformations). The following result is of paramount importance in the study of these approximations.

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Proposition 2.5.2. *Let K be an affine element and ρ_K the interpolation operator: $W(K) \to \underline{M}_k(K)$. Let moreover π_K be the L^2-projection on $D_k(K) = \operatorname{div}(\underline{M}_k(K))$. Then we have, for all $\underline{q} \in W(K)$,*

$$\operatorname{div}(\rho_K \underline{q}) = \pi_K \operatorname{div} \underline{q}. \tag{2.5.17}$$

Proof. Since $\operatorname{div}_K \underline{q} \in D_k(K)$ by definition, we only have to prove that

$$\int_K v \operatorname{div}(\rho_K \underline{q})\, dx = \int_K v \operatorname{div} \underline{q}\, dx, \ \forall v \in D_k(K). \tag{2.5.18}$$

Indeed,

$$\begin{aligned}\int_K v(\operatorname{div} \rho_K \underline{q} - \operatorname{div} \underline{q})\, dx = \int_K (\underline{q} - \rho_K \underline{q}) \cdot \underline{\operatorname{grad}}\, v\, dx \\ - \int_K (\underline{q} - \rho_K \underline{q}) \cdot \underline{n}\, v\, dx,\end{aligned} \tag{2.5.19}$$

and it is easy to check that, for all the possible choices of ρ_K, the right-hand side of (2.5.19) vanishes. □

Remark 2.5.2. The statement of Proposition 2.5.2 can also be expressed as

$$\begin{array}{ccc} W(K) & \xrightarrow{\operatorname{div}} & L^2(K) \\ {\scriptstyle \rho_K}\downarrow & & {\scriptstyle \pi_K}\downarrow \\ \underline{M}_k(K) & \xrightarrow{\operatorname{div}} & D_k(K) \end{array} \tag{2.5.20}$$

and is often called the "commuting diagram property" (see [177, 178]). We shall comment more on this property in Sect. 2.5.6. □

From Proposition 2.5.2, using Lemmas 2.1.7 and 2.1.8 and usual techniques, we easily have the following result.

Proposition 2.5.3. *Let K be an affine element and ρ_K the interpolation operator: $W(K) \to \underline{M}_k(K)$. There exists a constant c depending only on k and on the shape of K such that, for $1 \le m \le \phi(\underline{M}_k)$, we have*

$$\| \operatorname{div}(\underline{q} - \rho_K \underline{q}) \|_{0,K} \le c h_K^m \, | \operatorname{div} \underline{q} |_{m,K}, \tag{2.5.21}$$

where $\phi(\underline{M}_k) = k$ for $\mathcal{BDM}_k(K)$ or $\mathcal{BDM}_{[k]}$ and $\phi(\underline{M}_k) = k+1$ for the other choices.

Remark 2.5.3. Proposition 2.5.3 shows that choosing $\mathcal{RT}_{[k]}$, $\mathcal{BDFM}_{k+1}$ or $\mathcal{BDFM}_{[k+1]}$ leads to the same accuracy in $H(\operatorname{div}; K)$ as we have in $(L^2(K))^n$. This is not the case for $\mathcal{BDM}_k$ or $\mathcal{BDM}_{[k]}$ where the accuracy in $H(\operatorname{div}; K)$

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is of one order less than the accuracy in $(L^2(K))^n$. However, as we shall see in Chap. 7, the commuting diagram property is so strong that this drawback can be circumvented. □

Remark 2.5.4. For non-affine elements, the situation is more complicated. In particular, we have now to define $D_k(K)$ and $\mathcal{F}(D_k(\hat{K}))$, where $\hat{K}$ is the reference element and $\mathcal{F}$ is defined in (2.1.59). On the other hand, $\operatorname{div}(\underline{M}_k(K))$ will be $\mathcal{F}(J^{-1} \operatorname{div} \underline{M}_k(\hat{K}))$. Hence it is clear that Proposition 2.5.2 will not hold any more. Moreover, Proposition 2.5.3 does not hold (at least for $\mathcal{RT}_{[k]}$-elements; see again [366]). More comments on these issues will be given in Sect. 2.5.5. □

2.5.2 *Approximation Spaces for* $H(\mathbf{div}; \Omega)$

It is clear that the spaces defined in the previous sections can be used to define internal approximations of $H(\operatorname{div}; \Omega)$. The choice of degrees of freedom has obviously been made in order to ensure continuity of $\underline{q} \cdot \underline{n}$ at interfaces of elements. We can then define, for each choice of $\underline{M}_k(K)$, the space

$$\underline{M}_k(\Omega, \mathcal{T}_h) := \{\underline{q} \mid \underline{q} \in H(\operatorname{div}; \Omega),\ \underline{q}|_K \in \underline{M}_k(K)\}. \tag{2.5.22}$$

In a similar manner we have, in agreement with the notation (2.2.23),

$$\mathcal{L}^0(D_k, \mathcal{T}_h) := \{v \mid v \in L^2(\Omega),\ v|_K \in D_k(K)\}. \tag{2.5.23}$$

It is clear that for affine elements

$$\operatorname{div} \underline{M}_k(\Omega, \mathcal{T}_h) \subset \mathcal{L}^0(D_k, \mathcal{T}_h). \tag{2.5.24}$$

Moreover, we can now define a *global* interpolation operator from

$$W := H(\operatorname{div}; \Omega) \cap L^s(\Omega)^n \tag{2.5.25}$$

(for a fixed $s > 2$) into $\underline{M}_k(\Omega; \mathcal{T}_h)$ by simply setting

$$\Pi_h \underline{q}|_K = \rho_K(\underline{q}|_K). \tag{2.5.26}$$

By defining $P_h :=$ projection on $\mathcal{L}^0(D_k, \mathcal{T}_h)$ we have therefore the following commuting diagram:

$$\begin{array}{ccc} W & \xrightarrow{\operatorname{div}} & L^2(\Omega) \\ {\scriptstyle \Pi_h}\downarrow & & {\scriptstyle P_h}\downarrow \\ \underline{M}_k(\Omega, \mathcal{T}_h) & \xrightarrow{\operatorname{div}} & \mathcal{L}^0(D_k, \mathcal{T}_h) \end{array} \tag{2.5.27}$$

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This will imply in particular that

$$\operatorname{div} \underline{M}_k(\Omega; \mathcal{T}_h) = \mathcal{L}^0(D_k, \mathcal{T}_h). \tag{2.5.28}$$

Finally we have from Propositions 2.5.1 and 2.5.3 the following estimates for the interpolation operator Π_h.

Proposition 2.5.4. *Let $\mathcal{T}_h$ be a regular family of decompositions of Ω and let Π_h be defined as in (2.5.26). Then there exists a constant c independent of h such that*

$$\|\underline{q} - \Pi_h \underline{q}\|_{0,\Omega} \leq c h^m \ |q|_{m,\Omega} \tag{2.5.29}$$

for $1 \leq m \leq k+1$. Moreover,

$$\|\operatorname{div}(\underline{q} - \Pi_h \underline{q})\|_{0,\Omega} \leq c h^s \ |\operatorname{div} \underline{q}|_{s,\Omega}, \tag{2.5.30}$$

where $s \leq k$ for $\mathcal{BDM}_k$ or $\mathcal{BDM}_{[k]}$ and $s \leq k+1$ for the other choices of $\underline{M}_k$.

□

2.5.3 Approximations of $H(\underline{\mathrm{curl}}; \Omega)$

Now, we show how to use the definitions given in the previous subsection in order to construct conforming approximations of $H(\underline{\mathrm{curl}}; \Omega)$. First of all, we need to define an interpolation operator. We start with the description of the first case we discussed, namely the tetrahedral space $\mathcal{N}_k$.

We follow the theory developed in [8]; for the error estimate we refer to [248] and to the improved modification proposed in [85].

The main question for the definition of the interpolant concerns the regularity assumptions on the function to be interpolated. We have already seen that $H^1(\Omega)$ regularity does not allow for the existence of a *nodal* interpolant (since point-wise values are not defined in $H^1(\Omega)$) and that $H(\operatorname{div}; \Omega)$ regularity does not guarantee the existence of a *face* interpolant (essentially because it is not possible to evaluate the integral of $\underline{q} \cdot \underline{n}$ on a single face of K if $\underline{q} \cdot \underline{n}$ belongs only to $H^{-1/2}(\partial K)$). The case of the *edge* interpolant is more tricky than the previous ones since, according to the theory developed in the previous subsection, we have two families of degrees of freedom associated with the boundary of K: degrees of freedom defined in (2.3.39) (corresponding to the edges of K) and the ones defined in (2.3.40) (corresponding to the faces of K). In [8] it has been proved that if $\underline{\chi}$ belongs to the following space

$$X(K) := \{\underline{\chi} \mid \underline{\chi} \in (L^s(K))^3, \underline{\mathrm{curl}}\, \underline{\chi} \in (L^s(K))^3, \ (\underline{\chi} \wedge \underline{n})_{|\partial K} \in (L^s(K))^2\} \tag{2.5.31}$$

(for a fixed $s > 2$), then the moments defined in Proposition 2.3.5 make sense.

We are then in the position of defining the interpolation operator for the edge element spaces introduced in the previous subsection.

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Case 2.5.1 (Case $n = 2$). As it has been explained at the beginning of Sect. 2.5.3 (see also Remark 2.1.5), two dimensional approximations of $H(\mathrm{curl};\Omega)$ can be obtained from corresponding approximations of $H(\mathrm{div};\Omega)$ through a rotation by a right angle.

Case 2.5.2 (Case $n = 3$, tetrahedral elements).

(i) $\mathcal{N}_k(K) := (P_k(K))^3 \oplus [\underline{x} \wedge (\tilde{P}_k(K))^3]$.
$\sigma_K : X(K) \to \mathcal{N}_k(K)$ is defined by

$$\begin{cases} \displaystyle\int_e (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{t}\, p_k \, ds = 0 & \forall\, p_k \in P_k(e),\ \forall e \\ \displaystyle\int_f (\pi_t \underline{\chi} - \pi_t \sigma_K \underline{\chi}) \cdot \underline{\phi}_{k-1} \, d\sigma = 0 & \forall \underline{\phi}_{k-1} \in (P_{k-1}(f))^2,\ \forall f \\ \displaystyle\int_K (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{p}_{k-2} \, dx = 0 & \forall \underline{p}_{k-2} \in (P_{k-2}(K))^3. \end{cases} \tag{2.5.32}$$

(ii) $\mathcal{NC}_k(K) := (P_k(K))^3$.
$\sigma_K : X(K) \to \mathcal{NC}_k(K)$ is defined by

$$\begin{cases} \displaystyle\int_e (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{t}\, p_k \, ds = 0 & \forall\, p_k \in P_k(e),\ \forall e \\ \displaystyle\int_f (\pi_t \underline{\chi} - \pi_t \sigma_K \underline{\chi}) \cdot \underline{\phi}_{k-1} \, d\sigma = 0 & \forall \underline{\phi}_{k-1} \in \mathcal{RT}_{k-1}(f),\ \forall f \\ \displaystyle\int_K (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{p}_{k-2} \, dx = 0 & \forall \underline{p}_{k-2} \in \mathcal{RT}_{k-2}(K). \end{cases} \tag{2.5.33}$$

Case 2.5.3 (Case $n = 3$, cubic elements).

(i) $\mathcal{N}_{[k]}(K) := P_{k,k+1,k+1}(K) \times P_{k+1,k,k+1}(K) \times P_{k+1,k+1,k}(K)$.
$\sigma_K : X(K) \to \mathcal{N}_{[k]}(K)$ is defined by

$$\begin{cases} \displaystyle\int_e (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{t}\, p_k \, ds, & \forall\, p_k \in P_k(e),\ \forall e \\ \displaystyle\int_f (\pi_t \underline{\chi} - \pi_t \sigma_K \underline{\chi}) \cdot \underline{\phi}_{k-1} \, d\sigma, & \forall \underline{\phi}_{k-1} \in \mathcal{RT}_{[k-1]}(f),\ \forall f \\ \displaystyle\int_K (\underline{\chi} - \sigma_K \underline{\chi}) \cdot \underline{q}_{k-1} \, dx, & \forall \underline{q}_{k-1} \in \mathcal{RT}_{[k-1]}(K). \end{cases} \tag{2.5.34}$$

We now conclude this section with error estimates, in analogy to what has been done for $H(\mathrm{div};\Omega)$ at the end of Sect. 2.5. Let us denote by $N(K)$ any of the elements presented so far. More precisely, let $N_k(K)$ denote an approximation of order k of $H(\underline{\mathrm{curl}};\Omega)$ (i.e., $\mathcal{N}_k(K)$, $\mathcal{NC}_k(K)$, or $\mathcal{N}_{[k]}(K)$).

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An important difference between the error estimates for edge elements with respect to the ones presented in Sect. 2.5 for face elements is that, in general, stronger regularity assumptions are required. The following result has been proved in [85].

Proposition 2.5.5. *Let K be an affine element and σ_K be the interpolation operator $X(K) \to N_k(K)$. Then there exists a constant c depending only on k and on the shape of K, such that, for $1 < m \leq k+1$, for any $\underline{\chi} \in (H^m(K))^3$, we have*

$$\|\underline{\chi} - \sigma_K \underline{\chi}\|_{0,K} \leq c h_K^m |\underline{\chi}|_{m,K}. \tag{2.5.35}$$

Moreover, for $1/2 < s \leq 1$, we have ($p > 2$)

$$\|\underline{\chi} - \sigma_K \underline{\chi}\|_{0,K} \leq c h_K^s (|\underline{\chi}|_{s,K} + \|\underline{\mathrm{curl}}\, \underline{\chi}\|_{L^p(K)}). \tag{2.5.36}$$

□

Let us now characterize the space

$$E_k(K) := \underline{\mathrm{curl}}(N_k(K)). \tag{2.5.37}$$

Taking into account (2.1.92), we have

$$\underline{\mathrm{curl}}(\mathcal{N}) \subset P_k(K), \tag{2.5.38}$$

$$\underline{\mathrm{curl}}(\mathcal{NC}) \subset P_{k-1}(K), \tag{2.5.39}$$

$$\underline{\mathrm{curl}}(\mathcal{N}_{[k]}) \subset \mathcal{G}(\mathcal{RT}_{[k]}(K)), \tag{2.5.40}$$

with $\mathcal{G}$ defined in (2.1.69). On the other hand, arguing as in Lemma III.5.11 of [223] it is possible to show that

$$\underline{\mathrm{curl}}(\mathcal{N}) = P_k(K) \cap \underline{\mathrm{curl}}(H(\underline{\mathrm{curl}}; K)), \tag{2.5.41}$$

$$\underline{\mathrm{curl}}(\mathcal{NC}) = P_{k-1}(K) \cap \underline{\mathrm{curl}}(H(\underline{\mathrm{curl}}; K)), \tag{2.5.42}$$

$$\underline{\mathrm{curl}}(\mathcal{N}_{[k]}) = \mathcal{G}(\mathcal{RT}_{[k]}(K) \cap \underline{\mathrm{curl}}(H(\underline{\mathrm{curl}}; K)). \tag{2.5.43}$$

Arguing as in Propositions 2.5.2 and 2.5.3, we can get an estimate for the interpolation error in the $H(\underline{\mathrm{curl}}; \Omega)$ norm. Indeed, the following *commuting diagram* holds true

$$\begin{array}{ccc} X(K) & \xrightarrow{\underline{\mathrm{curl}}} & W(K) \\ \sigma_K \downarrow & & \rho_K \downarrow \\ N_k(K) & \xrightarrow{\underline{\mathrm{curl}}} & E_k(K) \end{array} \tag{2.5.44}$$

from which the next result can be deduced.

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Proposition 2.5.6. *Let K be an affine element and σ_K the interpolation operator: $X(K) \to N_k(K)$. Then there exists a constant c depending only on k and on the shape of K such that*

$$\|\underline{\mathrm{curl}}(\underline{\chi} - \sigma_K \underline{\chi})\|_{0,K} \leq c h_K^m |\underline{\mathrm{curl}}\, \underline{\chi}|_{m,K} \tag{2.5.45}$$

for $1 \leq m \leq \phi_N(k)$, where $\phi_N(k) = k+1$ for $\mathcal{N}$ or $\mathcal{N}_{[k]}$ and $\phi_N(k) = k$ for $\mathcal{NC}$. □

2.5.4 Approximation Spaces for $H(\underline{\mathrm{curl}}; \Omega)$

Let Ω be a domain in $\mathbb{R}^3$ (the two-dimensional case can be dealt with according to Remark 2.1.5).

The spaces introduced in the previous sections can be used to define internal approximations of $H(\underline{\mathrm{curl}}; \Omega)$. The choice of degrees of freedom has been made in such a way that the continuity of the trace γ_t is enforced across the inter-element boundaries, which is the natural condition for being conforming in $H(\underline{\mathrm{curl}}; \Omega)$, according to Proposition 2.1.3.

In analogy to what has been done in Sect. 2.5.2, we can define, for each choice of $N_k(K)$, the spaces

$$N_k(\Omega, \mathcal{T}_h) := \{\underline{\chi} \mid \underline{\chi} \in H(\underline{\mathrm{curl}}; \Omega),\ \underline{\chi}|_K \in N_k(K)\} \tag{2.5.46}$$

and

$$\mathcal{L}^0(E_k, \mathcal{T}_h) := \{\underline{q} \mid \underline{q} \in (L^2(\Omega))^3,\ \underline{q}|_K \in E_k(K)\}. \tag{2.5.47}$$

Given $\varepsilon > 0$, let us consider the space

$$X := H(\underline{\mathrm{curl}}; \Omega) \cap (H^{1/2+\varepsilon}(\Omega))^3, \tag{2.5.48}$$

so that the *global* interpolation operator

$$\Sigma_h \underline{\chi}|_K := \sigma(\underline{\chi}|_K) \tag{2.5.49}$$

can be defined (see (2.5.31)).

Given the space $W_0 := \underline{\mathrm{curl}}(X)$, then we have the following commuting diagram

$$\begin{array}{ccc} X & \xrightarrow{\underline{\mathrm{curl}}} & W_0 \\ {\scriptstyle \Sigma_h}\downarrow & & \downarrow{\scriptstyle \Pi_h} \\ N_k(\Omega, \mathcal{T}_h) & \xrightarrow{\underline{\mathrm{curl}}} & \mathcal{L}^0(E_k, \mathcal{T}_h) \end{array} \tag{2.5.50}$$

where Π_h is the face interpolant defined in (2.5.26).

The final error estimates are summarized in the following result.

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Proposition 2.5.7. *Let $\mathcal{T}_h$ be a regular family of decompositions of Ω and let Σ_h be defined as in (2.5.49). Then there exists a constant c independent of h such that, for $1 < m \leq k+1$,*

$$\|\underline{\chi} - \Sigma_h \underline{\chi}\|_{0,\Omega} \leq c h^m |\underline{\chi}|_{m,\Omega}. \tag{2.5.51}$$

Moreover, for $1/2 < s \leq 1$ and $p > 2$,

$$\|\underline{\chi} - \Sigma_h \underline{\chi}\|_{0,\Omega} \leq c h^s \left(|\underline{\chi}|_{s,\Omega} + \|\underline{\mathrm{curl}}\, \underline{\chi}\|_{L^p(\Omega)} \right). \tag{2.5.52}$$

Finally,

$$\|\underline{\mathrm{curl}}(\underline{\chi} - \Sigma_h \underline{\chi})\|_{0,\Omega} \leq c h^t \|\underline{\mathrm{curl}}\, \underline{\chi}\|_{t,\Omega}, \tag{2.5.53}$$

where $t \leq k+1$ for $\mathcal{N}_k$ or $\mathcal{N}_{[k]}$ and $t \leq k$ for $\mathcal{NC}_k$. □

2.5.5 Quadrilateral and Hexahedral Approximation of Vector-Valued Functions in $H(\mathrm{div};\Omega)$ and $H(\underline{\mathrm{curl}};\Omega)$

The results of Sect. 2.2.4 extend to vector valued functions. We refer the interested reader to [21], where necessary and sufficient conditions have been presented for optimal order approximations of functions in $H(\mathrm{div};\Omega)$ in two dimensions. The general situation is more complicated and only partial results exists so far. The most general result can be found in [19] where sufficient conditions are investigated and sharper results are shown in [191] for particular three-dimensional situations.

The theory of [21] applies to the approximation by means of vector-valued discrete functions defined on a reference square element $\hat{K}$ and mapped to the actual quadrilateral element K via the Piola transformation (2.1.69).

Given a smooth function $\underline{q} : \Omega \to \mathbb{R}^2$, we say that a finite element space family $\{X_h\}$ is optimally convergent in $L^2(\Omega)$ if

$$\inf_{\underline{p_h} \in X_h} \|\underline{q} - \underline{p_h}\|_{0,\Omega} = O(h^{k+1}), \tag{2.5.54}$$

where k refers to the polynomial degree of the reference finite element space. Similarly, the finite element space family $\{X_h\}$ is optimally convergent in the $H(\mathrm{div};\Omega)$ semi-norm if

$$\inf_{\underline{p_h} \in X_h} \|\mathrm{div}_h(\underline{q} - \underline{p_h})\|_{0,\Omega} = O(h^{k+1}), \tag{2.5.55}$$

where div_h is the divergence operator evaluated element by element.

Let $\widetilde{\mathcal{RT}}_{[k]}(\hat{K})$ be the subspace of co-dimension one of $\mathcal{RT}_{[k]}(\hat{K})$, where the two highest order fields $(\hat{x}^{k+1}\hat{y}^k, 0)$ and $(0, \hat{x}^k\hat{y}^{k+1})$ are replaced by the single field $(\hat{x}^{k+1}\hat{y}^k, -\hat{x}^k\hat{y}^{k+1})$. Then, for shape-regular families of quadrilateral meshes,

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condition (2.5.54) is valid if and only if the reference space contains $\widetilde{\mathcal{RT}}_{[k]}(\hat{K})$. Moreover, if the reference space contains $\widetilde{\mathcal{RT}}_{[k]}(\hat{K})$, then the following estimate holds true:

$$\inf_{\underline{p_h} \in \underline{X}_h} \|\underline{q} - \underline{p_h}\|_{0,\Omega} \leq c h^{k+1} |\underline{q}|_{k+1,\Omega}. \tag{2.5.56}$$

Finally, let $\tilde{Q}_k(\hat{K})$ be the subspace of co-dimension one of $Q_k(\hat{K})$ obtained by eliminating the higher order term $\hat{x}^k \hat{y}^k$. Then, condition (2.5.55) holds true if and only if the divergence of the reference space contains $\tilde{Q}_{k+1}(\hat{K})$. In this case the following estimate holds true:

$$\inf_{\underline{p_h} \in \underline{X}_h} \| \operatorname{div}(\underline{q} - \underline{p_h})\|_{0,\Omega} \leq c h^{k+1} | \operatorname{div} \underline{q}|_{k+1,\Omega}. \tag{2.5.57}$$

Remark 2.5.5. The reported results have dramatic consequences for the finite elements presented in Sect. 2.4. In particular, it turns out that none of the presented finite element families achieve optimal convergence in $H(\operatorname{div}; \Omega)$ on general quadrilateral meshes. Actually, $\mathcal{RT}_{[k]}$ is optimal in $L^2(\Omega)$ (since, of course, $\mathcal{RT}_{[k]}(\hat{K})$ contains $\widetilde{\mathcal{RT}}_{[k]}(\hat{K})$), but not in $H(\operatorname{div}; \Omega)$ (in particular, there is no convergence of the divergence for $k = 0$); while $\mathcal{BDM}_{[k]}$ and $\mathcal{BDFM}_{[k]}$ are suboptimal both in $L^2(\Omega)$ and $H(\operatorname{div}; \Omega)$. □

A possible cure to the pathologies outlined in Remark 2.5.5 has been presented in [21] where the family of spaces $\mathcal{ABF}$ is introduced. The basic idea is to add $H(\operatorname{div})$-conforming bubbles to the $\mathcal{RT}$ spaces so that optimal convergence properties can be achieved.

The results of this section, as it has been already observed, apply to finite element spaces which are defined on the reference element and mapped to the actual element by means of standard transformations. The suboptimal approximation orders have been observed in practical computations (see [20–22, 91]).

However, other finite element definitions are possible for which the (negative) results of this section might not apply. An example of such definitions is the non-conforming quadrilateral element presented in [330] which is constructed locally on the physical element or the reduced integration technique (interpreted as a local projection technique) presented in [94], where optimal convergence in $H(\operatorname{div}; \Omega)$ for the $\mathcal{RT}_{[k]}$ family is recovered.

2.5.6 *Discrete Exact Sequences*

We have introduced in Sect. 2.1.4 the exact sequence (2.1.104). We now show how this translates to some of the finite element approximations introduced above. Let us make a particular choice of finite elements approximating the functional spaces involved with (2.1.104). We consider a simplicial decomposition of a simply connected domain Ω in $\mathbb{R}^3$ and use the following finite elements: we take

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$\mathcal{L}^1_{k+1}$ (i.e., standard continuous piecewise polynomials of degree $k+1$) for the approximation of $H^1(\Omega)$, first kind Nédélec elements $\mathcal{N}_k$ for the approximation of $H(\underline{\mathrm{curl}};\Omega)$, Raviart–Thomas elements $\mathcal{RT}_k$ for the approximation of $H(\mathrm{div};\Omega)$, and discontinuous elements $\mathcal{L}^0_k$ for the approximation of $L^2(\Omega)$. Moreover, we consider the three sets of interpolation operators onto these spaces r_h, Σ_h, Π_h, respectively, and the L^2 projection P_h. With these particular choices, de Rham complex reads as follows:

$$\begin{array}{ccccccc} C^\infty(\Omega) & \xrightarrow{\underline{\mathrm{grad}}} & (C^\infty(\Omega))^3 & \xrightarrow{\underline{\mathrm{curl}}} & (C^\infty(\Omega))^3 & \xrightarrow{\mathrm{div}} & C^\infty(\Omega) \\ \downarrow{\scriptstyle r_h} & & \downarrow{\scriptstyle \Sigma_h} & & \downarrow{\scriptstyle \Pi_h} & & \downarrow{\scriptstyle P_h} \\ \mathcal{L}^1_{k+1} & \xrightarrow{\underline{\mathrm{grad}}} & \mathcal{N}_k & \xrightarrow{\underline{\mathrm{curl}}} & \mathcal{RT}_k & \xrightarrow{\mathrm{div}} & \mathcal{L}^0_k \end{array} \tag{2.5.58}$$

Diagram (2.5.58) has to be understood in the sense that the two lines are exact and the entire diagram commutes. Thus, for instance, we have that $\underline{\mathrm{grad}}\, r_h v = \Sigma_h \underline{\mathrm{grad}}\, v$ or $\Pi_h \underline{\mathrm{curl}}\, \underline{\chi} = \underline{\mathrm{curl}}\, \Sigma_h \underline{\chi}$. Some of these properties have been already recalled in this chapter (see, in particular, (2.5.27) and (2.5.50)) when analysing the finite element spaces. We refer the interested reader to [18, 33] where the general results are stated and where it has been shown that several other finite element choices are possible for the diagram to commute. We could indeed consider, for example, instead of (2.5.58),

$$\begin{array}{ccccccc} C^\infty(\Omega) & \xrightarrow{\underline{\mathrm{grad}}} & (C^\infty(\Omega))^3 & \xrightarrow{\underline{\mathrm{curl}}} & (C^\infty(\Omega))^3 & \xrightarrow{\mathrm{div}} & C^\infty(\Omega) \\ \downarrow{\scriptstyle r_h} & & \downarrow{\scriptstyle \Sigma_h} & & \downarrow{\scriptstyle \Pi_h} & & \downarrow{\scriptstyle P_h} \\ \mathcal{L}^1_{k+2} & \xrightarrow{\underline{\mathrm{grad}}} & \mathcal{NC}_{k+1} & \xrightarrow{\underline{\mathrm{curl}}} & \mathcal{BDM}_{k+1} & \xrightarrow{\mathrm{div}} & \mathcal{L}^0_k \end{array} \tag{2.5.59}$$

Remark 2.5.6. The information contained in Sect. 2.1.4 and in this section are by far not exhaustive of the connections between exterior calculus and finite element analysis. This active research area, not only proves useful for the analysis and the understanding of existing finite elements, but is also of fundamental importance for the design of new schemes. The reader who needs an introduction into this fascinating field is referred to the seminal papers [33, 34]. A more recent and succinct overview of finite element spaces constructed in the language of differential forms can be found in [12].

2.6 Explicit Basis Functions for $H(\mathrm{div};K)$ and $H(\underline{\mathrm{curl}};K)$ on Triangles and Tetrahedra

Although it is obviously possible to make explicit the previously defined spaces on a reference element and to transfer them on an arbitrary element by the Piola transformation (2.1.69) or (2.1.82), it is worth mentioning that there exist, on

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simplicial elements, general formulas using barycentric coordinates which are therefore totally general. We shall first introduce some notation.

- In the two-dimensional case, $\underline{x}_i, \underline{x}_j, \underline{x}_k$ are the three vertices of a triangle and $\lambda_i, \lambda_j, \lambda_k$ the associated barycentric coordinates.
- In the three-dimensional case, $\underline{x}_i, \underline{x}_j, \underline{x}_k, \underline{x}_l$ are the four vertices of a tetrahedron and $\lambda_i, \lambda_j, \lambda_k, \lambda_l$ the associated barycentric coordinates.
- We shall denote by $\underline{t}_{ij} = \underline{x}_j - \underline{x}_i$ the edge connecting $\underline{x}_i$ and $\underline{x}_j$ and similarly for other pairs of indices. We shall write l_{ij} the length of this vector.
- In the three-dimensional case, we denote f_{ijk} the face of the tetrahedron defined by the vertices $\underline{x}_i, \underline{x}_j, \underline{x}_k$ and similarly for other indices.
- The height of the triangle from $\underline{x}_k$ to the opposite edge $\underline{t}_{ij}$ will be denoted by h_k.
- The height of the tetrahedron from $\underline{x}_l$ to the opposite face f_{ijk} will be denoted by h_l.
- In a triangle, the outward normal to an edge t_{ij} is denoted by $\underline{n}_{ij}$. In a tetrahedron the outward normal to a face f_{ijk} is denoted by $\underline{n}_{ijk}$.

The gradient of the barycentric coordinates is related to the normals. In the two-dimensional case we have

$$\underline{\mathrm{grad}}\,\lambda_k = -\frac{1}{h_k}\underline{n}_{ij} \tag{2.6.1}$$

and the similar three-dimensional formula is

$$\underline{\mathrm{grad}}\,\lambda_l = -\frac{1}{h_l}\underline{n}_{ijk}. \tag{2.6.2}$$

When building basis functions, we shall distinguish between those associated with edge or face degrees of freedom and 'bubble' basis functions associated to degrees of freedom internal to K. We shall make explicit the lower order cases.

2.6.1 *Basis Functions for $H(\mathrm{div};K)$: The Two-Dimensional Case*

A basis for $\mathcal{BDM}_1(K)$ (6 degrees of freedom). Let us define for an edge $\underline{t}_{ij}$, $\underline{x}_k$ being the opposite vertex,

$$\phi_{i,ij} := \frac{1}{h_k}\,\underline{t}_{ki}\,\lambda_i \tag{2.6.3}$$

and

$$\phi_{j,ij} := \frac{1}{h_k}\,\underline{t}_{kj}\,\lambda_j. \tag{2.6.4}$$

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Let n_{ij} be the normal to $\underline{t}_{ij}$. We have

$$\phi_{i,ij} \cdot n_{ij} = \lambda_i \tag{2.6.5}$$

$$\phi_{j,ij} \cdot n_{ij} = \lambda_j . \tag{2.6.6}$$

Using all possible indices, we get six such functions, two for each edge. From (2.6.5) and (2.6.6), it is clear that we have thus obtained a basis for $\mathcal{BDM}_1(K)$. □

A basis for $\mathcal{RT}_0(K)$ (3 degrees of freedom). Summing (2.6.5) and (2.6.6), we obtain

$$\phi_{i,ij} \cdot n_{ij} + \phi_{j,ij} \cdot n_{ij} = 1. \tag{2.6.7}$$

Defining

$$\phi_{ij} := \phi_{i,ij} + \phi_{j,ij} \tag{2.6.8}$$

we therefore have a basis function for $\mathcal{RT}_0(K)$, associated with the edge $\underline{t}_{ij}$. It is easy to see that we have

$$\phi_{ij} = \frac{1}{h_k}(\underline{x} - \underline{x}_k). \tag{2.6.9}$$

which is a more standard way of writing this basis. □

A basis for $\mathcal{BDM}_2(K)$ (12 degrees of freedom). To make this construction, we need to increase the order of polynomials on the edges. We can indeed define on each edge,

$$\underline{t}_{m,ij} := (\underline{t}_{ki} + \underline{t}_{kj})/2 \tag{2.6.10}$$

and

$$\phi_{m,ij} := \frac{1}{h_k} \underline{t}_{m,ij} \lambda_i \lambda_j . \tag{2.6.11}$$

To get $\mathcal{BDM}_2(K)$, we add these three functions to those previously obtained in (2.6.3) and (2.6.4). This yields 3 degrees of freedom per edge.

To complete the construction, we also need to consider 'bubble' functions in the sense of $H(\mathrm{div}; K)$. This means that their normal component must vanish on ∂K. They can be built in an easy way from the quadratic functions

$$b_{ij} := \underline{t}_{ij} \lambda_i \lambda_j . \tag{2.6.12}$$

We have three such expressions, one associated with each edge. With the previously defined 9, we obtain a basis for $\mathcal{BDFM}_2(K)$. □

A basis for $\mathcal{BDFM}_1(K)$ (9 degrees of freedom) and $\mathcal{RT}_1(K)$ (8 degrees of freedom). A basis for $\mathcal{BDFM}_1(K)$ is readily obtained by suppressing the function defined in (2.6.10) from the previous construction. To get a basis for $\mathcal{RT}_1(K)$ we have to suppress a combination of the bubbles. This is allowed because there exists such a combination which is divergence-free and which, in a sense, does not

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contribute. However, this construction shows that, in a way, the slightly richer space $\mathcal{BDFM}_1(K)$ is respecting better the symmetries of the triangle. □

A basis for $\mathcal{BDFM}_3(K)$ (20 degrees of freedom). We follow the same lines. We can build 4 degrees of freedom on each edge, using for example (2.6.3) and (2.6.4), and

$$\phi_{\frac{1}{3},ij} = \frac{1}{h_k}\left(\frac{1}{3}\underline{t}_{ki} + \frac{2}{3}\underline{t}_{kj}\right), \tag{2.6.13}$$

$$\phi_{\frac{2}{3},ij} = \frac{1}{h_k}\left(\frac{2}{3}\underline{t}_{ki} + \frac{1}{3}\underline{t}_{kj}\right). \tag{2.6.14}$$

Bubbles can be generated from (2.6.12) using the following expression

$$b_{\alpha_1,\alpha_2,\alpha_3,ij} := \underline{t}_{ij}\lambda_i^{\alpha_i}\lambda_j^{\alpha_j}\lambda_k^{\alpha_k},\ (\alpha_i+\alpha_j+\alpha_k)=3. \tag{2.6.15}$$

We remark that $b_{1,2,0,ij} + b_{2,1,0,ij} = b_{ij}$ so that we still have the bubbles of $\mathcal{BDM}_1(K)$. Moreover, we note that the three functions of the form $b_{1,1,1,ij}$ are not linearly independent as we have

$$\underline{t}_{ij} + \underline{t}_{jk} + \underline{t}_{ki} = 0. \tag{2.6.16}$$

We should then select two linear combinations of these three functions to get a basis. □

A basis for $\mathcal{BDFM}_2(K)$ (17 degrees of freedom) and $\mathcal{RT}_2(K)$ (15 degrees of freedom). To obtain $\mathcal{BDFM}_2(K)$, the easiest way is to consider on the edges the basis functions for $\mathcal{BDM}_2(K)$ and the eight bubbles of $\mathcal{BDM}_3(K)$. For $\mathcal{RT}_2(K)$, we must again eliminate the divergence-free combination of bubbles. □

2.6.2 *Basis Functions for $H(\mathrm{div};K)$: The Three-Dimensional Case*

We can now consider the three-dimensional case.

A basis for $\mathcal{BDM}_1(K)$ (12 degrees of freedom) and $\mathcal{RT}_0(K)$ (4 degrees of freedom). On the face f_{ijk} we have three basis functions

$$\phi_{i,ijk} := \frac{1}{h_l}\,\underline{t}_{li}\,\lambda_i \tag{2.6.17}$$

$$\phi_{j,ijk} := \frac{1}{h_l}\,\underline{t}_{lj}\,\lambda_j \tag{2.6.18}$$

$$\phi_{k,ijk} := \frac{1}{h_l}\,\underline{t}_{lk}\,\lambda_k. \tag{2.6.19}$$

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Considering all four faces, we have obtained a basis for $\mathcal{BDM}_1(K)$. It is interesting to note that it employs the same edge-based basis functions as in the two-dimensional case. The sum

$$\phi_{ijk} := \phi_{i,ijk} + \phi_{j,ijk} + \phi_{k,ijk} = \frac{1}{h_l}(\underline{x} - \underline{x}_l) \tag{2.6.20}$$

is a basis function for $\mathcal{RT}_0(K)$. □

A basis for $\mathcal{BDFM}_1(K)$ (18 degrees of freedom) and $\mathcal{RT}_1(K)$ (4 degrees of freedom). As in the two-dimensional case, we can associate a 'bubble', function to each edge, using exactly the same formula (2.6.12) as in the two-dimensional case. We now have six such bubbles which we can use to build $\mathcal{BDFM}_1(K)$ by adding them to the basis of $\mathcal{BDM}_1(K)$. There exist three linear combinations of these bubbles which are divergence free and which can be eliminated to obtain $\mathcal{RT}_1(K)$. □

The reader should have now understood the mechanism and be able to move to higher degrees.

2.6.3 *Basis Functions for $H(curl; K)$: The Two-Dimensional Case*

As we have noted previously, in the two-dimensional case, the space $H(\underline{\text{curl}}; K)$ is essentially the same as $H(\text{div}; K)$. A basis for the discrete spaces can readily be obtained by a rotation of the vectors. However, we can write the basic construction in a slightly different way that will be more suitable for the extension to the three-dimensional case. Instead of (2.6.3) and (2.6.4), let us define for edge $\underline{t}_{ij}$

$$\psi_{i,ij} := l_{ij}\,\lambda_i\ \underline{\text{grad}}\,\lambda_j \tag{2.6.21}$$

and

$$\psi_{j,ij} := l_{ij}\,\lambda_j\ \underline{\text{grad}}\,\lambda_i\,. \tag{2.6.22}$$

Given that $\underline{\text{grad}}\,\lambda_i$ is a vector orthogonal to $\underline{t}_{kj}$ of length $1/h_i$ it is easy to check that those definitions correspond exactly to what we obtain when replacing in (2.6.3) and (2.6.4) the vectors $\underline{t}_{ki}$ and $\underline{t}_{kj}$ by their orthogonal. It is also easy to see that the tangential components of $\psi_{i,ij}$ and $\psi_{j,ij}$ along $\underline{t}_{ij}$ are respectively λ_i and $-\lambda_j$.

2.6.4 *Basis Functions for $H(\underline{curl}; K)$: The Three-Dimensional Case*

A basis for $\mathcal{NC}_1(K)$ (12 degrees of freedom) and $\mathcal{N}_0(K)$ (6 degrees of freedom). In order to define the lowest degree spaces, we define 12 basis functions associated to edges. The interesting fact is that we can still use on any edge $\underline{t}_{ij}$ the expressions

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(2.6.21) and (2.6.22). On the other edges, the vectors obtained in this way are either zero because, for example, λ_i is zero or orthogonal to the edge. It is thus clear that we have obtained a basis for $\mathcal{NC}_1(K)$. Summing the two expressions (up to the orientation), we get the basis for $\mathcal{N}_0(K)$

$$\psi_{ij} := l_{ij}\,(\lambda_j\ \underline{\mathrm{grad}}\,\lambda_i - \lambda_i\ \underline{\mathrm{grad}}\,\lambda_j). \tag{2.6.23}$$

□

We can move to higher degree elements by using similar constructions.

***H*(curl; *K*)-Bubbles.** We shall be interested in the $H(\underline{\mathrm{curl}}; K)$-bubbles which correspond to the internal degrees of freedom of $\mathcal{N}_k(K)$ and $\mathcal{NC}_k(K)$. We recall that from (2.3.46) and (2.3.50) these degrees of freedom are associated respectively to $(P_{k-2})^3$ and to $\mathcal{RT}_{k-2}(K) = (P_{k-2})^3 + \underline{x}P_{k-2}$. We thus suppose that $k \geq 2$ and we define on a tetrahedron, for the four faces defined by the choice of three barycentric coordinates $\lambda_i, \lambda_j, \lambda_k$,

$$\Phi_{ijk} = P_{k-2}(f_{ijk})\,\lambda_i\lambda_j\lambda_k\,\underline{n}_{ijk}. \tag{2.6.24}$$

For $k = 2$ this defines four functions which are associated with the four faces and which are the bubbles of $\mathcal{NC}_{k+1}(K)$. To obtain the bubbles of $\mathcal{N}_k(K)$ we must suppress the gradient of the standard bubble $\lambda_1\lambda_2\lambda_3\lambda_4$ which is

$$\lambda_1\lambda_2\lambda_3\,\underline{\mathrm{grad}}\,\lambda_4 + \lambda_1\lambda_2\lambda_4\,\underline{\mathrm{grad}}\,\lambda_3 + \lambda_1\lambda_3\lambda_4\,\underline{\mathrm{grad}}\,\lambda_2 + \lambda_2\lambda_3\lambda_4\,\underline{\mathrm{grad}}\,\lambda_1 \tag{2.6.25}$$

which is a combination of the four bubbles of (2.6.24) by (2.6.2).

For $k \geq 3$ we must add to the functions defined by (2.6.24) standard bubbles of the form

$$\underline{P}_{k-3}(K)\lambda_1\lambda_2\lambda_3\lambda_4 = \underline{P}_{k-3}(K)B_4(K), \tag{2.6.26}$$

which vanish totally on the boundary of K. This defines a basis for the bubbles of $\mathcal{NC}_k(K)$ and to obtain $\mathcal{N}_k(K)$ we need to remove the gradients of the standard bubbles. For $k = 3$, for example, we must remove the gradients of the four bubbles $\underline{P}_k(K)B_4(K)$.

2.7 Concluding Remarks

This chapter is evidently not a complete presentation of finite element approximation methods. It cannot be, unless it becomes a book by itself. Our aim was therefore to present examples of the most classical cases and to consider a construction for the less standard case $H(\mathrm{div}; \Omega)$ and $H(\underline{\mathrm{curl}}; \Omega)$. On the other hand, approximations of elasticity problems will also require special spaces. They will be described in due time. We however believe that the present chapter will then provide a sound basis for these developments.

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